

External Validity in Empirical Public Finance: With an Application to Taxable Income Elasticities ^{*}

Meng Hsuan Hsieh
Ross School of Business
University of Michigan
rexhsieh@umich.edu

Gerardo Sanz-Maldonado
Department of Economics
University of Michigan
sanzg@umich.edu

Joel Slemrod
Department of Economics
Ross School of Business
University of Michigan
jslemrod@umich.edu

August 1, 2026

Abstract

Estimates of the elasticity of taxable income (ETI) based on data from only one country implicitly assume nothing can be learned from the experience of other countries. This assumption is wrong, but rigorously relaxing it requires a principled framework to aggregate information from different contexts. We propose a general three-step empirical Bayes framework grounded in statistical decision theory that accomplishes this by modeling cross-country heterogeneity as a function of the structural determinants of the tax system. The framework determines endogenously how much each country's estimate should be informed by external evidence, placing more weight on cross-country evidence for countries whose within-country estimates are imprecise. We apply our framework to 27 OECD countries, characterize how ETI estimates should be revised taking into account evidence from other countries, and outline how to predict behavioral responses for countries that have not adopted a given tax.

^{*}We are grateful to Enrico Rubolino and Daniel Waldenström for providing data to us for our empirical work. We thank Xinyu Chen for careful and extensive data collection and cleaning. We thank the editor, Ben Lockwood, three anonymous referees, Isaiah Andrews, Will Boning, Ash Craig, Michael Devereux, Florian Gunsilius, Andreas Hagemann, Wojciech Kopczuk, Emmanuel Saez, and Jeff Smith for providing valuable feedback. We also thank participants in the Paris-London Public Economics Conference, the IFN Stockholm Conference on Taxation and Inequality, the Public Finance PhD Student Seminar at the University of Michigan, the Michigan Tax Invitational (M-TAXI), the University of Kentucky, the 2023 International Institute of Public Finance Annual Congress, the 2023 National Tax Association Annual Conference, and Oxford University for thoughtful comments during this paper's early stages.

1 Introduction

In this paper, we investigate the cross-country external validity of a central parameter in modern tax analysis—the elasticity of taxable income (ETI). The ETI has been the subject of an extensive empirical literature spanning many countries and time periods. The resulting estimates vary considerably, both across countries and across studies within the same country. This variation poses a practical challenge for policymakers. How informative, for example, is an estimate of the elasticity of taxable income in, say, Italy, for predicting the likely magnitude of the behavioral response to a rate change in Mexico? What light does the estimated elasticity of reported taxable wealth with respect to the annual wealth tax rate—estimated on Norwegian data—shed on the likely behavioral response if the United States were to enact a similar tax? ¹ At present, empirical public finance lacks a formalized framework to answer these questions.

To understand the consequences of this, consider the growing set of careful causal estimates of the ETI, derived from a combination of administrative data and estimation techniques such as bunching and instrumental variables. What makes these estimates credible is typically a combination of administrative microdata, deep institutional knowledge of the study setting’s tax system, and tools from the causal inference toolkit that yield internal validity. One natural way to think about the variance in these estimates is to treat them as separate, noisy attempts to recover a single underlying structural parameter—with cross-country differences reflecting sampling error rather than genuine heterogeneity. A notable issue with this approach is that, while it pools evidence across countries, the variation is simply too large and too systematic to be due only to sampling error (Rubolino and Waldenström, 2019; Neisser, 2021).

Alternatively, one may treat each country’s estimate as an unbiased measurement of a parameter that is entirely context-specific. The issue with this approach is that it imposes the unattractive restriction that each estimate stands alone: the ETI estimated from Danish data, for instance, says nothing about the ETI in New Zealand, even if the two countries share similar tax structures and social preferences. Yet statistical decision theory suggests one can do better: when estimating many related parameters simultaneously, ignoring the shared structure that governs their distribution means leaving potentially useful information on the table—the cross-country pattern of ETIs could improve any individual country’s estimate, and discarding it has a cost.

The economic theory of optimal tax systems provides some guidance on just what that structure might look like. Consider the view, due to Slemrod and Kopczuk (2002), that the ETI is not a primitive parameter, but rather depends on many aspects of the tax system such as the broadness of the tax base, the effectiveness of enforcement, and the availability of avoidance opportunities. In selecting the tax system parameters simultaneously with the effective progressivity of the tax system, a jurisdiction can be thought of as *choosing*

¹Within the scope of external validity, these questions focus on the issue of transportability: the problem of using causal evidence identified in one setting to draw valid inferences about a different target setting (Pearl and Bareinboim, 2011, 2014; Degtiar and Rose, 2023). Transportability is the counterpart of generalizability (Deaton and Cartwright, 2018; Findley et al., 2021); whereas generalizability asks whether an estimate generalizes beyond the context that produced it, transportability asks what must be measured, modeled, and assumed for evidence from one setting to inform decisions in another.

the ETI simultaneously with its choice of the tax rate schedule. This choice depends on the social preference for equality, which will lead to a higher top tax rate and a lower ETI, and on the social cost of implementing a broad-based, well-enforced tax system, which will tend to produce a lower top tax rate and a higher ETI. This implies that the cross-country variation in ETIs should be at least partially predictable as a function of these underlying determinants of the tax system.

We propose an econometric framework, grounded in the empirical Bayes (EB) approach (see, for example [Robbins, 1956](#); [Morris, 1983](#); [Walters, 2024](#)), for pooling evidence across contexts while respecting the structural relationship between the ETI and aspects of tax systems stipulated by optimal tax theory. We operationalize the EB framework using a three-stage procedure. In the first stage, we estimate each country’s ETI using only its own data—this tells us what we can learn from each country in isolation. In the second stage, we model how true ETIs vary across countries as a function of observable characteristics motivated by the optimal tax theory of [Slemrod and Kopczuk \(2002\)](#); this captures the systematic, (imperfectly) predictable component of cross-country heterogeneity. We let each characteristic enter through two separately identified channels: a within-country channel, identified from movements in that characteristic over time interacted with a country’s own tax variation, and a cross-country level channel, identified from differences in country characteristic averages. In the third stage, we combine these two sources of information optimally via EB shrinkage weights, where the weight placed on each country’s own estimate versus the cross-country prediction is determined by their relative precisions. Because both stages are estimated on the same panel, the optimal weight requires two corrections that we derive: a leverage term netting out the share of a country’s own data already embedded in the cross-country prediction, and an extrapolation penalty that returns weight to a country’s own estimate when its prediction must be extrapolated from a region of the covariate space it does not occupy. To our knowledge, both are new. In doing so we extend the classical compound-decision and parametric empirical Bayes tradition ([Robbins, 1956](#); [Morris, 1983](#); [Efron, 2019](#)) to a setting in which the prior mean is not a scalar, but is rather a structural function recovered from the pooled panel, linking our approach to recent work on covariate-powered shrinkage (e.g. [Armstrong et al., 2022](#); [Ignatiadis and Wager, 2022](#)). This framework has decision-theoretic foundations: the empirical Bayes estimator minimizes compound Bayes risk under squared error loss. The shrinkage weight is not a tuning parameter chosen by the analyst; it is the answer, derived from the data, to the question of how much a country should rely on external evidence.

We apply this framework to estimate the ETI for top income earners in 27 OECD countries from 1981 to 2014, using data on top income shares and tax rates drawn largely from [Rubolino and Waldenström \(2019\)](#). For the observable determinants of the ETI, we focus on the underlying features of a country that determine why it chooses the tax system it has, drawing on survey data from the World Values Survey and other sources to proxy for the social demand for redistribution and the social cost of achieving a progressive tax system. We deliberately do not explore direct measures of the broadness of the tax base or the extent of enforcement resources, as these are manifestations of deeper preferences and constraints rather than of the preferences and constraints themselves.

Our choice to estimate ETIs using consistent panel data across countries—rather than aggregating existing heterogeneous study estimates—reflects two considerations that have limited prior work in this space. One, as the meta-analysis of [Neisser \(2021\)](#) illustrates, is that existing studies differ in identification strategies, income concepts, and sample restrictions—making it difficult to separate underlying ETI heterogeneity from the impact of methodological choices. Another is that most countries lack the administrative data or within-country policy variation required to meet the standards of modern causal identification, so appropriate estimates only exist for a handful of countries. With a consistent identification strategy applied to comparable data, the same two unsatisfactory options from our motivating thought experiment reappear: estimate each country’s ETI in isolation—the segmented approach, which discards cross-country information in favor of treating each ETI estimate as context-specific—or pool all countries into a single common parameter, ignoring the observable characteristics that drive genuine ETI differences. Our consistent panel setting thus provides a clean empirical foundation for the EB framework: by holding the identification strategy fixed across countries, it ensures that observed cross-country differences in ETI estimates reflect genuine behavioral variation rather than artifacts of study design.

Estimating the ETI separately for each of our 27 OECD countries yields a simple cross-country average of 0.204, with the U.S. estimate of 0.240 modestly above the mean. The estimates display considerable dispersion, and many are statistically imprecise, in part reflecting limited within-country variation in top marginal tax rates. The cross-country structure speaks directly to this heterogeneity, linking each country’s estimated ETI to its observable characteristics. Most of the estimated coefficients on country characteristics are imprecisely estimated. Under our specification, which separates each characteristic’s within-country (dynamic) channel from its cross-country (level) channel, several characteristics nonetheless enter significantly: in the cross-country (level) channel, countries whose citizens place greater weight on government (rather than individual) responsibility, express more confidence in government, and view tax cheating as more justifiable have significantly lower ETIs, as do more populous countries; in the within-country (dynamic) channel, a greater emphasis on efficiency over equity raises the ETI. Together, the observable characteristics account for our finding that the US has a slightly higher ETI than a country-segmented estimation would suggest: the US lies on the elasticity-raising side of several of the attitudes that enter the structural relationship, placing relatively low weight on government responsibility, viewing tax cheating as relatively unjustifiable, and placing a low value on redistribution, although this is substantially counter-weighted by the large US population, which puts downward pressure on the ETI.

Having estimated the cross-country structure, we turn to a question that is central to transportability: how should we revise our understanding of a country’s ETI when we observe data from additional jurisdictions? Within the empirical Bayes framework, the answer has a clean structure. Because first-stage estimates are computed from within-country data alone, they are invariant to the arrival of new countries: learning new evidence of behavioral responses from new countries does not “update” our understanding of their magnitudes in existing countries. New data affect existing countries’ estimates entirely through two updated objects: (i) the estimated coefficient vector, which governs the predicted ETI as a function of observable country

characteristics, and (ii) the estimated residual variance, which measures the cross-country heterogeneity not captured by observables. We decompose the resulting revision to any country’s empirical Bayes estimate into two interpretable channels: a prior mean channel, operating through revised coefficients on observables, and a shrinkage channel, operating through a revised estimate of residual heterogeneity. We show that a new country increases the estimated residual heterogeneity if and only if its squared residual from the existing model exceeds the expected total variance—that is, if the new country is “surprising” relative to the existing cross-country pattern. This characterization of how data from a newly observed jurisdiction propagate to every existing estimate is, to our knowledge, new.

Finally, we distinguish between two forms of transportability that our framework can address. The main application—estimating country-specific ETIs for countries that all levy an income tax—involves optimally combining a within-country estimate with a cross-country prediction. A conceptually different problem arises when we seek to predict the behavioral response in a country that has not adopted the tax in question. In this case, there is no within-country estimate to shrink toward the cross-country prediction; the prediction rests entirely on the estimated structural relationship and the non-adopter’s observable characteristics. We outline how the empirical Bayes framework handles this as out-of-sample prediction, using wealth taxation as a motivating example. Because the adoption of a wealth tax is endogenous—crucially, jurisdictions with lower behavioral elasticities are, *ceteris paribus*, more likely to adopt a wealth tax, because the social cost due to behavioral response is lower—the average elasticity among adopters systematically understates the elasticity a non-adopter would face. To resolve this challenge, we augment our second-stage estimation procedure with a selection model, and propose a new selection-corrected empirical Bayes estimator that generalizes the classic Heckman selection model (Heckman, 1979) to allow us to estimate the structural relationship between observables and the ETI. Under this model, we illustrate a selection-bias correction procedure and show how to estimate valid prediction intervals. Although data limitations prevent us from presenting estimates of the elasticity of taxable wealth, the framework provides prediction intervals that reflect both the estimation uncertainty in the cross-country relationship and the heterogeneity not captured by observables.

Although we develop and apply our framework in the context of taxable income elasticities, the approach is general. Any setting where a researcher has estimated context-specific parameters and suspects they are governed by a shared structure is amenable to the same three-step procedure. First, one collects or produces estimates of the parameter of interest from each context, along with their associated standard errors—these are the inputs to the measurement equation. Second, one specifies a model for how the true parameters vary across contexts as a function of observable characteristics, guided by theory about why the parameter should differ across settings. Third, one computes the precision-weighted combination of each context’s own estimate and the cross-context prediction, where the weights are determined by the relative informativeness of the two sources. The result is a set of context-specific estimates that are disciplined by cross-context structure, without being forced to conform to it. The framework requires the analyst to take a stand on what observable characteristics belong in the second stage, but it does not require that these characteristics

explain all of the cross-context heterogeneity—any residual heterogeneity is estimated from the data and incorporated into the shrinkage weights. In this sense, our empirical application to the ETI is a proof of concept: the substantive findings about which country characteristics predict the ETI are not dispositive, but the framework itself is immediately portable to other settings where context-specific behavioral parameters are estimated in isolation and a structural basis for cross-context comparison exists.

The rest of this paper is structured as follows. Section 2 presents the general three-stage empirical Bayes framework, establishes its decision-theoretic foundations, and shows how it nests prior approaches. Section 3 describes our data and presents the empirical results, including country-specific ETI estimates under both the CRC and empirical Bayes specifications. Section 4 develops the extensions: learning from new data, prediction for non-adopters, and observing repeated observations from a single jurisdiction. Section 5 concludes.

2 Model and Theory

The central challenge is how to balance within-country evidence from each country’s own tax rate variation against cross-country evidence about how ETIs relate to observable characteristics. In this section, we formalize the three-stage EB framework, first establishing its decision-theoretic foundations and then showing how it nests several widely-used alternatives as special cases.

Our framework builds directly on the compound decision theory pioneered by Robbins (1951, 1956) and subsequent advances in empirical Bayes methods (Morris, 1983; Efron, 2019; Armstrong et al., 2022; Ignatiadis and Wager, 2022; Chen, 2026). The core insight is that, when we face the simultaneous estimation of N distinct, jurisdiction-specific ETIs, $\varepsilon_1, \dots, \varepsilon_N$, it is suboptimal to estimate each in isolation. The estimation problem amounts to seeking a decision rule $\delta = (\delta_1, \dots, \delta_N)$ that minimizes the compound Bayes risk

$$\mathcal{R}_N(\theta, \delta) = \mathbb{E}_\theta \left[\frac{1}{N} \sum_{i=1}^N (\delta_i(D_N) - \varepsilon_i)^2 \right], \quad (1)$$

where D_N is the full available dataset across all N jurisdictions, and θ parameterizes the shared cross-country structural relationship. By the law of iterated expectations, this compound risk is minimized when each individual decision rule δ_i is chosen to be the posterior mean $\mathbb{E}_\theta[\varepsilon_i \mid D_N]$. A fundamental insight of this theoretical framework is the bias-variance decomposition: any estimator’s squared error decomposes into its squared deviation from the posterior mean plus the irreducible posterior variance. Selecting the posterior mean therefore guarantees optimality across jurisdictions in the sense of minimizing (1).

By framing the estimation in this manner, we can rigorously define and operationalize external validity. By external validity, we mean that the structural relationship between the ETI and observable characteristics is stable across jurisdictions. We describe the three stages in turn below, and then discuss the framework’s properties, its relationship to existing approaches, and several conceptual issues that arise in its application.

All proofs and technical exposition are deferred to Appendix A.

2.1 Stage 1: Within-Country Identification

Consider the following panel data model for country i :

$$y_{it} = \alpha_i + \lambda_i t + \varepsilon_i R_{it} + \eta_{it}, \quad t = 1, \dots, T, \quad (2)$$

where y_{it} is the log of the top income share, $R_{it} \equiv \log(1 - \tau_{it})$ is the log net-of-tax rate, α_i and λ_i are country-specific intercepts and linear time trends, respectively, and η_{it} is a structural error term representing idiosyncratic shocks to the income share. This is a standard empirical specification for estimating the ETI. For each country i , we call (2) the segmented regression. Identification of ε_i relies strictly on within-country, over-time variation in the net-of-tax rate R_{it} after partialling out country fixed effects, linear trends (and potentially covariates).

2.2 Stage 2: Cross-Country Heterogeneity Modeling

In the second stage, we incorporate the insights of optimal tax theory (Slemrod and Kopczuk, 2002), positing that the true country-specific ETI, ε_i , is a structural function of observable country characteristics $Z_{i,t}$ that potentially varies over time. As a result, the ETI is a time-varying parameter. We put the following modelling assumptions on this parameter of interest.

Assumption 2.1 (Behavioral Elasticity). For each jurisdiction i ,

- (i) The country-specific behavioral elasticity varies over time according to:

$$\varepsilon_{it} = \varepsilon + b' Z_{it} + u_i^\varepsilon, \quad (3)$$

where ε is a constant, b is a vector of parameters reflecting response to time-varying characteristics Z_{it} , and u_i^ε is a time-invariant, jurisdiction-specific latent random effect representing unobservable structural heterogeneity.

- (ii) The error term u_i^ε of (3) takes the form

$$u_i^\varepsilon = \psi' \bar{Z}_i + \tilde{u}_i^\varepsilon, \quad (4)$$

where $\bar{Z}_i \equiv \frac{1}{T} \sum_{t=1}^T Z_{it}$ is the time average, and the structural projection error satisfies the normalization $\mathbb{E}[\tilde{u}_i^\varepsilon \mid \mathbf{Z}_i] = 0$ and $\text{Var}(\tilde{u}_i^\varepsilon \mid \mathbf{Z}_i) = s^2$.

Assumption 2.1 describes the true data-generating process for the ETI. Two features are of note in this data-generating process. First, controlling for $\bar{Z}_i$ absorbs the component of the latent structural heterogeneity u_i^ε that is correlated with the levels of the observed characteristics. This matters because governments

plausibly choose tax rates, and time reforms, on the basis of the same persistent characteristics that drive the elasticity; projecting u_i^ε onto $\bar{Z}_i$ accounts for that channel of correlation between the tax path and the elasticity, rather than assuming it away. It does not, however, dispose of the endogeneity problem entirely: any correlation between the tax path and the projection error $\tilde{u}_i^\varepsilon$ remains, and is ruled out by assumption in Assumption 2.2(i) below. Second, the parameter b captures how within-country changes in characteristics dynamically shift the behavioral response, whereas the level coefficient π , defined in (5) below, captures the same effect coming from cross-country level differences. These two channels are separately identified. The modelling decisions in Assumption 2.1 are also known as Mundlak correlated random effects (Mundlak, 1978; Chamberlain, 1984). In the panel data literature, this is known as the correlated random coefficients (CRC) model, which has been studied and discussed in, among others, Chamberlain (1982, 1984); Wooldridge (2010); Hastie and Tibshirani (1993); Beran (1995); Cai et al. (2000); Fan and Zhang (2008); Park et al. (2015). Finally, with $\ddot{Z}_{it} \equiv Z_{it} - \bar{Z}_i$ denoting the within-country deviation, Assumption 2.1 implies

$$\varepsilon_{it} = \varepsilon + b' \ddot{Z}_{it} + \pi' \bar{Z}_i + \tilde{u}_i^\varepsilon, \quad \pi \equiv b + \psi. \quad (5)$$

The level coefficient π is the object identified by cross-country differences in average characteristics.

We are now ready to describe what the segmented regressions of stage 1 estimate. Let boldface denote the vector of an entire panel of a variable, i.e. $\mathbf{R} = (R_{i,t})_{i,t}$. We maintain the following standard set of assumptions.

Assumption 2.2 (Strict Exogeneity). For each jurisdiction i :

- (i) Conditional on the full history of tax rates and covariates, the raw idiosyncratic shocks and the random effect projection errors are mean-independent of the regressors:

$$\mathbb{E}[\eta_{it} \mid \mathbf{R}, \mathbf{Z}] = 0, \quad \mathbb{E}[\tilde{u}_i^\varepsilon \mid \mathbf{R}, \mathbf{Z}] = 0. \quad (6)$$

- (ii) Conditional on $(\mathbf{R}, \mathbf{Z})$, the shocks to the dependent variable η_{it} are serially independent with constant variance $\text{Var}(\eta_{it} \mid \mathbf{R}, \mathbf{Z}) = \sigma_\eta^2$.
- (iii) $\{\eta_{it}\}_i$ and $\{\tilde{u}_i^\varepsilon\}_i$ are mutually independent, and both are independent across jurisdictions, conditional on $(\mathbf{R}, \mathbf{Z})$.
- (iv) Conditional on $(\mathbf{R}, \mathbf{Z})$, $\eta_{it} \sim \mathcal{N}(0, \sigma_\eta^2)$.

In light of this structure, we show the following identification result for segmented regressions. In the following, we write $\omega_i \equiv \sum_{t=1}^T (R_{it}^\perp)^2$ to denote country i 's identifying tax variation. $\mathbf{W}_i$ denotes the stacked $T \times (2K + 1)$ regressors matrix for country i (with rows $\mathbf{W}_{it}' = (R_{it}^\perp, ((\ddot{Z}_{it} R_{it})^\perp)', \bar{Z}_i' R_{it}^\perp)$; each block is the corresponding regressor obtained by substituting (5) into (2), residualized on the country fixed effects and trends; the level block reduces to $\bar{Z}_i' R_{it}^\perp$ because $\bar{Z}_i$ is time-invariant within country i), and

$\mathbf{Z}_i^* \equiv \omega_i^{-1} \mathbf{W}_i' \mathbf{R}_i^\perp$ to denote the tax-variation-weighted covariates, where the weights come from that we are exploiting within-country variation in the tax rates in estimating OLS regressions. For convenience, we use the equivalent form $\mathbf{Z}_i^* = \begin{pmatrix} 1 & \ddot{Z}_i^* & \bar{Z}_i \end{pmatrix}'$ for $\ddot{Z}_i^* \equiv \frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp (\ddot{Z}_{it} R_{it})^\perp$ and $\ddot{Z}_{it}$ are the demeaned observable factors, which comes from straightforward algebraic manipulations; the last block follows from $\sum_t (\bar{Z}_i R_{it}^\perp) R_{it}^\perp = \bar{Z}_i \omega_i$. See Lemma D.1 for details.

Theorem 2.3. Suppose Assumption 2.2(i)–(iii) holds. Then, conditional on $(\mathbf{R}, \mathbf{Z})$ and $\tilde{u}_i^\varepsilon$, the segmented estimate $\hat{\varepsilon}_i$ is an unbiased estimator of the true structural ETI evaluated at the tax-variation-weighted covariates $\mathbf{Z}_i^*$, denoted ε_i^* . As a consequence, we have

$$\hat{\varepsilon}_i = \varepsilon_i^* + v_i, \quad \mathbb{E}[v_i \mid \mathbf{R}, \mathbf{Z}, \tilde{u}_i^\varepsilon] = 0, \quad \text{Var}(v_i \mid \mathbf{R}, \mathbf{Z}) = \sigma_i^2 = \frac{\sigma_\eta^2}{\omega_i}, \quad (7)$$

where $v_i = \frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp \eta_{it}^\perp$ is the sampling error, independent of ε_i^* conditional on $(\mathbf{R}, \mathbf{Z})$. If, in addition, Assumption 2.2(iv) holds, then $v_i \mid \mathbf{R}, \mathbf{Z} \sim \mathcal{N}(0, \sigma_i^2)$.

A consequence of Theorem 2.3 is that the interpretation of segmented estimates of the ETI is not directly comparable across countries. By design, segmented estimates use only within-country variations, and the estimates are not informed by the cross-country evidence that we have available. Furthermore, we implicitly ignore any residual heterogeneity among estimates $\hat{\varepsilon}_i$ that is not captured beyond the observable factors Z_i . Residual heterogeneity after modelling Z_i need to be estimated directly, which is itself a nontrivial problem.

To model cross-country heterogeneity, we employ a hierarchical model in the tradition of EB frameworks (e.g. Robbins, 1956; Morris, 1983) to model the distribution of true ETIs ε_i^* across countries.

Assumption 2.4 (Hierarchical model). The true ETI estimand, ε_i^* , is drawn from a normal distribution conditional on the tax-variation-weighted covariates $\mathbf{Z}_i^*$:

$$\varepsilon_i^* \mid \mathbf{Z}_i^* \sim \mathcal{N}((\mathbf{Z}_i^*)' \beta, s^2). \quad (8)$$

Furthermore, the parameters $(\varepsilon_1^*, \dots, \varepsilon_N^*)$ and the measurement errors $(v_1, \dots, v_N)$ are mutually independent across countries, conditional on observable factors.

Following the empirical Bayes literature, s^2 in (8) captures cross-country variations in the true ETI estimand beyond what a set of observed factors captures. We refer to this as residual structural heterogeneity. Combining the measurement equation (7) with the hierarchical model (8) yields the marginal distribution of the *observed* ETI estimates:

$$\hat{\varepsilon}_i \mid \mathbf{Z}_i^* \sim \mathcal{N}((\mathbf{Z}_i^*)' \beta, s^2 + \sigma_i^2). \quad (9)$$

The total variance of the observed estimates, $s^2 + \sigma_i^2$, decomposes cleanly into two sources: the true, underlying structural heterogeneity (s^2) and the statistical sampling error (σ_i^2). We will hereafter refer to $\theta = (\beta, s^2)$ as parameters estimated in the second stage of our method.

In (8), the prior mean is $(\mathbf{Z}_i^*)'\beta$, which is based on the structural assumptions in Assumption 2.4. To estimate β , based on the structural assumptions in Assumption 2.4, we employ the CRC estimator; this is a regression estimated on the pooled panel data, and takes the form

$$\hat{\beta}^{CRC} = \mathbf{M}^{-1} \sum_{j=1}^N \sum_{s=1}^T \mathbf{W}_{js} y_{js}^\perp, \quad \text{where} \quad \mathbf{M} = \sum_{j=1}^N \sum_{s=1}^T \mathbf{W}_{js} \mathbf{W}_{js}' . \quad (10)$$

In Appendix A, we show that this pooled panel CRC estimator consistently estimates β under the maintained assumptions above plus some regularity conditions. In practice, β is estimated with substantial precision because it pools observations from all countries in the data, avoiding the severe degrees-of-freedom issues of estimating β solely from the N country-level estimates $\hat{\varepsilon}_i$.

Intuitively, the pooled CRC regression exploits all tax variations in the panel: any within-country variation in $\ddot{Z}_{it}$ interacted with tax changes identifies b , while cross-country differences in average characteristics $\bar{Z}_i$ interacted with tax variation identify the level coefficients π from (4). This is where our method departs from standard meta-regression, which regresses the N noisy estimates $\hat{\varepsilon}_i$ on country characteristics and therefore cannot separately identify b or π in our notation, nor does it exploit within-country tax variations. Furthermore, under Assumption 2.4(iii), the variance of the estimator $\hat{\beta}^{CRC}$ can be shown to be

$$\mathbf{V}_\beta \equiv \text{Var}(\hat{\beta}^{CRC}) = \mathbf{V}_\beta^{\text{struct}} + \mathbf{V}_\beta^{\text{sampling}}, \quad \text{where} \quad \mathbf{V}_\beta^{\text{struct}} = s^2 \mathbf{M}^{-1} \left(\sum_{j=1}^N \omega_j^2 \mathbf{Z}_j^* (\mathbf{Z}_j^*)' \right) \mathbf{M}^{-1}, \quad \mathbf{V}_\beta^{\text{sampling}} = \sigma_\eta^2 \mathbf{M}^{-1}. \quad (11)$$

In this expression, the residual heterogeneity term s^2 needs to be estimated; in Appendix B, we discuss how to estimate this via a likelihood estimation procedure.²

2.3 Stage 3: Optimal Shrinkage and Prediction

The third stage optimally combines the within-country (segmented, stage 1) and cross-country estimates (CRC, stage 2). Let the structural prediction be $\hat{\varepsilon}_i^{CRC} = (\mathbf{Z}_i^*)' \hat{\beta}^{CRC}$, and define the prediction error

$$E_i \equiv \hat{\varepsilon}_i^{CRC} - \varepsilon_i^* = (\mathbf{Z}_i^*)' (\hat{\beta}^{CRC} - \beta) - \tilde{u}_i^\varepsilon,$$

where the second equality uses $\varepsilon_i^* = (\mathbf{Z}_i^*)'\beta + \tilde{u}_i^\varepsilon$. All moments of (v_i, E_i) below are taken conditional on $(\mathbf{R}, \mathbf{Z})$, so that $\mathbf{M}$, ω_i , and the leverage entries h_{ij} are fixed constants; the only sources of randomness are the structural draws $\tilde{u}_j^\varepsilon$ and the $\eta_{js}^\perp$. Because the pooled OLS estimator $\hat{\beta}^{CRC}$ utilizes the data of country i , the measurement error v_i and the prediction error E_i are correlated. To minimize the expected squared error

²We also note that it is possible to allow the residual heterogeneity to be heteroskedastic, so that it varies by country i . We provide details of this extension in Appendix D.4.

compound Bayes risk, we restrict our estimator to the class

$$\widehat{\varepsilon}_i^{EB} = w_i \widehat{\varepsilon}_i + (1 - w_i) \widehat{\varepsilon}_i^{CRC}. \quad (12)$$

This restriction is without loss of optimality within the Gaussian model: because $(\widehat{\varepsilon}_i, \widehat{\varepsilon}_i^{CRC}, \varepsilon_i^*)$ are jointly Gaussian conditional on $(\mathbf{R}, \mathbf{Z})$ under Assumption 2.2(iv), the posterior mean of ε_i^* is affine in the estimates $(\widehat{\varepsilon}_i, \widehat{\varepsilon}_i^{CRC})$, and imposing that the combination be unbiased for ε_i^* (i.e. the weights sum to one) pins it down to the form (12).

To formally introduce the optimal weights, we define the leverage matrix $\mathbf{H}$ with entries $h_{ij} \equiv \omega_j (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^*$. The entries h_{ij} measure how strongly country j 's data moves the CRC prediction of the ETI of country i ; analogously, h_{ii} measures how important country i 's own data is in the CRC prediction of its own ETI.

Theorem 2.5 (Optimal Leverage-Corrected Weight). Define $A_i \equiv (1 - h_{ii})^2 + \sum_{j \neq i} h_{ij}^2$. Under Assumption 2.2, the expected squared error risk-minimizing shrinkage weight is³

$$w_i^* = \frac{s^2 A_i}{s^2 A_i + \sigma_i^2 (1 - h_{ii})} \equiv \frac{(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* + s^2 (1 - 2h_{ii})}{(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* + s^2 (1 - 2h_{ii}) + \sigma_i^2 (1 - h_{ii})}. \quad (13)$$

Theorem 2.5 departs from the typical EB shrinkage weight of the form $s^2 / (s^2 + \sigma_i^2)$ (e.g. in Morris, 1983) in two ways. The first is the factor $(1 - h_{ii})$ multiplying σ_i^2 in the denominator, which addresses the fact that both stage 1 and stage 2 estimators are applied to the same sample. Because country i 's data enter the pooled regression, the stage 1 measurement error v_i and the prediction error E_i are mechanically correlated. The factor $(1 - h_{ii})$ removes exactly that share, so the weight reflects only the *incremental* information in the segmented estimate beyond what the cross-country prediction has already absorbed. To our knowledge, this correction is a feature novel to our framework.

The second is the extrapolation penalty A_i . When country i 's observable factors $\mathbf{Z}_i^*$ lies in the interior of support of all observable factors, no single country is influential for the prediction, so $A_i \approx 1$ and w_i^* is determined by other terms in the expression. However, when country i is systematically different from other countries in terms of their observable factors, then the CRC prediction for country i must be extrapolated from a region of the covariate space where those countries are observed but country i is not. In this case, the leverage terms h_{ij} grow in magnitude and A_i diverges, driving $w_i^* \rightarrow 1$. Therefore, the optimal weight then discounts the CRC prediction and relies almost entirely on country i 's own segmented estimate $\widehat{\varepsilon}_i$. This feature safeguards the analyst from being overconfident about the estimates' external validity when they might be driven by out-of-sample extrapolation. We discuss this point in more detail in Remark 2.7.

³Alternatively, expressing this in terms of the full covariance matrix for estimated β , $\mathbf{V}_\beta$, yields

$$w_i^* = \frac{(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + s^2 (1 - 2h_{ii}) - h_{ii} \sigma_i^2}{(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + s^2 (1 - 2h_{ii}) + \sigma_i^2 (1 - h_{ii})}. \quad (14)$$

2.4 Discussion of Our Approach

2.4.1 Properties of the Estimator

Remark 2.6 (Behavior of Optimal Weights). If the chosen observable factors explain the cross-country variations in the ETI poorly, i.e. $s^2 \rightarrow \infty$, then the optimal weights $w_i^* \rightarrow 1$, and the EB estimator collapses to the segmented estimate $\hat{\varepsilon}_i$. The cross-country hierarchical model is not useful for explaining cross-country heterogeneity in the ETI. Conversely, as $s^2 \rightarrow 0$, the full-sample weight $w_i^* = s^2 A_i / [s^2 A_i + \sigma_i^2 (1 - h_{ii})]$ converges to 0 for every country with $h_{ii} < 1$, because the extrapolation penalty A_i remains bounded while the numerator is proportional to s^2 . The full-sample EB estimator therefore collapses to the structural prediction by the CRC. $\diamond$

Remark 2.7 (Outlier Countries Measured By Observable Factors). Suppose the values of observable factors of country i , $\mathbf{Z}_i^*$, are “very far away” from the support of the other countries’ observable factors. It is worth noting that this outlyingness alone does not determine the weight; we show, in Lemma D.2(iv), that the extrapolation identity is bounded by functions of leverage term

$$(1 - h_{ii})^2 \leq A_i \leq (1 - h_{ii})^2 + \bar{r}_i h_{ii} (1 - h_{ii}), \quad \bar{r}_i \equiv \max_{j \neq i} \frac{\omega_j}{\omega_i},$$

so A_i can diverge only through the off-diagonal leverages h_{ij} for $j \neq i$, and only if country i has little identifying variation in tax rates compared to other countries.

As such, two regimes follow. If the outlying country i carries little identifying variation, then the CRC prediction must be extrapolated from the other countries’ data, hence the off-diagonal leverages grow large, A_i diverges, and $w_i^* \rightarrow 1$ regardless of s^2 . The EB estimator then pulls country i ’s estimate away from the structural prior mean prediction $(\mathbf{Z}_i^*)' \hat{\beta}$ and towards the segmented ETI $\hat{\varepsilon}_i$.

If, instead, the outlying country i has substantial identifying variation in tax rates, the pooled CRC regression fits its region of the covariate space using chiefly the country’s own observation: the self-leverage h_{ii} increases, the off-diagonal leverages vanish, and $A_i \rightarrow 0$. The extrapolation penalty then reduces w_i^* relative to the first regime. United States, in our empirical illustration (Section 3), is in this second regime: its characteristics lie far from the OECD mean, yet it has one of the *smallest* extrapolation penalties in the sample ($A_i = 0.126$) precisely because its self-leverage is the largest ($h_{ii} = 0.892$)—its CRC prediction is anchored by its own data, not extrapolated from the other countries’. Its overall weight, $w_i^* = 0.438$, is nonetheless near the middle of the distribution, because the extrapolation penalty is offset by the relative precision of its own segmented estimate. $\diamond$

Remark 2.8 (Bias-Variance Tradeoffs in the Structural Model). The choice of covariates $\mathbf{Z}_i^*$ governs a classic bias-variance trade-off. We can decompose the mean squared error of the empirical Bayes estimator for country i as

$$\mathbb{E} \left[(\hat{\varepsilon}_i^{EB} - \varepsilon_i^*)^2 \right] = (1 - w_i)^2 \mathbb{E} \left[((\mathbf{Z}_i^*)' \hat{\beta} - \varepsilon_i^*)^2 \right] + w_i^2 \sigma_i^2. \quad (15)$$

Letting $g(\mathbf{Z}_i^*) = \mathbb{E}[\varepsilon_i^* | \mathbf{Z}_i^*]$ denote the true conditional mean, the prior prediction mean squared error is

$$\mathbb{E} \left[((\mathbf{Z}_i^*)' \hat{\beta} - \varepsilon_i^*)^2 \right] = ((\mathbf{Z}_i^*)' \beta^* - g(\mathbf{Z}_i^*))^2 + (\mathbf{Z}_i^*)' \text{Var}(\hat{\beta}) \mathbf{Z}_i^* + s^2, \quad (16)$$

hence

$$\mathbb{E} \left[(\hat{\varepsilon}_i^{EB} - \varepsilon_i^*)^2 \right] = (1 - w_i)^2 \left[((\mathbf{Z}_i^*)' \beta^* - g(\mathbf{Z}_i^*))^2 + (\mathbf{Z}_i^*)' \text{Var}(\hat{\beta}) \mathbf{Z}_i^* + s^2 \right] + w_i^2 \sigma_i^2. \quad (17)$$

If a model is under-specified, the specification bias and s^2 increase, raising w_i^* and placing more weight on $\hat{\varepsilon}_i$. If a model is over-specified, the estimation variance $(\mathbf{Z}_i^*)' \text{Var}(\hat{\beta}) \mathbf{Z}_i^*$ increases, which also raises w_i^* . $\diamond$

Remark 2.9 (Agnostic to Identification Strategy in Stage 1). Our three-stage framework treats the stage 1 estimates $\hat{\varepsilon}_i$ as primitives; what we require is that each country's estimate be approximately Gaussian and unbiased for its own elasticity estimand, $\hat{\varepsilon}_i = \varepsilon_i^* + v_i$ with mean-zero sampling error v_i . Our stages 2 and 3 do not require anything about *how* stage 1 estimates are obtained. In our application we identify $\hat{\varepsilon}_i$ from within-country variations in statutory tax rates, but estimates from administrative microdata, bunching designs, or structural vector autoregressions and local projections identified by external (narrative) instruments—such as the approach in [Mertens and Montiel Olea \(2018\)](#)—would serve equally well as inputs to stage 1 of our proposed framework. The main function of stage 2 model is to discipline cross-country heterogeneity and provide a structural prediction via the CRC model, which we use for shrinkage and for out-of-sample prediction. Fundamentally, the questions of transportability and prediction for jurisdictions with little within-country variation in statutory tax rates still require a model that links the estimand to observables. Finally, our reliance on within-country variations in statutory tax rates in forming stage 1 estimates reflects data availability: credible, long narrative series and external instruments exist for only a handful of countries, whereas the cross-country estimates of stage 2 requires comparable data for all the countries we consider in our sample. $\diamond$

2.4.2 Relationship to Existing Approaches

Remark 2.10. Our three-stage EB framework nests several widely-used econometric approaches as special cases. Understanding this nesting structure reveals the implicit assumptions that simpler approaches place on the sources of heterogeneity, the statistical information they discard, and the role of estimation uncertainty. Specifically, the relationship between the three-stage empirical Bayes estimator and classical estimators depends on three dimensions: (i) whether country-specific covariates $\mathbf{Z}_i^*$ are included, (ii) whether residual cross-country heterogeneity s^2 is permitted to be non-zero, and (iii) whether the estimation uncertainty $\mathbf{V}_\beta$ is accounted for in the optimal shrinkage weights. Table 1 summarizes this nesting structure. The main takeaway is that fixed-effects meta-analysis, random-effects meta-analysis, and meta-regression can be recovered from our estimator by restricting the covariates, the residual heterogeneity, or the treatment of parameter uncertainty; put differently, our approach *estimates* the objects that a conventional meta-study

holds fixed by assumption. $\diamond$

Table 1: Nesting Structure of the Three-Stage Framework

Approach	Obs. Factors $\mathbf{Z}_i^*$	Residual Heterogeneity s^2	Uncertainty $\mathbf{V}_\beta$	Shrinkage Weight w_i
Fixed-Effects Meta-Analysis	Intercept only	$s^2 = 0$	Irrelevant for w_i	0
Random-Effects Meta-Analysis	Intercept only	$s^2 > 0$	Irrelevant for w_i	$\frac{s^2}{s^2 + \sigma_i^2}$
Meta-Regression	Covariates included	$s^2 > 0$	Irrelevant for w_i	$\frac{s^2}{s^2 + \sigma_i^2}$
Correlated Random Coefficients	Covariates included	$s^2 = 0$	Irrelevant for w_i	0
Three-Stage EB (this paper)	Covariates included	$s^2 \geq 0$	$\mathbf{V}_\beta$ (accounted for)	w_i^* (13)

Remark 2.11 (Empirical Bayes and Minimax Linear Estimation). The shrinkage rules derived here have a natural interpretation in the language of statistical decision theory, and in particular relate closely to the minimax linear estimators recently used to study transportability and evidence aggregation (Ishihara and Kitagawa, 2024; Yata, 2024; Montiel Olea et al., 2026). Consider the elementary transportability problem of using an estimate $\widehat{\varepsilon}_j$ from country j to predict the elasticity estimand ε_i^* in country i , restricting attention to linear rules $\omega \widehat{\varepsilon}_j$. Writing the transportation bias as $b \equiv \varepsilon_j^* - \varepsilon_i^*$, the mean-squared error of $\omega \widehat{\varepsilon}_j$ trades shrinkage off against this bias exactly as in the compound-risk calculation of (1). The minimax approach takes an ex-ante bound $|b| \leq B$ as primitive and selects the ω minimizing the worst-case risk over that bound; our empirical Bayes weight w_i is the average-case counterpart, replacing the worst-case bound B with the residual heterogeneity s^2 that we estimate from the cross-section of countries. Both rules shrink the source estimate toward zero (or, with covariates, toward the structural prediction $(\mathbf{Z}_i^*)' \beta$) by an amount that increases in the bias or heterogeneity relative to the sampling variance. The two are therefore complementary: the minimax rule is preferable when the analyst is unwilling to model the cross-country distribution and instead wishes to bound the bias and obtain worst-case guarantees, whereas the empirical Bayes rule exploits the estimability of that distribution when, as here, many comparable jurisdictions are available. A worst-case analogue of our second stage could replace the Gaussian prior in (8) with, for instance, a Lipschitz bound on how ε_i^* varies with $\mathbf{Z}_i^*$, which would bound the transportation bias directly; we view this as an alternative in settings with few jurisdictions. $\diamond$

2.4.3 Further Remarks

Remark 2.12 (Leave-one-out Estimation). The EB estimate for country i is a weighted average of its segmented estimate $\widehat{\varepsilon}_i$ and the CRC prediction $\mathbf{Z}_i' \widehat{\beta}$, where the latter is estimated using data from all N countries, including country i itself. For an exercise in external validity—where the goal is to ask what the experience of other countries implies about country i ’s ETI—it may seem more natural to estimate the Stage 2 hyperparameters (β, s^2) using all countries except country i , yielding a leave-one-out (LOO) estimator $\widehat{\theta}^{(-i)}$. The LOO approach is conceptually more rigorous for transportability: it ensures that the structural prediction for country i is constructed entirely from external evidence. It also circumvents the need to introduce corrections in the optimal weights formula because the CRC estimates and segmented

estimates would no longer mechanically correlated. The practical difference between using our proposed approach above or the LOO approach is negligible if the set of countries used in our study is large; we present the theory in full in Appendix D.2. $\diamond$

Remark 2.13 (Equivalence to Joint Maximum Likelihood Estimation). Under the Gaussian model, the sequential three-stage empirical Bayes procedure is algebraically equivalent to a single-step joint maximization of the parameters and latent effects. The statement and proof of this equivalence is provided in Appendix A.3. $\diamond$

3 Estimating the Top Income ETI

We now turn to our application of the econometric methodology addressing external validity—explaining the top income share as a function of the top tax rate. In what follows, we say more about the model specification, describe our data sources, and present estimates from our methodology.

3.1 Model Specification

In this section we describe the model we propose for the simultaneous choice a country makes, explicitly, about the top tax rate and, implicitly, about the elasticity of taxable income.

First, we recognize, as suggested by the theory of optimal taxation, that the top tax rate is likely to be a function of both the social value of redistribution and the elasticity of taxable income. In turn, according to the logic in Slemrod and Kopczuk (2002) the elasticity of taxable income is itself implicitly a chosen parameter, subject to influence by the social preference for redistribution as well as factors that affect the government’s ability to institute a broad-based and well-enforced tax system. This characterization of the ETI and top marginal tax rates suggests that both policy instruments are driven by a set of common exogenous factors. These factors constitute the components of $Z_{i,t}$.

Because our estimation relies on panel data, we need to make a decision about how to handle country-specific heterogeneity in the ETI and the top share of income. We allow for country fixed effects in the outcome variable, as well as linear time trends that vary across countries. These are meant to capture secular trends in top income shares that differ across countries, as well as the country-specific average top income share. That said, such modeling decisions are not strictly necessary, although useful in our setting, and our framework extends to alternative specifications. Note that our specification assumes a common intercept across countries in the relationship between ETIs and country characteristics. This assumption is necessary for the cross-sectional comparisons that we would like to make. In making this modeling decision, our assumption is not that observables capture all country-specific heterogeneity in the ETI—the framework explicitly allows residual heterogeneity s^2 —but that the systematic component of that heterogeneity is a common function of the tax rate and the elements of $Z_{i,t}$: we require that different countries with the same observable characteristics have the same *expected* ETI.

Combining this discussion with the model from Section 2, using all the notation introduced earlier, we arrive at our estimating equation

$$\ln Y_{i,t} = \alpha_i + \lambda_i t + \varepsilon R_{i,t} + \sum_{k \in \mathcal{W}} b_k (\ddot{Z}_{it,k} R_{i,t}) + \sum_{k=1}^K \pi_k (\bar{Z}_{i,k} R_{i,t}) + \underbrace{\tilde{u}_i^\varepsilon R_{i,t} + \eta_{i,t}}_{\equiv \tilde{\eta}_{i,t}} \quad (18)$$

where $Y_{i,t}$ is the top 1% share of reported taxable income and $R_{i,t} \equiv \log(1 - \tau_{i,t})$ is the log of the top net-of-tax rate. There are K many observable characteristics. Each observable characteristic enters the ETI through their interactions with the net-of-tax rate term. The within-country deviation $\ddot{Z}_{it,k} \equiv (Z_{it,k} - \bar{Z}_{i,k})/s_k$ —where $\bar{Z}_{i,k} = \frac{1}{T} \sum_t Z_{it,k}$ is country i 's over-time mean and s_k is the standard deviation—carries the identifying content for the *within* coefficient b_k , while the (cross-country-centered and standardized) country level $\bar{Z}_{i,k}$ carries the *level* coefficient π_k of (5). The within channel is included only for the set $\mathcal{W}$ of characteristics that vary over time within countries.⁴ We allow for a country fixed effect α_i and a country-specific linear time trend $\lambda_i t$ in the top income share. With this setup, the estimated coefficients map back to our linear characterization of the expected ETI from Section 2: ε is the ETI of a country whose characteristics equal the cross-country averages, b_k measures the effect on a country's ETI of a one-standard-deviation within-country change in characteristic k , and π_k measures the same effect but of a one-standard-deviation difference in its cross-country level.

3.2 Data

We implement the econometric model described in the previous section with data for the top percentile share of pre-tax reported income, the marginal tax rate for the top percentile earner, and indices of country characteristics that are contained in $Z_{i,t}$.

Our analysis relies on an aggregate panel of 27 countries for the years 1981 through 2014, constructed and analyzed by Rubolino and Waldenström (2019). Data on the top percentile shares of national “fiscal” income come from the World Inequality Database (WID.world, 2022). The measure of fiscal income from the WID is intended to capture the component of taxable income that is reported in income tax filings. This definition of income is attractive for the estimation of the ETI under its interpretation as a measure of the fiscal externality caused by the behavioral response of taxpayers to changes in the marginal tax rate. One drawback of using this measure is that it includes realized capital gains, which are in many cases not subject to the same tax rate as ordinary income. This raises omitted variable concerns in so far as changes in the top tax rate are correlated with changes in the capital gains rate. Rubolino and Waldenström (2019) note that, for a subset of countries where capital gains-adjusted top shares are available, exclusion of capital gains in the top income share does not meaningfully change estimates of the ETI. When estimates of the top income share are missing from the WID data, we impute using linear interpolation. Country-year observations with

⁴In our data, as we discuss in the next subsection, there are no within-country variation for two of the observable factors we employ; hence, this notational complexity is introduced.

missing top income share values at the beginning or end of the sample period are excluded from our analysis.

Top tax rates are taken from [Rubolino and Waldenström \(2019\)](#), and are supplemented by data from the OECD tax table I.1 for more recent years ([OECD, 2023](#)) and information from the World Tax Indicators from the Andrew Young School for Policy Studies ([Andrew Young School of Policy Studies, 2010](#)). For both longitudinal and cross-sectional consistency, we focus on the statutory marginal rate applicable to the top bracket. After taking logs of the top share of reported income and the marginal net-of-tax rate, the interpretation of the coefficient ε is the elasticity of taxable income.

For the elements $X_{i,t}$ that capture the perceived benefits of redistribution as well as the costs of enforcing a broad and well-enforced tax base, we first construct indicators based on survey responses from the World Values Survey (WVS). These indicators are constructed as follows. Each question is associated with an index. First, we pull the individual survey responses from the WVS website. Then, for each question, we calculate the weighted mean of the responses within each country and wave. We assign this value to the index for all years within each survey wave. For the years in between survey waves, or any years where the question was not asked in a particular country, we linearly interpolate based on the values from the prior time the question was asked and the next time the question was asked. Any early-year missing values are carried back from the first time the question was asked, and any tail-year missing values are carried forward from the last time the question was asked. In the rare cases where a particular survey question was never asked in a country⁵, we assign the average survey value across countries. The mean and standard deviation for each of the constructed indexes in our analysis sample are reported in Table 2. Before estimation, each index is standardized to have mean zero and unit standard deviation, so that the coefficients reported below are interpretable as the effect of a one-standard-deviation change; Table 2 reports the indices prior to standardization.

3.2.1 Social Attitudes toward Redistribution

Following the logic of [Slemrod and Kopczuk \(2002\)](#), we look for measures of the social demand for tax progressivity and for measures of the social cost of achieving progressivity. We discuss each in turn below.

Our first indicator comes from a Likert scale that assesses respondents' beliefs that engaging in redistribution is an essential part of democracy. We believe that this question most directly elicits citizens' preferences about redistribution and the social value of redistributive taxation. The question is posed as follows:

Question 1. Many things are desirable, but not all of them are essential characteristics of democracy. Please tell me for each of the following things how essential you think it is as a characteristic of democracy. Use this scale where 1 means “not at all an essential characteristic of democracy” and 10 means it definitely is “an essential characteristic of democracy”

⁵This occurs in Belgium and Ireland for the "Redistribution is Essential" and "Government Right to Collect Information" indices. It also occurs in France for the Ethnic Fractionalization Index.

- a. Governments tax the rich and subsidize the poor.

Although Question 1 appeals to respondents' ideal role of government, the next pair of questions are posed in a way that appeals to the proper role of government relative to the status quo. They reflect fundamental attitudes about the public's desired level of government intervention and the perceived fairness of market outcomes. Question 2.a is particularly interesting because we believe that it captures the public's views regarding the relative importance of equity versus efficiency, which is an important object for the study of optimal taxes; we refer to the index created by this question as "Efficiency over Equity." We will refer to the second as "Government over Own Responsibility." The questions are worded as follows:

Question 2. Now I'd like you to tell me your views on various issues. How would you place your views on this scale? 1 means you agree completely with the statement on the left; 10 means you agree completely with the statement on the right; and if your views fall somewhere in between, you can choose any number in between.

- a. Incomes should be made more equal (1) vs. We need larger income differences as incentives for individual effort (10)
- b. People should take more responsibility to provide for themselves (1) vs. Government should take more responsibility to ensure that everyone is provided for (10)

3.2.2 Measures of the Social Cost of Redistribution

We turn next to measures of the social cost of achieving a progressive tax system. One aspect of this is the prevalence of tax evasion. As a measure of a country's attitudes toward evasion, we consider the average responses to the following question, with the idea that the higher is people's willingness to evade taxes, the higher will be the cost of achieving tax progressivity.

Question 3. Please tell me for each of the following actions whether you think it can always be justified (10), never be justified (1), or something in between, using this card.

- Cheating on taxes if you have a chance.

We then consider direct survey evidence on confidence in government, which arguably lowers the cost of implementing a progressive tax system:

Question 4. I am going to name a number of organizations. For each one, could you tell me how much confidence you have in them: is it a great deal of confidence, quite a lot of confidence, not very much confidence, or none at all?

- a. The government

The WVS codes responses to this question from 1 (a great deal of confidence) to 4 (none at all). We reverse-code the country-wave mean as $4 - \overline{\text{response}}$, so that our index runs from 0 to 3 and higher values indicate *greater* confidence in government.

Next, we construct a measure to capture respondents' beliefs about privacy. Beliefs about privacy matter for the social cost of achieving progressivity because government's access to taxpayer information facilitates the implementation of an equitable tax system. The less the public believes that the government has a right to access private information, the more difficult it may be to implement, for example, robust third-party information reporting programs. Our measure is based on the following question:

Question 5. Do you think that your country's government should or should not have the right to do the following?

a. Collect information about anyone living in this country without their knowledge

As with the preceding question, the WVS codes responses from 1 (the government definitely should have the right) to 4 (definitely should not). We again reverse-code the country-wave mean as $4 - \overline{\text{response}}$, so that our index runs from 0 to 3 and higher values indicate greater public support for the government's right to collect this information.

We also consider two non-survey-based country characteristics. One is the resident population, on the grounds that smaller countries will likely have more open economies, and more open economies will tend to have higher ETIs, due to its facilitation of migration of factors and other tax bases. We use the log of population in 2000, ignoring its drift over time. The other is a measure of ethnic diversity. As discussed in [Alesina and Ferrara \(2005\)](#), ethnic diversity has been found to have an inverse relationship with the size of the social safety net and the amount of redistribution. To capture this dynamic, we make use of the Historical Index of Ethnic Fractionalization ([Drazanova, 2019](#)), which is available for all but one country over the entire time period.

It is important to note that, in choosing country characteristics, we purposely have focused on the underlying features of a country that determine *why* it chooses the tax system it has, as opposed to *how* these choices are implemented. For this reason, we deliberately do not make use of direct measures of the broadness of the tax base such as the extent of resources devoted to tax enforcement, or how extensive is the coverage of third-party information. Instead, we study, for example, the attitudes of the population toward sharing information with the government, which affects how much third-party, or first-party, information a tax authority decides to make use of. Thus, our approach is different, though complementary, to the approach taken in [Kleven \(2014\)](#) to explain why Scandinavian countries can levy relatively high taxes and maintain a relatively low ETI, which focuses on the *how* question, in particular the broadness of the tax base and the coverage of third-party information.

3.3 Benchmarks

If the goal of our analysis were to simply describe the nature and extent of ETI heterogeneity across countries, one might want to run “segmented” regressions separately for each country, and potentially for separate time periods. Indeed, this is the exercise at the heart of [Rubolino and Waldenström \(2019\)](#). As a benchmark for our approach, we regress our preferred measure of the logged top share of income against the log net-of-tax rate for each of the countries in our data set. The resulting estimates are reported in the first panel of Table 4, along with heteroskedasticity-and-auto-correlation robust standard errors. Keep in mind that, as discussed above, these estimates are biased to the extent that tax rates are chosen in part based on policy makers’ perceptions of the ETI, and these perceptions are correlated with the true ETI.⁶ The US estimate is 0.240, slightly higher than the overall world average of 0.204, and broadly in line with the evidence summarized in [Saez, Slemrod, and Giertz \(2012\)](#). These estimates are also broadly in line with those presented by [Rubolino and Waldenström \(2019\)](#). Note that many of the segmented ETI estimates are quite imprecise, due largely to the limited amount of within-country variation in some countries for the top marginal tax rate. This suggests that a pooled cross-country estimation may yield improved precision, depending on how much one can learn about one country’s elasticity from other countries’ experiences.

As discussed earlier, estimating the ETI from separate segmented regressions for each country presumes a stark answer to the question of how much can be learned from analysis of data in other countries: nothing at all. At the other extreme, one could estimate a pooled regression assuming that there is a common ETI for all countries, essentially treating other countries’ data as providing additional information about a common elasticity. If we do that while allowing for country fixed effects and country-specific linear time trends for the top taxable income, we estimate an ETI of 0.245, which is not far from the simple average of cross-country segmented elasticities, 0.204. But this approach completely ignores that we observe country characteristics that, according to theory, affect a country’s true ETI.

⁶We estimate elasticities using ordinary least squares. [Rubolino and Waldenström \(2019\)](#), who share our approach, test whether the results are robust to the use of an instrumental variables approach like [Gruber and Saez \(2002\)](#), and find similar results across OLS and 2SLS specifications.

Table 2: Summary Statistics of Sample

Country	Statistic	Top 1% Share of Fiscal Income	Top Marginal Tax Rate	Population in 2000 (thousands)	Redistribution Essential to Democracy	Efficiency over Equity	Government over Own Responsibility	Confidence in Government	Justifiable to Cheat on Taxes	Government Right to Collect Info.	Ethnic Fractional- ization Index
Australia	Mean	0.070	0.496	19.053	6.050	5.478	5.136	1.150	2.338	1.027	0.151
T = 34	St. Dev.	0.015	0.048	0.000	0.002	0.301	0.090	0.085	0.419	0.000	0.066
Belgium	Mean	0.090	0.576	10.495	6.336	5.734	4.928	1.134	3.566	0.817	0.573
T = 24	St. Dev.	0.005	0.089	0.000	0.000	0.157	0.153	0.000	0.350	0.000	0.013
Canada	Mean	0.121	0.301	31.100	6.128	6.162	4.507	1.331	2.177	0.778	0.713
T = 34	St. Dev.	0.022	0.029	0.000	0.081	0.559	0.507	0.016	0.159	0.000	0.009
Chile	Mean	0.177	0.454	16.295	6.561	4.999	6.625	1.570	1.934	0.811	0.462
T = 34	St. Dev.	0.017	0.078	0.000	0.191	0.981	0.294	0.237	0.170	0.000	0.014
Colombia	Mean	0.196	0.360	39.817	5.642	5.865	6.132	1.255	1.796	0.997	0.646
T = 18	St. Dev.	0.013	0.066	0.000	0.000	0.439	0.208	0.105	0.148	0.000	0.004
Czech Republic	Mean	0.075	0.298	10.220	4.743	5.930	4.956	1.185	2.187	0.411	0.333
T = 22	St. Dev.	0.024	0.113	0.000	0.000	0.913	0.651	0.258	0.405	0.000	0.040
Denmark	Mean	0.054	0.345	5.337	6.293	6.616	4.394	1.530	2.138	0.910	0.094
T = 30	St. Dev.	0.004	0.074	0.000	0.000	0.201	0.216	0.040	0.280	0.000	0.044
Finland	Mean	0.064	0.393	5.169	6.978	5.465	4.552	1.295	2.469	1.153	0.131
T = 29	St. Dev.	0.021	0.075	0.000	0.002	1.007	0.404	0.116	0.413	0.000	0.004
France	Mean	0.091	0.544	61.137	6.308	5.132	4.306	1.055	2.939	0.758	0.256
T = 34	St. Dev.	0.008	0.082	0.000	0.058	0.162	0.320	0.007	0.286	0.000	0.000
Germany	Mean	0.114	0.505	82.188	7.175	5.289	4.880	1.045	2.466	0.574	0.153
T = 32	St. Dev.	0.015	0.048	0.000	0.096	1.044	0.893	0.153	0.323	0.000	0.022
Greece	Mean	0.058	0.467	11.120	6.644	4.523	5.690	0.832	2.921	0.646	0.103
T = 34	St. Dev.	0.009	0.085	0.000	0.000	0.127	0.245	0.028	0.391	0.000	0.033

continued on next page

Country	Statistic	Top 1% Share of Fiscal Income	Top Marginal Tax Rate	Population in 2000 (thousands)	Redistribution Essential to Democracy	Efficiency over Equity	Government over Own Responsibility	Confidence in Government	Justifiable to Cheat on Taxes	Government Right to Collect Info.	Ethnic Fractional- ization Index
Hungary	Mean	0.066	0.383	10.098	7.386	4.991	6.212	1.139	2.437	0.534	0.143
T = 27	St. Dev.	0.028	0.114	0.000	0.065	0.780	0.572	0.232	0.597	0.000	0.048
Ireland	Mean	0.079	0.479	3.792	6.336	5.959	4.600	1.223	2.667	0.817	0.135
T = 34	St. Dev.	0.020	0.077	0.000	0.000	0.520	0.279	0.000	0.351	0.000	0.023
Italy	Mean	0.082	0.511	57.719	6.712	5.862	5.634	1.041	2.252	0.985	0.085
T = 29	St. Dev.	0.011	0.096	0.000	0.000	0.096	0.164	0.006	0.204	0.000	0.015
Japan	Mean	0.097	0.476	126.729	6.639	5.663	6.879	1.163	1.447	0.746	0.015
T = 30	St. Dev.	0.014	0.118	0.000	0.060	0.273	0.197	0.042	0.060	0.000	0.002
Korea	Mean	0.087	0.452	46.839	7.482	5.959	6.240	1.368	1.622	0.745	0.025
T = 34	St. Dev.	0.018	0.112	0.000	0.020	0.693	1.714	0.088	0.082	0.000	0.029
Netherlands	Mean	0.060	0.599	15.908	6.439	5.987	4.813	1.281	2.690	1.270	0.169
T = 34	St. Dev.	0.006	0.082	0.000	0.142	0.250	0.239	0.026	0.421	0.000	0.098
New Zealand	Mean	0.078	0.409	3.802	5.469	5.333	4.925	1.082	2.210	1.137	0.375
T = 34	St. Dev.	0.017	0.113	0.000	0.000	0.080	0.152	0.240	0.152	0.000	0.068
Norway	Mean	0.072	0.465	4.492	6.609	5.629	4.996	1.624	2.729	0.594	0.093
T = 34	St. Dev.	0.026	0.103	0.000	0.038	0.334	0.421	0.089	0.451	0.000	0.032
Poland	Mean	0.090	0.419	38.530	6.142	6.839	5.733	1.158	2.520	0.375	0.039
T = 34	St. Dev.	0.034	0.092	0.000	0.188	0.773	0.342	0.175	0.192	0.000	0.013

continued on next page

Country	Statistic	Top 1% Share of Fiscal Income	Top Marginal Tax Rate	Population in 2000 (thousands)	Redistribution Essential to Democracy	Efficiency over Equity	Government over Own Responsibility	Confidence in Government	Justifiable to Cheat on Taxes	Government Right to Collect Info.	Ethnic Fractional- ization Index
Portugal	Mean	0.076	0.468	10.336	5.989	4.834	4.804	1.029	3.006	0.758	0.077
T = 25	St. Dev.	0.017	0.111	0.000	0.000	0.339	0.276	0.014	0.714	0.000	0.066
Slovenia	Mean	0.067	0.474	1.967	5.981	4.722	6.187	1.208	2.157	0.519	0.222
T = 21	St. Dev.	0.007	0.042	0.000	0.131	0.909	0.469	0.225	0.301	0.000	0.020
Spain	Mean	0.091	0.479	40.016	6.633	5.141	6.189	1.152	2.310	0.967	0.615
T = 32	St. Dev.	0.011	0.070	0.000	0.164	0.185	0.304	0.143	0.358	0.000	0.038
Sweden	Mean	0.071	0.623	8.872	6.379	5.871	3.994	1.399	2.279	0.789	0.185
T = 33	St. Dev.	0.020	0.134	0.000	0.019	0.577	0.762	0.088	0.238	0.000	0.021
Switzerland	Mean	0.095	0.392	7.267	6.887	4.603	3.689	1.541	2.300	0.725	0.323
T = 34	St. Dev.	0.010	0.099	0.000	0.003	0.276	0.507	0.105	0.170	0.000	0.027
United Kingdom	Mean	0.114	0.453	59.522	6.116	5.863	4.763	0.999	2.376	1.240	0.356
T = 34	St. Dev.	0.028	0.082	0.000	0.031	0.558	0.369	0.023	0.263	0.000	0.024
United States	Mean	0.173	0.384	282.158	5.016	6.168	4.074	1.245	2.005	0.967	0.430
T = 34	St. Dev.	0.038	0.065	0.000	0.010	0.561	0.634	0.046	0.162	0.000	0.059
All	Mean	0.092	0.454	37.407	6.336	5.578	5.179	1.223	2.368	0.817	0.256
N = 820	St. Dev.	0.039	0.118	56.060	0.625	0.805	0.989	0.225	0.546	0.232	0.203

Notes: For each of the questions defined in Section 3.2, this table displays the average and standard deviation of the survey indicators used in our analysis sample. Sample counts reflect the impact of interpolation and extrapolation. Note that we interpolate, but do not extrapolate, the missing top 1% share values. Survey indicators are both interpolated for between-wave years, and extrapolated for beginning and end years. Population counts are kept constant across years based on their values in 2000. Sources: Top income shares and population counts are from [WID.world \(2022\)](#). Top marginal tax rates are from [Rubolino and Waldenström \(2019\)](#), supplemented with data from [OECD \(2023\)](#) and [Andrew Young School of Policy Studies \(2010\)](#). Survey indices are calculated based on World Values Survey data.

3.4 CRC Estimation

In this section we present the results of the pooled analysis according to the CRC model outlined in Section 2, based on the specification described in Section 3.1. Identification of the coefficients β operates through two distinct channels. The level coefficients π are identified by cross-country differences in average characteristics $\bar{Z}_i$ interacted with each country's tax variation; the within coefficients b are identified by over-time movements of $\ddot{Z}_{it}$ within a country, interacted with its own tax variation. The intercept of the ETI equation is interpreted as the predicted ETI of a country that is “average” with respect to the observable country characteristics. With this interpretation in mind, we note that our model is able to explain, for example, why the United States has an ETI that differs from Norway's, as a function of the difference between the observable characteristics of the two countries.

Table 3 shows the results of estimating the determinants of a country's ETI, with analytic standard errors reported.⁷ In this table, the explanatory variables are standardized, so the coefficients should be interpreted as the estimated effect on the ETI of a one standard deviation change in the corresponding characteristic. The first-listed coefficient reported in this table is the intercept of the ETI equation, interpreted as the average-profile ETI—the predicted ETI of a country at the sample-average characteristics—and is estimated at 0.2286. This differs slightly from the mean of the country-specific CRC predictions in Table 4 (0.202) because the within-channel term $\ddot{Z}_i^*$ is tax-variation-weighted and so is not exactly mean-zero across countries. The remaining coefficients separate, for each characteristic, its within-country (dynamic, b) channel from its cross-country (level, π) channel, and capture how the explanatory factors explain the cross-country variation of country ETIs.

Because the survey indices vary little within countries over time, the within-country channel b is imprecisely estimated for most characteristics, though not all, and the cross-country predictions are carried mainly by the level channel π . Several coefficients are nonetheless statistically significant. In the level channel, greater emphasis on government (relative to individual) responsibility, greater confidence in government, a higher perceived justifiability of tax cheating, and larger population are each associated with significantly lower ETIs; in the within channel, a greater emphasis on efficiency over equity is associated with a significantly higher ETI, a higher perceived justifiability of tax cheating with a modestly higher ETI, and greater confidence in government with a modestly lower one. The justifiability of tax cheating thus enters with opposite signs in the two channels: countries that are on average more tolerant of cheating have lower ETIs, whereas within a country, periods of greater tolerance coincide with a modestly higher elasticity.

The second column of Table 3 records, for each characteristic, the direction of the effect on the ETI anticipated by the discussion of Section 3.2: a characteristic that raises the social demand for progressivity, or lowers the social cost of achieving it, is expected to lower a country's ETI, because such a country

⁷The estimation sample has 98 country-year observations, spanning 13 countries, for which either the top income share or the marginal tax rate is unavailable. Most of these exclusions (81 observations) reflect gaps in top income share variable: observations are lost where a country's series begins after 1981 (e.g., Belgium and Colombia in the 1980s) or ends before 2014 (e.g., Portugal after 2005, and Denmark, Finland, Hungary, Italy, and Japan in the early 2010s). The remaining exclusions reflect missing top marginal tax rates, which are confined almost entirely to the Czech Republic, Hungary, and Slovenia in the 1980s and early 1990s.

Table 3: CRC Regression Estimates

Variable	Predicted Sign	Estimate	SE
ETI Intercept ($\hat{\varepsilon}$)	—	0.2286***	(0.0435)
<i>Within channel: $\ddot{Z}_{it} \times \log(1 - \tau)$ (dynamic b)</i>			
Redistribution is essential	—	−0.0018	(0.0093)
Efficiency over Equity	+	0.0809***	(0.0116)
Govt vs. Own Responsibility	—	−0.0052	(0.0090)
Confidence in Government	—	−0.0269*	(0.0139)
Justifiability: Cheating on Taxes	+	0.0223**	(0.0103)
Ethnic Fractionalization	+	−0.0035	(0.0409)
<i>Level channel: $\bar{Z}_i \times \log(1 - \tau)$ (cross-country $\pi=b+\psi$)</i>			
Redistribution is essential	—	−0.0651	(0.0606)
Efficiency over Equity	+	−0.0048	(0.0382)
Govt vs. Own Responsibility	—	−0.1508***	(0.0526)
Confidence in Government	—	−0.1732***	(0.0568)
Justifiability: Cheating on Taxes	+	−0.1788***	(0.0587)
Govt Right: Collect Information	—	−0.0357	(0.0469)
Ethnic Fractionalization	+	0.0756	(0.0579)
Log Population	—	−0.1037**	(0.0427)
Observations		820	
R ²		0.9364	

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Coefficient estimates are from the regression specification detailed in (18). Explanatory variables are standardized before regression. Each characteristic enters through a within-country (dynamic, b) and a cross-country (level, π) channel; population and the government-right-to-collect-information index have no within-country variation and enter through the level channel only. HAC standard errors are presented. The “Predicted Sign” column records the direction of the effect on the ETI anticipated by the discussion of Section 3.2: a characteristic that raises the social demand for progressivity, or lowers the social cost of achieving it, is expected to reduce a country’s ETI, because such a country will invest more in the base-broadening and enforcement measures that limit behavioral responsiveness (Slemrod and Kopczuk, 2002); a characteristic that lowers the demand for progressivity or raises its cost is expected to raise the ETI. Because the prediction concerns the structural relationship between a characteristic and the elasticity in (3), the same sign is anticipated for both the within and level channels. No sign is predicted for the intercept.

will invest more in the base-broadening and enforcement measures that limit behavioral responsiveness; a characteristic that lowers the demand for progressivity, or raises its cost, is expected to raise the ETI. Eleven of the fourteen point estimates carry their predicted sign. Of the three that do not, two—the within coefficient on ethnic fractionalization and the level coefficient on efficiency over equity—are economically negligible and statistically indistinguishable from zero. The only significant coefficient of unanticipated sign is the level coefficient on the justifiability of cheating on taxes, noted above.

The negative level coefficient on population is consistent with the idea that smaller countries are on average more open, and that openness increases the ETI by facilitating the migration of tax bases. The level coefficient on ethnic fractionalization enters with a positive point estimate, as theory would suggest, though it is not statistically significant. The significant negative level coefficients on government-over-own responsibility, confidence in government, and the justifiability of tax cheating indicate that the cross-country variation in ETIs is accounted for by this broader set of governance and tax-morale attitudes—together with country size—rather than by the value placed on redistribution alone. We caution that the underlying survey questions ask respondents about their ideal level of government intervention relative to the status quo, as opposed to their underlying views, which complicates a structural interpretation of the individual signs.

We now have all of the ingredients needed to produce the estimated ETIs for each country using the CRC method. The predicted ETI and standard error for each country, based on the regression presented in (18), are presented in column (b) of Table 4. Note that these estimates are in some cases substantially different from those reported in column (a) of Table 4. Take, for example, the United States. The estimated ETI for the US using only US data, i.e. according to the segmented estimation procedure, is 0.240, but using the CRC model we estimate the average US ETI to be 0.317, about a third higher than the estimate from the segmented procedure. This difference reflects the fact that our estimation strategy eliminates the bias in the estimated ETI in the segmented estimates. More importantly, in the CRC method, we require each country's estimated ETI to follow the estimated linear function of the country characteristics we have selected.

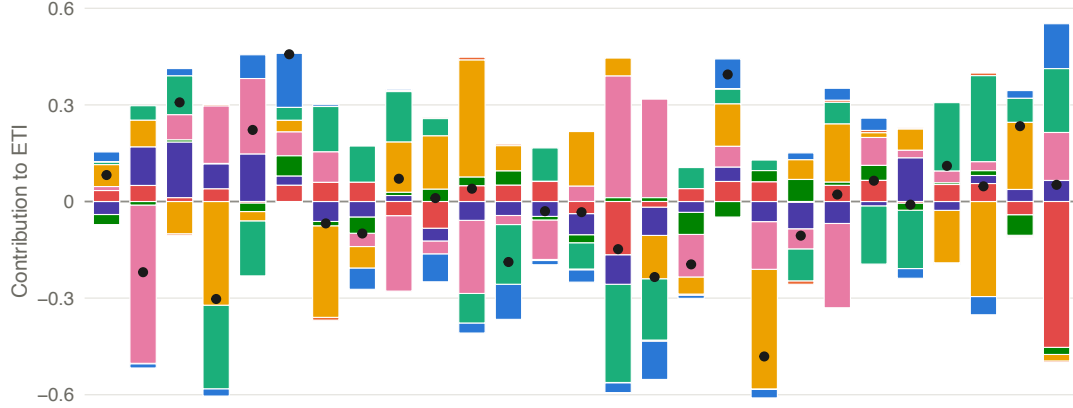
Figure 1 depicts how the country characteristics combine to produce each country's estimated ETI. The components above the zero line represent the role of factors that put upward pressure on a country's ETI, and the components below the zero line are factors that put downward pressure on the ETI. The black dots depict the net effect of all factors on the divergence of a country's CRC estimate from the average-profile ETI $\hat{\epsilon}$. Consider the case of the United States, shown in the far-right column. Its estimated ETI via our method is modestly above the average over all countries, and this prediction is the net product of large opposing forces. Several characteristics push the US ETI up: the largest upward contribution, about 0.20, comes from its comparatively low emphasis on government (relative to individual) responsibility; the low perceived justifiability of tax cheating and the low public value placed on redistribution each add roughly 0.15; and the country's relatively high ethnic fractionalization contributes a further 0.07. These upward forces are largely counter-weighted by the role of country size: the relatively large US population lowers the predicted ETI by about 0.45. On net, these factors leave the US only slightly above the cross-country average.

Inspection of Figure 1 shows that the CRC estimates are shaped by several characteristics rather than any single one. A country's population is among the most influential: a relatively small population pushes up the ETI in Finland, the Netherlands, and New Zealand, while a large population pulls it down in the US, the United Kingdom, and Japan. Citizens' views about whether redistribution is essential also shape the predictions: less-than-average agreement pushes up the ETI in Colombia, the Czech Republic, and New Zealand, while greater-than-average agreement pushes it down in Germany, Hungary, and Korea. Relatively high ethnic fractionalization increases the CRC-estimated ETI in Belgium, Canada, Chile, Colombia, and Spain.

Figure 1: Contribution of Factors to Deviations from the Average-Profile ETI $\hat{\varepsilon} = 0.229$

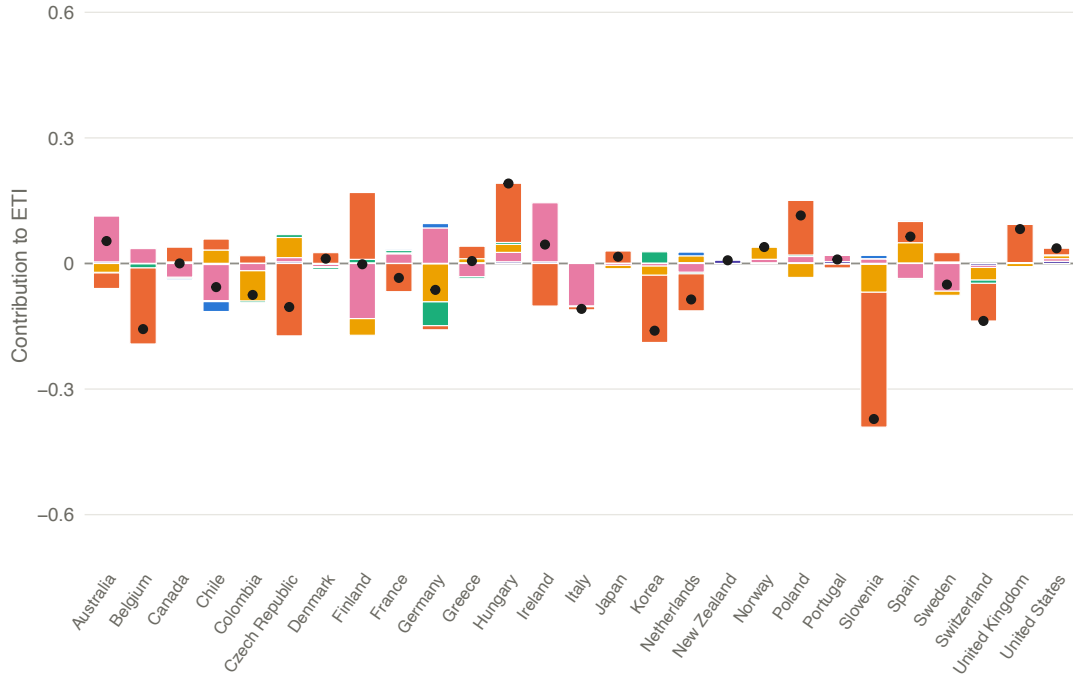
A. Level channel: cross-country differences in average characteristics

Bars are the level contributions; black dots are their sum. Invariant to reform timing.



B. Within channel: tax-variation-weighted within-country deviations

Same vertical scale as panel A. Depends on when each country reformed.



■ Redistribution is essential ■ Efficiency over equity ■ Gov. over own responsibility ■ Confidence in government
■ Justif. of cheating on taxes ■ Gov. right to collect info ■ Ethnic fractionalization ■ Logged population in 200C

Notes: This figure depicts the role that country-level factor deviations play in explaining the departure of country-specific CRC-predicted ETIs from the average-profile ETI $\hat{\varepsilon} = 0.229$, the intercept of Table 3, separately for the within and across channels. Each bar from panel A is the total contribution of characteristic k to country i 's deviation, $c_{ik} = \hat{\pi}_k \bar{Z}_{i,k}$, summing the cross-country (level) channels using the coefficients shown in Table 3. Panel B shows the equivalent decomposition but for the within channel, $l_{ik} = \hat{b}_k \ddot{Z}_{i,k}^*$. Here $\ddot{Z}_{i,k}^*$ is the tax-variation-weighted within-country deviation defined in Section 2. The components above the zero line represent the role of factors that put upward pressure on a country's ETI, and the components below the zero line are factors that put downward pressure on the ETI. The black points depict the net effect of all factors, so that in the first figure $\sum_k c_{ik} = \hat{\varepsilon}_i^{CRC} - \hat{\varepsilon} - \sum_k \hat{b}_k \ddot{Z}_{i,k}^*$ and for the second figure $\sum_k l_{ik} = \hat{\varepsilon}_i^{CRC} - \hat{\varepsilon} - \sum_k \hat{\pi}_k \bar{Z}_{i,k}$.

3.5 Empirical Bayes Estimation Results

The final step in our method is the empirical Bayes procedure. Column (c) of Table 4 presents the results of estimating that model. Recall from Section 2.3 that the EB estimator is a weighted average of what we would estimate from (only) each country’s own data and what we would predict based on the country’s characteristics and the cross-country structural relationship modeled via the CRC approach. The weight in the weighted average depends on the relative precision of the two estimation procedures. The final column of Table 4 shows the estimated shrinkage factor for each country. Table 5 shows all the components of the shrinkage factor used in computing them. Figure 2 visually depicts all of these estimates.

A few aspects of Table 4 are worth highlighting. First, as our construction dictates, the EB estimate for each country is bounded by the segmented and CRC estimates, lying closer to the segmented estimate the higher is the shrinkage factor (which is itself larger the more identifying tax variation the country contributes to its own within-country estimate). , an economically reassuring feature of the EB estimates is that they are largely positive and of a sensible magnitude. Whereas eight of the twenty-seven segmented estimates are negative—an economically implausible result—just six of the empirical Bayes estimates remain below zero; the shrinkage procedure reverses the sign of two of these eight estimates (Denmark and Greece). The most extreme case is Belgium, whose segmented estimate of -0.961 is the largest negative value among all the countries’ estimates; because that estimate is imprecise, however, Belgium receives a shrinkage weight of only 0.073 , so the empirical Bayes estimator discounts the country’s own data almost entirely and reports -0.207 , close to Belgium’s structural prediction of -0.148 . Across all countries the empirical Bayes estimates lie in the range $[-0.213, 0.635]$, with a mean of 0.204 , comfortably within the range of credible elasticities in the literature. The EB approach disciplines the noisy segmented estimates toward structural predictions that are consistent with economic theory.

The six countries whose EB estimates remain non-positive—Belgium, Chile, Korea, the Netherlands, Norway, and Slovenia—can be understood via the results in Table 5. Korea illustrates the role of precision: it contributes ample identifying tax variation, so its model-implied sampling variance is small, hence it receives a high shrinkage weight of 0.540 , leading the EB estimator to defer largely to Korea’s own (negative) estimate. Belgium, by contrast, has a very imprecise own estimate and receives a weight of only 0.073 , so the EB estimator relies almost entirely on its structural prediction—itsself negative (-0.148)—rather than on Belgium’s noisy -0.961 . Korea thus illustrates an important property of the procedure: a negative but precisely estimated value using only own-country evidence is preserved, whereas an imprecise one, as in Belgium, is pulled toward the structural prediction.

Finally, for the US, the empirical Bayes estimate is 0.283 , a modest upward revision relative to the segmented estimate of 0.240 and to the body of ETI estimates documented in Saez et al. (2012). The upward revision is consistent with our data and estimation results: the US lies on the elasticity-raising side of several structural characteristics—placing relatively low weight on government responsibility, viewing tax cheating as relatively unjustifiable, and placing a low value on redistribution—which the estimated cross-country

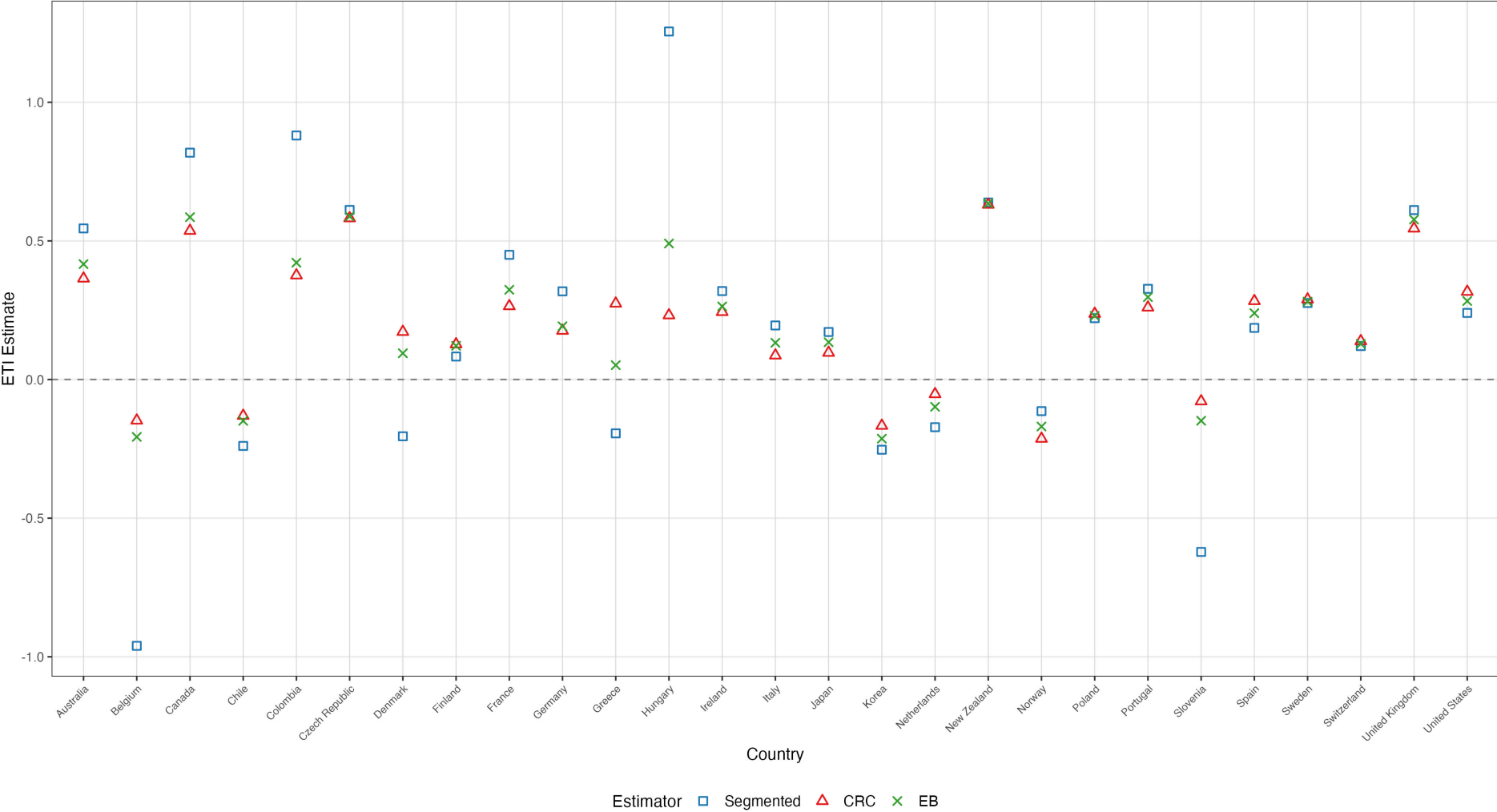
relationship associates with higher elasticities, although this is substantially tempered by the downward pull of its large population. At the same time, the US has the highest leverage in the cross-country CRC regression: its own data largely determine its CRC prediction, and the optimal weights discount the CRC predictions by taking leverage into account. What we deem significant here is the direction of the revision: a higher elasticity implies a lower revenue-maximizing top marginal tax rate, so incorporating the cross-country evidence using our framework points to greater US behavioral responsiveness than within-country evidence alone would indicate.

Table 4: ETI Estimates: (a) Segmented, (b) CRC (Mundlak within/between), and (c) Three-Step EB

Country	(a)		(b)		(c)		Shrinkage Factor
	Point Estimate	SE	Point Estimate	SE	Point Estimate	SE	
Australia	0.545	(0.162)	0.364	(0.096)	0.416	(0.154)	0.288
Belgium	-0.961	(0.211)	-0.148	(0.195)	-0.207	(0.295)	0.073
Canada	0.818	(0.334)	0.537	(0.110)	0.586	(0.246)	0.174
Chile	-0.239	(0.158)	-0.131	(0.333)	-0.149	(0.270)	0.169
Colombia	0.880	(0.263)	0.376	(0.154)	0.421	(0.248)	0.091
Czech Republic	0.612	(0.377)	0.582	(0.221)	0.588	(0.262)	0.220
Denmark	-0.205	(0.184)	0.172	(0.172)	0.095	(0.191)	0.205
Finland	0.083	(0.363)	0.127	(0.128)	0.121	(0.193)	0.133
France	0.450	(0.117)	0.265	(0.127)	0.324	(0.159)	0.318
Germany	0.318	(0.310)	0.176	(0.292)	0.192	(0.244)	0.113
Greece	-0.194	(0.135)	0.274	(0.218)	0.052	(0.162)	0.475
Hungary	1.255	(0.153)	0.232	(0.250)	0.491	(0.219)	0.253
Ireland	0.319	(0.323)	0.244	(0.153)	0.264	(0.154)	0.266
Italy	0.195	(0.035)	0.087	(0.114)	0.133	(0.130)	0.423
Japan	0.171	(0.202)	0.097	(0.102)	0.135	(0.130)	0.507
Korea	-0.253	(0.046)	-0.167	(0.099)	-0.213	(0.100)	0.540
Netherlands	-0.172	(0.110)	-0.053	(0.121)	-0.098	(0.154)	0.383
New Zealand	0.639	(0.137)	0.631	(0.041)	0.635	(0.093)	0.534
Norway	-0.114	(0.378)	-0.213	(0.097)	-0.169	(0.154)	0.442
Poland	0.221	(0.130)	0.237	(0.034)	0.230	(0.103)	0.478
Portugal	0.327	(0.069)	0.260	(0.107)	0.297	(0.095)	0.556
Slovenia	-0.622	(0.198)	-0.078	(0.225)	-0.149	(0.236)	0.129
Spain	0.186	(0.133)	0.283	(0.134)	0.239	(0.147)	0.450
Sweden	0.276	(0.073)	0.289	(0.029)	0.281	(0.065)	0.629
Switzerland	0.121	(0.086)	0.138	(0.076)	0.129	(0.146)	0.556
United Kingdom	0.611	(0.081)	0.545	(0.080)	0.577	(0.108)	0.483
United States	0.240	(0.158)	0.317	(0.116)	0.283	(0.174)	0.438
Mean	0.204		0.202		0.204		0.345

Notes: The table reports elasticity of taxable income (ETI) estimates for the $N = 27$ countries in the estimation sample. Column (a) reports the first-stage segmented estimates $\hat{\varepsilon}_i$ from country-specific regressions of the outcome on the log net-of-tax rate R_{it} with a country-specific intercept and linear time trend, with heteroskedasticity- and autocorrelation-consistent (Newey–West) standard errors. Column (b) reports the cross-country correlated random coefficients (CRC) predictions $\hat{\varepsilon}_i^{CRC} = (\mathbf{Z}_i^*)' \hat{\beta}^{CRC}$ from the pooled Stage 2 regression (18), with analytic standard errors $\sqrt{(\mathbf{Z}_i^*)' \mathbf{V}_{\beta} \mathbf{Z}_i^*}$. Column (c) reports the three-step empirical Bayes estimates $\hat{\varepsilon}_i^{EB} = w_i \hat{\varepsilon}_i + (1 - w_i) \hat{\varepsilon}_i^{CRC}$ defined in (12), with standard errors equal to the square root of the corresponding posterior variance. The final column reports the leverage-corrected shrinkage weight w_i of (13); w_i is the weight placed on country i 's own segmented estimate, so values near one indicate reliance on the within-country estimate and values near zero indicate shrinkage toward the CRC prediction. The bottom row reports the cross-country mean of each column.

Figure 2: ETI Estimates Across Methods: Segmented, CRC, and Empirical Bayes



Notes: This figure shows country-level ETI estimates from the three estimation methods described in the text. Each tick on the horizontal axis represents a country in our study. Points show the point estimate from each method: segmented regression (column (a) of Table 4), CRC method (column (b) of Table 4), and empirical Bayes shrinkage (column (c) of Table 4). The empirical Bayes estimate always lies between the segmented and CRC estimates.

Table 5: Components of the Shrinkage Weight

Country	ω_i	σ_i^2	h_{ii}	A_i	w_i^*
Australia	0.157	0.063	0.110	1.052	0.288
Belgium	0.014	0.694	0.054	2.412	0.073
Canada	0.042	0.231	0.103	2.037	0.174
Chile	0.050	0.197	0.241	1.422	0.169
Colombia	0.023	0.433	0.054	1.910	0.091
Czech Republic	0.050	0.195	0.169	2.131	0.220
Denmark	0.081	0.122	0.098	1.321	0.205
Finland	0.049	0.201	0.061	1.358	0.133
France	0.175	0.056	0.169	1.016	0.318
Germany	0.037	0.262	0.129	1.360	0.113
Greece	0.345	0.028	0.393	0.730	0.475
Hungary	0.126	0.078	0.162	1.036	0.253
Ireland	0.155	0.063	0.148	0.912	0.266
Italy	0.329	0.030	0.235	0.781	0.423
Japan	0.472	0.021	0.594	0.405	0.507
Korea	0.843	0.012	0.659	0.218	0.540
Netherlands	0.283	0.035	0.471	0.532	0.383
New Zealand	1.012	0.010	0.759	0.125	0.534
Norway	0.312	0.031	0.539	0.536	0.442
Poland	0.861	0.011	0.850	0.073	0.478
Portugal	0.915	0.011	0.609	0.245	0.556
Slovenia	0.039	0.249	0.099	1.555	0.129
Spain	0.370	0.026	0.646	0.359	0.450
Sweden	2.124	0.005	0.755	0.090	0.629
Switzerland	0.362	0.027	0.509	0.780	0.556
United Kingdom	0.651	0.015	0.541	0.302	0.483
United States	0.307	0.032	0.892	0.126	0.438
Mean					0.345

Notes: This table reports the country-specific inputs to the optimal empirical Bayes shrinkage weight of Theorem 2.5, $w_i^* = \hat{s}^2 A_i / (\hat{s}^2 A_i + \sigma_i^2 (1 - h_{ii}))$. $\omega_i = \sum_t (R_{it}^\perp)^2$ is the identifying (residualized) net-of-tax-rate variation contributed by country i ; $\sigma_i^2 = \sigma_\eta^2 / \omega_i$ is its sampling variance; h_{ii} is its self-leverage in the pooled CRC regression; and $A_i = (1 - h_{ii})^2 + \sum_{j \neq i} h_{ij}^2$ is the extrapolation penalty. The structural-shock variance and the residual cross-country heterogeneity are common to all countries and estimated at $\hat{\sigma}_\eta^2 = 0.0098$ and $\hat{s}^2 = 0.0214$ ($\hat{s} = 0.146$), respectively.

4 Extensions to the Empirical Bayes Framework

In this section we extend the foundational empirical Bayes model presented in Section 2 to handle several key issues related to external validity and transportability. First, we characterize what can be learned about the ETI when data from a new jurisdiction is added to the analysis sample. Second, we discuss how the framework can be utilized to make predictions about the ETI of a country that has not adopted a tax policy, using wealth taxation as an application. Finally, we provide a formal extension for meta-analyses that contain multiple reported studies from the same jurisdiction. All results stated in this section are proven in Appendix C.

4.1 Learning From New Data in the Empirical Bayes Framework

Section 2 described how to revise a country's ETI estimate by an optimal procedure in the sense of minimizing compound risk. The three-stage procedure provides a fundamentally different answer than segmented regressions or pooled OLS, and has clear theoretical foundations. Because Stage 1 estimates are computed from within-country data alone, they are inherently invariant to the arrival of new countries. The revision of an individual jurisdiction's predicted ETI operates entirely through hyperparameter updating (Stage 2) and shrinkage recalibration (Stage 3).

Throughout this section, we explicitly define the structural function linking observables to the ETI: $\varepsilon_i = g(Z_i, u_i; \theta)$, where g is the structural function, Z_i are observables, u_i is unobserved heterogeneity, and $\theta = (\beta, s^2)$ are the structural hyperparameters from our earlier notations. We denote $\hat{\theta}_N = (\hat{\beta}_N, \hat{s}_N^2)$ as the estimate based on data from N countries. The residual of the new country under the N -country model is defined as $r_{N+1}^N \equiv \hat{\varepsilon}_{N+1} - g(Z_{N+1}, 0; \hat{\theta}_N)$.

We formally frame the learning process as a sequential compound decision problem over the structural parameter space Θ . At stage N , the researcher observes the ensemble of data $D_N = \{(\hat{\varepsilon}_i, Z_i, \sigma_i^2)\}_{i=1}^N$ and seeks to solve the compound estimation problem by finding a vector of decision rules $\delta = (\delta_1, \dots, \delta_N)$ that minimizes the compound Bayes risk

$$\mathcal{R}_N(\theta, \delta) = \mathbb{E}_\theta \left[\frac{1}{N} \sum_{i=1}^N (\delta_i(D_N) - \varepsilon_i)^2 \right]. \quad (19)$$

When new data arrives, the learning process refines the posterior distribution over the structural parameter space Θ . Because all countries are bound by a shared structural relationship $g(\cdot)$, a single new observation shifts the structural parameters (β) and updates our confidence in the predictive power of observable factors (s^2). This sequential formulation highlights a crucial identifying assumption regarding the stability of the underlying economic environment across contexts.

Assumption 4.1 (External Validity and Parameter Stability). The structural parameters $\theta = (\beta, s^2)$ governing the distribution of the true ETIs via the function $g(\cdot)$ are stable across the expanding sample. Formally,

for all N , $\theta_{N+1} = \theta_N \equiv \theta_0$, and the distribution $\mathbb{P}(\varepsilon_{N+1} \mid Z_{N+1}, \theta_0)$ follows the same structural functional form as the previous N jurisdictions.

Under Assumption 4.1, the arrival of country $N + 1$'s data allows us to update our estimates of the hyperparameters. We provide these characterizations below.

Proposition 4.2 (Conditional Independence and Information Channels). Under the hierarchical model defined by $g(\cdot)$ with known and stable hyperparameters θ_0 , data from country $N + 1$ is completely uninformative about the latent parameter ε_i for any $i \neq N + 1$, conditional on the unit's own data $X_i = (\hat{\varepsilon}_i, Z_i, \sigma_i^2)$. In the empirical setting where θ_0 is unknown and must be estimated via $\hat{\theta}$, country $N + 1$'s data alters the estimates for existing countries strictly and exclusively through the improved estimation of the shared hyperparameters $\theta \in \Theta$.

Proposition 4.3 (Hyperparameter Update). Let the marginal residual of the new country under the linear structural model $g(Z_i, u_i; \theta) = Z_i' \beta + u_i$ be defined as $r_{N+1}^N \equiv \hat{\varepsilon}_{N+1} - Z_{N+1}' \hat{\beta}_N$. The hyperparameter update for the coefficient vector β satisfies the approximation:

$$\Delta \hat{\beta} \approx \left[\sum_{i=1}^N \frac{Z_i Z_i'}{\hat{s}_N^2 + \sigma_i^2} \right]^{-1} \frac{Z_{N+1} \cdot r_{N+1}^N}{\hat{s}_N^2 + \sigma_{N+1}^2}. \quad (20)$$

Furthermore, the variance component s^2 updates according to the following strict condition:

$$\Delta \hat{s}^2 > 0 \quad \text{if and only if} \quad (r_{N+1}^N)^2 > \hat{s}_N^2 + \sigma_{N+1}^2. \quad (21)$$

Having updated the hyperparameters, we can trace how this structural learning percolates down to the individual country estimates. The revision to any existing country's empirical Bayes estimate operates through two distinct channels: a prior mean channel, which shifts the structural anchor (via the CRC model in our setup), and a shrinkage channel, which recalibrates the weight placed on within-country data.

Proposition 4.4 (EB Revision Decomposition). Let $w_i^N \equiv \frac{\hat{s}_N^2}{\hat{s}_N^2 + \sigma_i^2}$ be the shrinkage weight for country i at stage N . Under Assumptions 4.1 and A.4, the revision to the Empirical Bayes estimate for any existing country $i \leq N$ decomposes into:

$$\Delta \hat{\varepsilon}_i^{EB} = (1 - w_i^N)(\mathbf{Z}_i^*)' \Delta \hat{\beta} + \Delta w_i \left(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}_N \right) + O_p(N^{-2}), \quad (22)$$

where the change in the shrinkage weight satisfies the exact relation:

$$\Delta w_i = \frac{\sigma_i^2 \Delta \hat{s}^2}{(\hat{s}_N^2 + \sigma_i^2)(\hat{s}_{N+1}^2 + \sigma_i^2)}. \quad (23)$$

Consequently, the weight change Δw_i shares the same sign as the variance update $\Delta \hat{s}^2$.

In addition to shifting the point estimates, the new evidence also updates our uncertainty about existing countries.

Proposition 4.5 (Posterior Variance Revision). The corresponding change in the posterior variance for country i upon observing country $N + 1$ is characterized by:

$$\Delta s_i^2 = \frac{\sigma_i^4(\Delta \hat{s}^2)}{(\hat{s}_N^2 + \sigma_i^2)(\hat{s}_{N+1}^2 + \sigma_i^2)} . \quad (24)$$

It is worth noting that in Proposition 4.5, Δs_i^2 shares the same sign as $\Delta \hat{s}^2$. The economic intuition behind the result is that a highly “surprising” new country (one that increases estimated heterogeneity, $\Delta \hat{s}^2 > 0$) uniformly increases the posterior uncertainty for all existing countries; conversely, a strictly conforming new country ($\Delta \hat{s}^2 < 0$) decreases the posterior uncertainty.

4.2 External Validity Between Adopters and Non-Adopters—With an Application to Wealth Taxation

We turn now to a setting where only some jurisdictions levy a given tax, and those that do so have systematically different tax systems. What can we learn about the elasticity of the tax base with respect to the tax rate for countries that do not levy such a tax, but may be contemplating adopting such a tax? This setting applies to an annual wealth tax. In 1990, twelve European (mostly OECD) countries levied an annual tax on net wealth. By 2018, only four—France, Norway, Spain, and Switzerland—levied such a tax. Analyses of its likely impact for a non-adopter like the United States have often cited evidence drawn from these remaining adopters.

Following the logic of the previous sections, we can suppose that a wealth tax is adopted in those countries in which the expected net social benefit is relatively high, the determinants of which include a relatively low behavioral response and therefore efficiency cost, as well as a relatively high demand for tax progressivity. This suggests that, on average, adopters have a lower elasticity of taxable wealth (ETW) than non-adopters. To formalize this, we introduce a self-selection component to our model. Let $D_i \in \{0, 1\}$ be the adoption indicator. We model adoption as a structural threshold crossing:

$$D_i = \mathbb{1}\{B(X_i) - C(\mathbf{Z}_i^*, \varepsilon_i^*) > 0\} , \quad (25)$$

where $B(X_i)$ is the social benefit and $C(\mathbf{Z}_i^*, \varepsilon_i^*)$ is the implementation cost, which strictly depends on the true latent potential ETW, ε_i^* . Under a linear approximation of the net benefit $B(X_i) - C(\mathbf{Z}_i^*, \varepsilon_i^*)$, we can write this threshold crossing as

$$D_i = \mathbb{1}\{(\mathbf{W}_i^*)' \gamma + \eta_i > 0\} , \quad (26)$$

where $\mathbf{W}_i^*$ is a vector of aggregated covariates predicting adoption (which may incorporate elements of

X_i and $\mathbf{Z}_i^*$, γ is a parameter vector, and $\eta_i \sim \mathcal{N}(0, 1)$ is the unobserved selection shock.⁸ We define the structural equation for the potential elasticity of taxable wealth (ETW) that would be realized if jurisdiction i levies a wealth tax, denoted by ε_i^* , as

$$\varepsilon_i^* = \mu(\mathbf{Z}_i^*; \beta) + \tilde{u}_i^\varepsilon \quad (27)$$

where $\mu(\mathbf{Z}_i^*; \beta) \equiv (\mathbf{Z}_i^*)' \beta$ is the structural conditional mean, and $\tilde{u}_i^\varepsilon \sim \mathcal{N}(0, s^2)$ is the residual heterogeneity term that maps to our earlier constructions. Because adoption D_i depends on ε_i^* , it follows that the structural shock $\tilde{u}_i^\varepsilon$ is correlated with the selection indicator D_i . Therefore, the expected behavioral response among adopters is different from the expected response among non-adopters, i.e.,

$$\mathbb{E}[\varepsilon_i^* \mid \mathbf{Z}_i^*, D_i = 1] \neq \mathbb{E}[\varepsilon_i^* \mid \mathbf{Z}_i^*, D_i = 0] . \quad (28)$$

This is the selection bias. Consequently, naively taking the simple average of ε_i^* across the $D_i = 1$ sample to predict the response for a non-adopter (where $D_j = 0$) will systematically underestimate the deadweight loss that would arise for the non-adopter.

Selection on unobservables is explicitly captured by the correlation between the country-specific shocks $(\tilde{u}_i^\varepsilon, \eta_i)$. To resolve this, we assume they are jointly normal

$$\begin{pmatrix} \tilde{u}_i^\varepsilon \\ \eta_i \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} s^2 & \rho s \\ \rho s & 1 \end{pmatrix} \right), \quad (29)$$

where $\rho \in [-1, 1]$. If $\rho < 0$, countries with higher elasticities (larger $\tilde{u}_i^\varepsilon$) have lower selection shocks η_i and are less likely to adopt, reflecting self-selection based on unobservable cost optimization. For the adopter sample ($D_i = 1$), we observe a noisy estimate $\hat{\varepsilon}_i = \varepsilon_i^* + v_i$, where $v_i \mid \varepsilon_i^*, D_i \sim \mathcal{N}(0, \sigma_i^2)$ is the sampling/measurement error, independent of $(\tilde{u}_i^\varepsilon, \eta_i)$.

Proposition 4.6 (Identification of Selection-Correction Parameters). Let $\theta = (\beta, s^2, \rho, \gamma)$ be the parameter vector. Under the bivariate normal selection model (29),

1. The first-stage selection parameter γ is identified under the standard normalization $\text{Var}(\eta_i) = 1$.
2. The structural mean coefficients β and the covariance $\theta_{\rho s} \equiv \rho s$ are identified from the conditional mean of adopters:

$$\mathbb{E}[\hat{\varepsilon}_i \mid \mathbf{Z}_i^*, \mathbf{W}_i^*, D_i = 1] = (\mathbf{Z}_i^*)' \beta + \rho s \lambda((\mathbf{W}_i^*)' \gamma), \quad (30)$$

⁸Setting the variance of the unobserved shock to one is a standard and necessary normalization in binary choice models to pin down the scale of the parameter vector γ . Because we only observe the binary indicator D_i rather than the latent net benefit, the scale of the latent index is under-identified. Specifically, the adoption probability is $\mathbb{P}(D_i = 1 \mid \mathbf{W}_i^*) = \Phi((\mathbf{W}_i^*)' \gamma / \sigma)$, where $\sigma^2 = \text{Var}(\eta_i)$. Because the data can only identify the ratio γ / σ , setting $\sigma = 1$ resolves this scale under-identification.

where $\lambda(x) = \phi(x)/\Phi(x)$ is the inverse Mills Ratio, provided the design matrix

$$\mathbf{M}_{\text{select}} \equiv \mathbb{E} \left[\begin{pmatrix} \mathbf{Z}_i^* \\ \lambda((\mathbf{W}_i^*)'\gamma) \end{pmatrix} \begin{pmatrix} (\mathbf{Z}_i^*)' & \lambda((\mathbf{W}_i^*)'\gamma) \end{pmatrix} \middle| D_i = 1 \right] \quad (31)$$

has full column rank.

3. The structural heterogeneity variance s^2 and selection correlation ρ are separately identified from the conditional variance of adopters

$$\text{Var}(\hat{\varepsilon}_i \mid \mathbf{Z}_i^*, \mathbf{W}_i^*, D_i = 1) = \sigma_i^2 + s^2(1 - \rho^2 \delta((\mathbf{W}_i^*)'\gamma)) , \quad (32)$$

where $\delta(x) = \lambda(x)(\lambda(x) + x) \in (0, 1)$ and the measurement variance σ_i^2 is known, provided $\delta((\mathbf{W}_i^*)'\gamma)$ varies across the adopter population.

Proposition 4.6 establishes a rigorous result for selection correction (Heckman, 1979) in the context of EB estimation, and is a generalization of the sample correction procedure to settings with heterogeneous, noisy estimates. Based on this, we provide the following result that characterizes optimal EB weights in the selection model.

Theorem 4.7 (Selection-Corrected Empirical Bayes Shrinkage). For any adopter country i ($D_i = 1$), the exact EB estimator for its true ETI is

$$\hat{\varepsilon}_{i,\text{EB}} \equiv \mathbb{E}[\varepsilon_i^* \mid \hat{\varepsilon}_i, \mathbf{Z}_i^*, \mathbf{W}_i^*, D_i = 1] = (1 - B_i)\hat{\varepsilon}_i + B_i((\mathbf{Z}_i^*)'\beta) + \frac{\rho s B_i}{\sqrt{1 - \rho^2(1 - B_i)}} \lambda(\alpha_i(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)) , \quad (33)$$

where $B_i = \frac{\sigma_i^2}{s^2 + \sigma_i^2}$, the updated selection argument is

$$\alpha_i(e) = \frac{(\mathbf{W}_i^*)'\gamma + \frac{\rho s}{s^2 + \sigma_i^2} e}{\sqrt{1 - \rho^2(1 - B_i)}} , \quad (34)$$

and the corresponding posterior variance is

$$\text{Var}(\varepsilon_i^* \mid \hat{\varepsilon}_i, \mathbf{Z}_i^*, \mathbf{W}_i^*, D_i = 1) = s^2 B_i \left(1 - \frac{\rho^2 B_i}{1 - \rho^2(1 - B_i)} \delta(\alpha_i(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)) \right) . \quad (35)$$

Theorem 4.7 shows how selection-correction and EB shrinkage are linked. The first two terms of (33) represent standard EB shrinkage, while the third term shifts the shrinkage target by correcting for selection truncation. We provide the proofs of these results in Appendix C.2. Furthermore, in Appendix C.2.2, we show how to generate valid predictions intervals for the ETW for countries who did not adopt such a tax. These valid prediction intervals require special construction because non-adoption is itself informative about the latent elasticity; hence, a non-adopter's counterfactual ETW is generally not Gaussian once the

appropriate selection adjustments are made. This is a novel contribution that results from our modelling framework.

Alas, the requisite data for assessing the elasticity of taxable wealth are much scarcer compared to those for the ETI. Because we lack sufficient comparable cross-country measures of taxable wealth (Advani and Tarrant, 2021), we do not present empirical estimates of ETW in this paper, but rather offer this framework to guide future out-of-sample policy predictions in this and other contexts.

4.3 Repeated Observations and Meta-Analysis

A common feature of the empirical public finance literature is the multiplicity of estimates from the same jurisdiction. Meta-analyses, such as Neisser (2021), routinely pool N total estimates where $k \ll N$ observations originate from a single heavily-studied country, such as the United States. The standard empirical Bayes model outlined in Section 2 assumes a single identifying estimate per jurisdiction. When multiple estimates per country exist, they are likely correlated, as they reflect the same underlying institutional environment, even if they leverage different datasets, tax reforms, or empirical specifications. Simply treating k US studies as k independent draws from the global distribution $\mathcal{N}(Z'_{US}\beta, s^2)$ overstates the cross-sectional information content and ignores within-country structural heterogeneity.

We can extend our framework from Section 2 to accomodate this feature. This provides a formalized mechanism to weigh the k existing studies from a specific country against the cross-country evidence when constructing an optimally shrunk ETI estimate. Let $i \in \{1, \dots, C\}$ index the distinct countries, and let $j \in \{1, \dots, k_i\}$ index the individual studies within country i , so the total number of studies is $N = \sum_{i=1}^C k_i$. We assume the following structure:

1. Global Structural Distribution: The true country-level ETI estimand for country i , ε_i^* , is drawn from the global cross-country distribution:

$$\varepsilon_i^* \mid \mathbf{Z}_i^* \sim \mathcal{N}((\mathbf{Z}_i^*)'\beta, s^2) . \quad (36)$$

2. Within-Country Structural Heterogeneity: The true ETI measured by study j in country i , denoted ε_{ij}^* , is heterogeneous and follows the form:

$$\varepsilon_{ij}^* \mid \varepsilon_i^* \sim \mathcal{N}(\varepsilon_i^*, \varsigma^2) . \quad (37)$$

3. Measurement Equation: The observed estimate $\widehat{\varepsilon}_{ij}$ is measured with statistical sampling error:

$$\widehat{\varepsilon}_{ij} \mid \varepsilon_{ij}^* \sim \mathcal{N}(\varepsilon_{ij}^*, \sigma_{ij}^2) , \quad (38)$$

where σ_{ij}^2 is the known sampling variance of study j .

The parameter s^2 captures structural variance *across* countries, whereas ς^2 captures structural variance *within* countries across studies. This formalization allows us to provide a direct link between the empirical Bayes framework and classical meta-analyses.

Proposition 4.8 (Nested Empirical Bayes Estimator). Define the precision-weighted average of the study-level estimates within country i as

$$\bar{\varepsilon}_i = \frac{\sum_{j=1}^{k_i} \frac{\hat{\varepsilon}_{ij}}{\varsigma^2 + \sigma_{ij}^2}}{\sum_{j=1}^{k_i} \frac{1}{\varsigma^2 + \sigma_{ij}^2}}, \quad \text{with associated variance} \quad V_i = \left(\sum_{j=1}^{k_i} \frac{1}{\varsigma^2 + \sigma_{ij}^2} \right)^{-1}. \quad (39)$$

The optimal empirical Bayes estimate (the posterior mean) of the country-level structural parameter ε_i^* is a convex combination

$$\hat{\varepsilon}_i^{EB} = \mathbb{E}[\varepsilon_i^* \mid \hat{\varepsilon}_i, \mathbf{Z}_i^*] = (1 - W_i) \cdot (\mathbf{Z}_i^*)' \beta + W_i \cdot \bar{\varepsilon}_i, \quad (40)$$

where $W_i = \frac{s^2}{s^2 + V_i}$. Furthermore, the optimal estimate for a specific study j within country i exhibits a secondary layer of shrinkage

$$\hat{\varepsilon}_{ij}^{EB} = \mathbb{E}[\varepsilon_{ij}^* \mid \hat{\varepsilon}_i, \mathbf{Z}_i^*] = (1 - w_{ij}) \hat{\varepsilon}_i^{EB} + w_{ij} \hat{\varepsilon}_{ij}, \quad (41)$$

where $w_{ij} \equiv \frac{\varsigma^2}{\varsigma^2 + \sigma_{ij}^2}$.

Proposition 4.8 explicitly answers how to balance a dense country-specific literature against a sparse global literature. When predicting the ETI for the United States ($i = US$), the weight placed on the k_{US} existing US studies relative to the cross-country structural model is governed by W_{US} . If the within-country structural heterogeneity ς^2 is large (meaning different US studies measure systematically different objects), the pooled variance V_{US} remains bounded away from zero even if $k_{US} \rightarrow \infty$. In this scenario, W_{US} remains strictly less than 1, implying that the cross-country heterogeneity (s^2) remains informative for predicting the US ETI, despite an abundance of US studies. Conversely, if ς^2 is small, the collection of US studies provides a highly precise estimate of ε_{US} ($V_{US} \rightarrow 0$), driving the shrinkage weight W_{US} to be approximately 1.

5 Conclusion

We develop a new empirical Bayes framework for estimating the ETI that utilizes data from many countries, recognizing that behavioral responses vary systematically in ways suggested by the theory of optimal taxation. Because the ETI depends on, among other things, the social preference for equality and the social cost of implementing a broad, well-enforced tax system, we estimate a correlated random coefficients model that separates the within-country and cross-country channels through which those determinants act, and that, through the country-level terms, absorbs the component of unobserved elasticity heterogeneity correlated with the persistent characteristics on which governments set tax rates. We then use empirical Bayes shrink-

age to optimally combine these structural predictions with segmented, within-country estimates, which is a standard way of estimating the ETI.

Applying this framework to 27 OECD countries, we find that several structural characteristics are significant predictors of the ETI in the cross-country (level) channel—the public’s confidence in government and its views on the proper scope of government responsibility, the perceived justifiability of tax cheating, and country size—while a greater emphasis on efficiency over equity raises the ETI through the within-country (dynamic) channel. The United States has a somewhat higher predicted ETI than the cross-country average, a prediction that nets several elasticity-raising attitudes against the downward pull of its large population. By applying empirical Bayes shrinkage, our method produces country-specific ETI estimates that borrow strength from the cross-country sample, yielding estimates that are structurally disciplined. We demonstrate that this framework can be extended in a number of practical ways: allow sequentially incorporating new country data; allow aggregating repeated observations from a single jurisdiction; and providing out-of-sample predictions for jurisdictions that have not yet adopted a specific tax policy (e.g. wealth tax).

We do not offer the specific ETI estimates presented here as the definitive parameters for each country. Rather, we offer the three-step empirical Bayes framework as a rigorous method for distilling the lessons to be learned from empirical evidence across settings. While meta-analyses aggregate existing estimates, we generally cannot separate true parameter heterogeneity from differences in research design. Our approach resolves this by taking within-country, segmented estimates obtained from the same empirical specification, and explicitly modelling cross-country variation in ETIs as a function of observed characteristics. This yields a shrinkage representation that, as we argue, is the decision-theoretically optimal way to combine local and global evidence: each country’s estimate is pulled toward the structural prediction by exactly the amount that minimizes compound Bayes risk under squared error loss.

A natural next step in this agenda is to expand the set of factors used to explain why tax systems and behavioral elasticities vary. Additionally, extending the framework beyond the OECD to developing countries would introduce greater variation in the factors affecting tax systems and observed elasticities, improving the predictive power of the model, albeit with the challenge of incorporating potentially lower-quality administrative data.

References

- ADVANI, A. AND H. TARRANT (2021): “Behavioural Responses to a Wealth Tax,” *Fiscal Studies*, 42, 509–537. (Cited on page 38.)
- ALESINA, A. AND E. L. FERRARA (2005): “Ethnic Diversity and Economic Performance,” *Journal of Economic Literature*, 43, 762–800. (Cited on page 18.)
- ANDREW YOUNG SCHOOL OF POLICY STUDIES (2010): *Andrew Young School World Tax Indicators*, 1. (Cited on pages 16 and 22.)
- ARMSTRONG, T. B., M. KOLESÁR, AND M. PLAGBORG-MØLLER (2022): “Robust Empirical Bayes Confidence Intervals,” *Econometrica*, 90, 2917–2954. (Cited on pages 2 and 5.)
- ARNOLD, B. C., R. J. BEAVER, A. AZZALINI, N. BALAKRISHNAN, A. BHAUMIK, D. K. DEY, C. M. CUADRAS, J. M. SARABIA, B. C. ARNOLD, AND R. J. BEAVER (2002): “Skewed Multivariate Models Related to Hidden Truncation and/or Selective Reporting,” *TEST*, 11, 7–54. (Cited on page A-23.)
- AZZALINI, A. (2005): “The Skew-Normal Distribution and Related Multivariate Families,” *Scandinavian Journal of Statistics*, 32, 159–188. (Cited on page A-23.)
- AZZALINI, A. AND A. CAPITANIO (2013): *The Skew-Normal and Related Families*, Cambridge University Press (CUP). (Cited on page A-23.)
- BERAN, R. (1995): “Prediction in Random Coefficient Regression,” *Journal of Statistical Planning and Inference*, 43, 205–213. (Cited on page 7.)
- CAI, Z., J. FAN, AND R. LI (2000): “Efficient Estimation and Inferences for Varying-Coefficient Models,” *Journal of the American Statistical Association*, 95, 888–902. (Cited on page 7.)
- CHAMBERLAIN, G. (1982): “Multivariate Regression Models for Panel Data,” *Journal of Econometrics*, 18, 5–46. (Cited on page 7.)
- (1984): “Chapter 22 Panel Data,” in *Handbook of Econometrics*, Elsevier, vol. 2, 1247–1318. (Cited on pages 7 and R-7.)
- CHEN, J. (2026): “Empirical Bayes When Estimation Precision Predicts Parameters,” *Econometrica*, 94, 305–340. (Cited on page 5.)
- CHETTY, R. AND N. HENDREN (2018): “The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates,” *Quarterly Journal of Economics*, 133, 1163–1228. (Cited on page R-9.)
- DEATON, A. AND N. CARTWRIGHT (2018): “Understanding and Misunderstanding Randomized Controlled Trials,” *Social Science & Medicine*, 210, 2–21. (Cited on page 1.)

- DEGTIAR, I. AND S. ROSE (2023): “A Review of Generalizability and Transportability,” *Annual Review of Statistics and Its Application*, 10, 501–524. (Cited on page 1.)
- DRAZANOVA, L. (2019): “Historical Index of Ethnic Fractionalization Dataset (HIEF),” *Harvard Data-verse*. (Cited on page 18.)
- EFRON, B. (2019): “Bayes, Oracle Bayes, and Empirical Bayes,” *Statistical Science*, 34, 177–201. (Cited on pages 2 and 5.)
- FAN, J. AND W. ZHANG (2008): “Statistical Methods with Varying Coefficient Models,” *Statistics and Its Interface*, 1, 179. (Cited on page 7.)
- FINDLEY, M. G., K. KIKUTA, AND M. DENLY (2021): “External Validity,” *Annual Review of Political Science*, 24, 365–393. (Cited on page 1.)
- GHOSH, I. AND K. T. NG (2025): “Hidden Truncation Models: Theory and Applications,” *Wiley Interdisciplinary Reviews Computational Statistics*, 17. (Cited on page A-23.)
- GRUBER, J. AND E. SAEZ (2002): “The Elasticity of Taxable Income: Evidence and Implications,” *Journal of Public Economics*, 84, 1–32. (Cited on page 19.)
- HASTIE, T. AND R. TIBSHIRANI (1993): “Varying-Coefficient Models,” *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 55, 757–779. (Cited on page 7.)
- HECKMAN, J. J. (1979): “Sample Selection Bias as a Specification Error,” *Econometrica*, 153–161. (Cited on pages 4 and 37.)
- IGNATIADIS, N. AND S. WAGER (2022): “Confidence Intervals for Nonparametric Empirical Bayes Analysis,” *Journal of the American Statistical Association*, 117, 1149–1166. (Cited on pages 2 and 5.)
- ISHIHARA, T. AND T. KITAGAWA (2024): “Evidence Aggregation for Treatment Choice,” *arXiv preprint arXiv:2108.06473*. (Cited on pages 13, R-13, and R-14.)
- KLEVEN, H. J. (2014): “How Can Scandinavians Tax So Much?” *Journal of Economic Perspectives*, 28, 77–98. (Cited on pages 18 and R-5.)
- MERTENS, K. AND J. L. MONTIEL OLEA (2018): “Marginal Tax Rates and Income: New Time Series Evidence,” *Quarterly Journal of Economics*, 133, 1803–1884. (Cited on pages 12 and R-15.)
- MONTIEL OLEA, J. L., C. QIU, AND J. STOYE (2026): “Decision Theory for Treatment Choice Problems with Partial Identification,” *Review of Economic Studies*, rdag015. (Cited on pages 13, R-13, and R-14.)
- MORRIS, C. N. (1983): “Parametric Empirical Bayes Inference: Theory and Applications,” *Journal of the American Statistical Association*, 78, 47–55. (Cited on pages 2, 5, 8, 10, A-36, R-7, and R-9.)

- MUNDLAK, Y. (1978): “On the Pooling of Time Series and Cross Section Data,” *Econometrica*, 46, 69–85. (Cited on pages 7 and R-7.)
- NEISSER, C. (2021): “The Elasticity of Taxable Income: A Meta-Regression Analysis,” *Economic Journal*, 131, 3365–3391. (Cited on pages 1, 3, and 38.)
- NEYMAN, J. AND E. L. SCOTT (1948): “Consistent Estimates Based on Partially Consistent Observations,” *Econometrica*, 16, 1–32. (Cited on page A-11.)
- OECD (2023): “*Taxing Wages 2023*,” 669. (Cited on pages 16 and 22.)
- PARK, B. U., E. MAMMEN, Y. K. LEE, AND E. R. LEE (2015): “Varying Coefficient Regression Models: A Review and New Developments,” *International Statistical Review*, 83, 36–64. (Cited on page 7.)
- PEARL, J. AND E. BAREINBOIM (2011): “Transportability of Causal and Statistical Relations: A Formal Approach,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 25, 247–254. (Cited on page 1.)
- (2014): “External Validity: From Do-Calculus to Transportability Across Populations,” *Statistical Science*, 29, 579 – 595. (Cited on page 1.)
- ROBBINS, H. (1951): “Asymptotically Subminimax Solutions of Compound Decision Problems,” in *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, University of California Press, 131–148. (Cited on page 5.)
- (1956): “An Empirical Bayes Approach to Statistics,” in *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, University of California Press, vol. 1, 157–163. (Cited on pages 2, 5, 8, R-7, and R-9.)
- RUBOLINO, E. AND D. WALDENSTRÖM (2019): “Trends and Gradients in Top Tax Elasticities: Cross-Country Evidence, 1900–2014,” *International Tax and Public Finance*, 26, 457–485. (Cited on pages 1, 2, 15, 16, 19, and 22.)
- SAEZ, E., J. SLEMROD, AND S. H. GIERTZ (2012): “The Elasticity of Taxable Income with Respect to Marginal Tax Rates: A Critical Review,” *Journal of Economic Literature*, 50, 3–50. (Cited on pages 19 and 28.)
- SLEMROD, J. AND W. KOPCZUK (2002): “The Optimal Elasticity of Taxable Income,” *Journal of Public Economics*, 84, 91–112. (Cited on pages 1, 2, 6, 14, 16, 24, R-2, and R-4.)
- STEIN, C. M. (1981): “Estimation of the Mean of a Multivariate Normal Distribution,” *Annals of Statistics*, 9, 1135–1151. (Cited on page R-9.)

- VAN DER VAART, A. W. (2000): *Asymptotic Statistics*, vol. 3, Cambridge University Press. (Cited on page A-42.)
- WALTERS, C. R. (2024): “Empirical Bayes Methods in Labor Economics,” in *Handbook of Labor Economics*, ed. by C. Dustmann and T. Lemieux, Elsevier, vol. 5, forthcoming. (Cited on page 2.)
- WID.WORLD (2022): “World Inequality Database,” . (Cited on pages 15 and 22.)
- WOOLDRIDGE, J. M. (2010): *Econometric Analysis of Cross Section and Panel Data*, MIT Press. (Cited on page 7.)
- YATA, K. (2024): “Optimal Decision Rules under Partial Identification,” *arXiv preprint arXiv:2111.04926*. (Cited on pages 13, R-13, and R-14.)

Appendix:
**External Validity in Empirical Public Finance: With an Application to
Taxable Income Elasticities**

Table of Contents

A	Proofs of Results from Section 2	A-1
A.1	Proof of Theorem 2.3	A-3
A.2	Proof of Theorem 2.5	A-3
A.3	Proof of the Equivalence of Three-Step Estimation to a Joint MLE	A-7
B	Estimation Details	A-12
C	Proofs of Results from Section 4	A-16
C.1	Learning from New Data	A-16
C.2	ETW and Non-Adopters	A-20
C.3	Nested Empirical Bayes Estimator (Repeated Observations)	A-26
D	Additional Results	A-28
D.1	Technical Results	A-28
D.2	Leave-One-Out Estimation for Stage 2	A-35
D.3	Equivalence and Asymptotic Similarity of LOO and Full-Sample EB	A-37
D.4	Heteroskedastic Residual Heterogeneity	A-40
E	Monte Carlo Simulation	A-44
E.1	Baseline Exercise	A-44
E.2	Credibility of the Feasible Estimates	A-45

A Proofs of Results from Section 2

This appendix provides rigorous proofs for the propositions and theorems stated in Section 2.

Throughout the appendix, $i, j \in \{1, \dots, N\}$ index jurisdictions and t, s index time periods. panels are balanced with T periods per country. The scalar $\omega_i \equiv \sum_{t=1}^T (R_{it}^\perp)^2$ measures country i 's identifying tax variation, $\mathbf{Z}_i^*$ is the tax-variation-weighted covariate profile, $h_{ij} \equiv \omega_j (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^*$ are leverage entries, and $A_i \equiv (1 - h_{ii})^2 + \sum_{j \neq i} h_{ij}^2$ is the extrapolation penalty. The variance s^2 denotes cross-country structural heterogeneity, $\sigma_i^2 = \sigma_\eta^2 / \omega_i$ the Stage 1 sampling variance, $\theta \equiv (\beta', s^2)'$ the hyperparameter vector, and $\pi \equiv b + \psi$ is the composite Mundlak coefficient. Unless stated otherwise, $O_p(\cdot)$ and $o_p(\cdot)$ orders are taken along the cross-sectional dimension $N \rightarrow \infty$, holding the panel length T fixed, and all finite-sample distributional statements are conditional on $(\mathbf{R}, \mathbf{Z})$.

To establish the consistency and asymptotic normality of our estimators, we require the following regularity conditions. We separate the conditions by the stage of the procedure at which they operate, and we flag which conclusions require the optional Gaussianity of Assumption 2.2(iv)—exact finite-sample distributional statements—and which hold under moment conditions alone—all asymptotic statements.

Assumption A.1 (Regularity conditions for Stage 1 Segmented Regressions). Let $\tilde{R}_{it}$ denote the log net-of-tax rate after partialling out fixed effects and trends. The following conditions hold:

- (i) **Identification:** $\sum_{t=1}^T (\tilde{R}_{it})^2 > 0$ for all i .
- (ii) **No Dominance:** $\max_{t \leq T} \tilde{R}_{it}^2 / \sum_{s=1}^T \tilde{R}_{is}^2 \rightarrow 0$ as $T \rightarrow \infty$.
- (iii) **Bounded Moments:** The structural errors have uniformly bounded moments $\mathbb{E}[|\eta_{it}|^{4+\iota}] \leq C < \infty$ for some $\iota > 0$.

Part (ii) is exactly the Lindeberg-type negligibility condition for the weighted error sum $v_i = \omega_i^{-1} \sum_t R_{it}^\perp \eta_{it}^\perp$: it requires that no single year carries a non-vanishing share of the identifying variation in the limit (as $T \rightarrow \infty$), and together with part (iii) it delivers the asymptotic normality of v_i / σ_i via the Lindeberg–Feller theorem (Lemma D.5).

Assumption A.2 (Regularity conditions for Stage 2 CRC regressions). Let $\mathbf{W}_{it}$ and $\mathbf{Z}_i^*$ be as defined in Lemma D.1.

- (i) **Positive Definite Limit:** The design matrix converges to a positive-definite limit, i.e.

$$\Sigma_W = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \mathbb{E}[\mathbf{W}_{it} \mathbf{W}_{it}'] \succ 0. \quad (\text{A.1})$$

- (ii) **Fourth moment bound on regressors:** $\mathbb{E}[\|\mathbf{W}_{it}\|^4] \leq C < \infty$.
- (iii) **CLT for In-Sample Shocks:** $\frac{1}{\sqrt{N}} \sum_{i=1}^N \sum_{t=1}^T \mathbf{W}_{it} \eta_{it}^\perp \xrightarrow{d} \mathcal{N}(0, \sigma_\eta^2 \Sigma_W)$.

- (iv) **CLT for Random Effects:** $\frac{1}{\sqrt{N}} \sum_{i=1}^N \omega_i \mathbf{Z}_i^* \tilde{u}_i^\varepsilon \xrightarrow{d} \mathcal{N}(0, s^2 \Sigma_{Z^*})$, where $\Sigma_{Z^*} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \omega_i^2 \mathbf{Z}_i^* (\mathbf{Z}_i^*)'$.
- (v) **Bounded Panels and Variation:** The common panel length satisfies $2 \leq T < \infty$, and the identifying variations satisfy $0 < \underline{\omega} \leq \omega_i \leq \bar{\omega} < \infty$ for all i , and the weighted covariate profiles are uniformly bounded, $\sup_i \|\mathbf{Z}_i^*\| \leq C < \infty$.
- (vi) **Well-Conditioned Design:** There exists $c > 0$ such that $\lambda_{\min}(\mathbf{M}) \geq cN$ with probability approaching one.

Note that Assumption A.2(iv) holds automatically under Assumption A.2(i) since $\mathbf{M}/N \rightarrow \Sigma_W \succ 0$.

Assumption A.3 (Regularity conditions for MLE). Let $\theta = (\beta', s^2)' \in \Theta \subset \mathbb{R}^K \times [0, \bar{s}^2]$ parameterize the marginal model (9).

- (i) **Compactness and Interiority:** Θ is compact and the true $\theta_0 = (\beta'_0, s_0^2)'$, with $s_0^2 > 0$, lies in the interior of Θ .
- (ii) **Bounded Covariates and Variances:** $\sup_i \|\mathbf{Z}_i^*\| \leq C < \infty$ and $0 < \underline{\sigma}^2 \leq \sigma_i^2 \leq \bar{\sigma}^2 < \infty$.
- (iii) **Identification:** $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \succ 0$, and the sampling variances $\{\sigma_i^2\}$ do not degenerate to a single point mass in the limit or, if they do, the common value is known; either condition makes θ_0 the unique maximizer of the limiting expected marginal log-likelihood.
- (iv) **Hessian Stability:** The standardized Hessian of the marginal log-likelihood converges uniformly on Θ to its expectation at rate $O_p(N^{-1/2})$.

Assumption A.4 (Additional MLE regularity conditions). The sequence of covariate profiles and parameters satisfies:

- (i) **Positive Definite Average Fisher Information:** The average expected Fisher Information matrix at stage N :

$$\mathbf{I}_N(\theta) \equiv \frac{1}{N} \sum_{i=1}^N \mathbb{E} \left[-\nabla_{\theta\theta}^2 \ell_i(\theta) \mid \mathbf{Z}_{1:N}^*, \sigma_{1:N}^2 \right] \quad (\text{A.2})$$

is positive definite and satisfies $\lambda_{\min}(\mathbf{I}_N(\theta)) \geq c > 0$ uniformly on Θ for all $N \geq N_0$.

- (ii) **Uniform Convergence of Hessian:** The empirical Hessian converges uniformly to its expectation:

$$\sup_{\theta \in \Theta} \left\| -\frac{1}{N} \sum_{i=1}^N \nabla_{\theta\theta}^2 \ell_i(\theta) - \mathbf{I}_N(\theta) \right\| = O_p(N^{-1/2}). \quad (\text{A.3})$$

A.1 Proof of Theorem 2.3

Proof. By Lemma D.1(i), the projected outcome satisfies $y_{it}^\perp = \beta' \mathbf{W}_{it} + \tilde{u}_i^\varepsilon R_{it}^\perp + \eta_{it}^\perp$ exactly. Substituting this into the Stage 1 segmented OLS estimator yields

$$\hat{\varepsilon}_i = \frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp y_{it}^\perp = \frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp \left(\beta' \mathbf{W}_{it} + \tilde{u}_i^\varepsilon R_{it}^\perp + \eta_{it}^\perp \right). \quad (\text{A.4})$$

Since the coefficient vectors do not depend on t ,

$$\begin{aligned} \hat{\varepsilon}_i &= \beta' \left(\frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp \mathbf{W}_{it} \right) + \tilde{u}_i^\varepsilon \left(\frac{1}{\omega_i} \sum_{t=1}^T (R_{it}^\perp)^2 \right) + \frac{1}{\omega_i} \sum_{t=1}^T R_{it}^\perp \eta_{it}^\perp \\ &= \beta' \mathbf{Z}_i^* + \tilde{u}_i^\varepsilon + v_i, \end{aligned}$$

where the first equality in the second line uses the weighted covariate identity $\frac{1}{\omega_i} \mathbf{W}_i' \mathbf{R}_i^\perp = \mathbf{Z}_i^*$ from Lemma D.1(ii), the middle term simplifies because $\frac{1}{\omega_i} \sum_t (R_{it}^\perp)^2 = 1$, and we define $v_i \equiv \frac{1}{\omega_i} \sum_t R_{it}^\perp \eta_{it}^\perp$. By the projection absorption property (Lemma D.1(iii)) with $\mathbf{c} = \mathbf{R}_i^\perp$, the sampling error has the equivalent shock representation $v_i = \frac{1}{\omega_i} \sum_t R_{it}^\perp \eta_{it}$.

We now condition on $(\mathbf{R}, \mathbf{Z}, \tilde{u}_i^\varepsilon)$. The first two terms of the decomposition are fixed under this conditioning—this is the point of conditioning on the latent effect: $\varepsilon_i^* = \beta' \mathbf{Z}_i^* + \tilde{u}_i^\varepsilon$ is the estimand, not a source of noise. For the third term, Assumption 2.2(i) and (iii) give $\mathbb{E}[\eta_{it} \mid \mathbf{R}, \mathbf{Z}, \tilde{u}_i^\varepsilon] = \mathbb{E}[\eta_{it} \mid \mathbf{R}, \mathbf{Z}] = 0$, where the first equality uses the independence of the shocks from the random effects. Hence

$$\mathbb{E}[\hat{\varepsilon}_i \mid \mathbf{R}, \mathbf{Z}, \tilde{u}_i^\varepsilon] = \beta' \mathbf{Z}_i^* + \tilde{u}_i^\varepsilon \equiv \varepsilon_i^*, \quad (\text{A.5})$$

so the segmented regression estimator is conditionally unbiased for the targeting estimand. For the variance, the shock representation and Assumption 2.2(ii) give

$$\text{Var}(v_i \mid \mathbf{R}, \mathbf{Z}) = \frac{1}{\omega_i^2} \sum_{t=1}^T (R_{it}^\perp)^2 \text{Var}(\eta_{it} \mid \mathbf{R}, \mathbf{Z}) = \frac{\sigma_\eta^2}{\omega_i^2} \omega_i = \frac{\sigma_\eta^2}{\omega_i} \equiv \sigma_i^2. \quad (\text{A.6})$$

Finally, under Assumption 2.2(iv) the raw shocks are conditionally Gaussian, so v_i , a fixed linear combination of them, satisfies $v_i \mid \mathbf{R}, \mathbf{Z} \sim \mathcal{N}(0, \sigma_i^2)$ exactly; without part (iv), Lemma D.5 gives $v_i/\sigma_i \xrightarrow{d} \mathcal{N}(0, 1)$ as $T \rightarrow \infty$. The independence of v_i from ε_i^* conditional on $(\mathbf{R}, \mathbf{Z})$ follows from Assumption 2.2(iii), since v_i is a function of the shocks alone while ε_i^* is a function of $\tilde{u}_i^\varepsilon$ alone. This completes the proof. $\square$

A.2 Proof of Theorem 2.5

Proof. First, we note that under Assumption 2.2(iv), conditional on $(\mathbf{R}, \mathbf{Z})$ the vector $(\hat{\varepsilon}_i, \hat{\varepsilon}_i^{CRC}, \varepsilon_i^*)$ is jointly Gaussian: each coordinate is an affine function of the jointly Gaussian collection $(\{\tilde{u}_j^\varepsilon\}_j, \{\eta_{jt}\}_{j,t})$.

For jointly Gaussian variables, the conditional expectation of ε_i^* given the two signals is affine in them, so the Bayes rule lies in the affine class. Moreover, both signals are conditionally unbiased for ε_i^* up to the common prior mean, and imposing that the combination reproduce ε_i^* in expectation forces the coefficients to sum to one. Hence the convex-combination form (12) spans the Bayes rule, and minimizing risk over w_i recovers it.

Step 1: Error decompositions Substituting the reduced form of Lemma D.1(i) into the pooled OLS formula and using $\sum_s \mathbf{W}_{js} R_{js}^\perp = \omega_j \mathbf{Z}_j^*$ (Lemma D.1(ii)),

$$\hat{\beta}^{CRC} - \beta = \mathbf{M}^{-1} \sum_{j=1}^N \sum_{s=1}^T \mathbf{W}_{js} \left(\tilde{u}_j^\varepsilon R_{js}^\perp + \eta_{js}^\perp \right) = \mathbf{M}^{-1} \sum_{j=1}^N \left(\omega_j \mathbf{Z}_j^* \tilde{u}_j^\varepsilon + \sum_{s=1}^T \mathbf{W}_{js} \eta_{js}^\perp \right). \quad (\text{A.7})$$

Substituting this into $E_i = (\mathbf{Z}_i^*)' (\hat{\beta}^{CRC} - \beta) - \tilde{u}_i^\varepsilon$ and recognizing the leverage entries $h_{ij} = \omega_j (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^*$ yields

$$E_i = (h_{ii} - 1) \tilde{u}_i^\varepsilon + \sum_{j \neq i} h_{ij} \tilde{u}_j^\varepsilon + \underbrace{(\mathbf{Z}_i^*)' \mathbf{M}^{-1} \sum_{j=1}^N \sum_{s=1}^T \mathbf{W}_{js} \eta_{js}^\perp}_{\equiv e_i^\eta}. \quad (\text{A.8})$$

Step 2: Variance of the prediction error The random effects $\{\tilde{u}_j^\varepsilon\}$ and the shocks $\{\eta_{js}\}$ are mutually independent by Assumption 2.2(iii), so the variance of (A.8) splits into a structural part and a sampling part with no cross term. For the structural part, the $\tilde{u}_j^\varepsilon$ are independent across j with common variance s^2 , hence

$$\text{Var} \left((h_{ii} - 1) \tilde{u}_i^\varepsilon + \sum_{j \neq i} h_{ij} \tilde{u}_j^\varepsilon \right) = s^2 \left[(1 - h_{ii})^2 + \sum_{j \neq i} h_{ij}^2 \right] = s^2 A_i.$$

For the sampling part, let $\mathbb{W}$ denote the stacked design and $\boldsymbol{\eta}$ the stacked raw shock vector. By the projection absorption identity (Lemma D.1(iii) applied column-wise), $e_i^\eta = (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbb{W}' \boldsymbol{\eta}$, where the *raw* shocks appear; since $\text{Var}(\boldsymbol{\eta} \mid \mathbf{R}, \mathbf{Z}) = \sigma_\eta^2 \mathbf{I}$,

$$\text{Var}(e_i^\eta) = \sigma_\eta^2 (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbb{W}' \mathbb{W} \mathbf{M}^{-1} \mathbf{Z}_i^* = \sigma_\eta^2 (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_i^* = \frac{\sigma_\eta^2 h_{ii}}{\omega_i} = h_{ii} \sigma_i^2,$$

using $\mathbb{W}' \mathbb{W} = \mathbf{M}$ and the definitions of h_{ii} and σ_i^2 . Adding the two parts, $\text{Var}(E_i) = s^2 A_i + h_{ii} \sigma_i^2$.

Step 3: Covariance between measurement and prediction errors By the shock representations $v_i = \omega_i^{-1} (\mathbf{R}_i^\perp)' \boldsymbol{\eta}_i$ and $e_i^\eta = (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbb{W}' \boldsymbol{\eta}$, and the independence of v_i from every $\tilde{u}_j^\varepsilon$, the only non-vanishing

contribution to $\text{Cov}(v_i, E_i)$ runs through the shocks of country i itself:

$$\text{Cov}(v_i, E_i) = \text{Cov}(v_i, e_i^\eta) = \frac{\sigma_\eta^2}{\omega_i} (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{W}_i' \mathbf{R}_i^\perp = \frac{\sigma_\eta^2}{\omega_i} (\mathbf{Z}_i^*)' \mathbf{M}^{-1} (\omega_i \mathbf{Z}_i^*) = \sigma_\eta^2 (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_i^* = h_{ii} \sigma_i^2,$$

where the second equality again invokes Lemma D.1(ii). The covariance is positive whenever $h_{ii} > 0$.

Step 4: Risk minimization The combined estimator's error is $\hat{\varepsilon}_i^{EB} - \varepsilon_i^* = w_i v_i + (1 - w_i) E_i$, so the conditional risk is the quadratic

$$R_i(w) = w^2 \sigma_i^2 + (1 - w)^2 \text{Var}(E_i) + 2w(1 - w) \text{Cov}(v_i, E_i).$$

Differentiating with respect to w and setting the derivative to zero,

$$\begin{aligned} \frac{dR_i}{dw} &= 2w\sigma_i^2 - 2(1 - w) \text{Var}(E_i) + 2(1 - 2w) \text{Cov}(v_i, E_i) = 0 \\ \iff w \left[\sigma_i^2 + \text{Var}(E_i) - 2 \text{Cov}(v_i, E_i) \right] &= \text{Var}(E_i) - \text{Cov}(v_i, E_i). \end{aligned}$$

The bracketed coefficient equals $\text{Var}(v_i - E_i) > 0$ (it can vanish only in the degenerate case $v_i = E_i$ almost surely), so the second-order condition holds strictly and the unique minimizer is

$$w_i^* = \frac{\text{Var}(E_i) - \text{Cov}(v_i, E_i)}{\text{Var}(v_i) + \text{Var}(E_i) - 2 \text{Cov}(v_i, E_i)} = \frac{(\mathbf{s}^2 A_i + h_{ii} \sigma_i^2) - h_{ii} \sigma_i^2}{\sigma_i^2 + (\mathbf{s}^2 A_i + h_{ii} \sigma_i^2) - 2h_{ii} \sigma_i^2} = \frac{\mathbf{s}^2 A_i}{\mathbf{s}^2 A_i + \sigma_i^2 (1 - h_{ii})}.$$

Alternatively, we can express this weight in terms of the full covariance matrix $\mathbf{V}_\beta$. First, substituting $\mathbf{s}^2 A_i = (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* + \mathbf{s}^2 (1 - 2h_{ii})$ into both numerator and denominator yields the representation (13) in terms of $\mathbf{V}_\beta^{\text{struct}}$. Second, since

$$(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* = \mathbf{s}^2 \sum_{j=1}^N h_{ij}^2 + h_{ii} \sigma_i^2 \quad (\text{A.9})$$

by direct calculations, the full covariance satisfies $(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* = (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* + h_{ii} \sigma_i^2$ by (A.9), so we may eliminate the structural quadratic form in favor of the full one: the numerator becomes $(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + \mathbf{s}^2 (1 - 2h_{ii}) - h_{ii} \sigma_i^2$ and the denominator becomes the numerator plus $\sigma_i^2 (1 - h_{ii})$, which equals $(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + \mathbf{s}^2 (1 - 2h_{ii}) + \sigma_i^2 (1 - 2h_{ii})$. Hence, we have

$$w_i^* = \frac{(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + \mathbf{s}^2 (1 - 2h_{ii}) - h_{ii} \sigma_i^2}{(\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + \mathbf{s}^2 (1 - 2h_{ii}) + \sigma_i^2 (1 - 2h_{ii})}.$$

This completes the proof of Theorem 2.5. □

For completeness, we characterize the excess risk if we ignore the leverage term.

Proposition A.5 (Exact Excess Risk of Ignoring Leverage). Under Assumption 2.2, the conditional risk of the convex combination (12) is the strictly convex quadratic

$$R_i(w) = w^2\sigma_i^2 + (1-w)^2 \left(s^2 A_i + h_{ii}\sigma_i^2 \right) + 2w(1-w) h_{ii}\sigma_i^2 = R_i(w_i^*) + (w-w_i^*)^2 \left[s^2 A_i + \sigma_i^2(1-h_{ii}) \right],$$

with minimized value

$$R_i(w_i^*) = \frac{\sigma_i^2 s^2 A_i + h_{ii}(1-h_{ii})\sigma_i^4}{s^2 A_i + \sigma_i^2(1-h_{ii})}.$$

In particular, the excess risk of the naive Morris weight $w_i^M \equiv s^2/(s^2 + \sigma_i^2)$ is

$$R_i(w_i^M) - R_i(w_i^*) = \frac{s^4 \sigma_i^4 \left[h_{ii}(1-h_{ii}) - \sum_{j \neq i} h_{ij}^2 \right]^2}{(s^2 + \sigma_i^2)^2 \left[s^2 A_i + \sigma_i^2(1-h_{ii}) \right]} \geq 0. \quad (\text{A.10})$$

Under Assumption A.2, the excess risk is $O_p(N^{-2})$ per country, so the leverage correction is a finite-sample refinement in well-balanced designs; however, for countries whose covariate profiles command non-negligible leverage— h_{ii} or $\sum_{j \neq i} h_{ij}^2$ bounded away from zero, as under extrapolative profiles or small N —the gap is first-order.

Proof of Proposition A.5. From Step 4 of the proof of Theorem 2.5, the conditional risk is the strictly convex quadratic $R_i(w) = w^2\sigma_i^2 + (1-w)^2 \text{Var}(E_i) + 2w(1-w) \text{Cov}(v_i, E_i)$ with second derivative $2[\sigma_i^2 + \text{Var}(E_i) - 2 \text{Cov}(v_i, E_i)] = 2[s^2 A_i + \sigma_i^2(1-h_{ii})] + 2h_{ii}\sigma_i^2 - 2h_{ii}\sigma_i^2$. Since a quadratic equals its minimum plus half its second derivative times the squared distance to the minimizer, and here half the second derivative is $\sigma_i^2 + \text{Var}(E_i) - 2 \text{Cov}(v_i, E_i) = s^2 A_i + \sigma_i^2(1-h_{ii})$, we have

$$R_i(w) = R_i(w_i^*) + (w - w_i^*)^2 \left[s^2 A_i + \sigma_i^2(1-h_{ii}) \right] \quad \text{for all } w.$$

For the minimized risk, the standard formula for the optimum of this quadratic gives

$$R_i(w_i^*) = \frac{\text{Var}(v_i) \text{Var}(E_i) - \text{Cov}(v_i, E_i)^2}{\text{Var}(v_i) + \text{Var}(E_i) - 2 \text{Cov}(v_i, E_i)} = \frac{\sigma_i^2(s^2 A_i + h_{ii}\sigma_i^2) - h_{ii}^2\sigma_i^4}{s^2 A_i + \sigma_i^2(1-h_{ii})} = \frac{\sigma_i^2 s^2 A_i + h_{ii}(1-h_{ii})\sigma_i^4}{s^2 A_i + \sigma_i^2(1-h_{ii})}.$$

It remains to compute the distance between the naive and optimal weights. Putting both over a common denominator,

$$\begin{aligned} w_i^M - w_i^* &= \frac{s^2}{s^2 + \sigma_i^2} - \frac{s^2 A_i}{s^2 A_i + \sigma_i^2(1-h_{ii})} = \frac{s^2 [s^2 A_i + \sigma_i^2(1-h_{ii})] - s^2 A_i(s^2 + \sigma_i^2)}{(s^2 + \sigma_i^2) [s^2 A_i + \sigma_i^2(1-h_{ii})]} \\ &= \frac{s^2 \sigma_i^2 [(1-h_{ii}) - A_i]}{(s^2 + \sigma_i^2) [s^2 A_i + \sigma_i^2(1-h_{ii})]} = \frac{s^2 \sigma_i^2 [h_{ii}(1-h_{ii}) - \sum_{j \neq i} h_{ij}^2]}{(s^2 + \sigma_i^2) [s^2 A_i + \sigma_i^2(1-h_{ii})]}, \end{aligned}$$

where the final equality uses $(1-h_{ii}) - A_i = (1-h_{ii}) - (1-h_{ii})^2 - \sum_{j \neq i} h_{ij}^2 = h_{ii}(1-h_{ii}) - \sum_{j \neq i} h_{ij}^2$.

Substituting this into the quadratic expansion, we get

$$R_i(w_i^M) - R_i(w_i^*) = (w_i^M - w_i^*)^2 \left[s^2 A_i + \sigma_i^2 (1 - h_{ii}) \right] = \frac{s^4 \sigma_i^4 \left[h_{ii}(1 - h_{ii}) - \sum_{j \neq i} h_{ij}^2 \right]^2}{(s^2 + \sigma_i^2)^2 \left[s^2 A_i + \sigma_i^2 (1 - h_{ii}) \right]},$$

which is (A.10). For the rate claim, Lemma D.2(v) gives $h_{ii} = O_p(1/N)$ and $\sum_{j \neq i} h_{ij}^2 = O_p(1/N)$ uniformly in i , so the squared bracket is $O_p(N^{-2})$ while the denominator converges to $(s^2 + \sigma_i^2)^2 (s^2 + \sigma_i^2) > 0$; hence the excess risk is $O_p(N^{-2})$. When instead the leverage quantities remain bounded away from zero—small N , a dominant jurisdiction, or an extrapolative covariate profile—the bracket is bounded away from zero and the excess risk is of the same order as the risk itself. This completes the proof. $\square$

A.3 Proof of the Equivalence of Three-Step Estimation to a Joint MLE

We formalize the equivalence result below.

Proposition A.6 (Equivalence to Marginal MLE and the Degeneracy of Joint MLE). Maintain the Gaussian hierarchical model (Assumptions 2.2 and 2.4) with known sampling variances σ_i^2 . Write $\mu_i(\beta) = (\mathbf{Z}_i^*)' \beta$ for the prior mean and $w_i^M = s^2 / (s^2 + \sigma_i^2)$ for the posterior-mean (Morris) weight. Let $\ell_c(\beta, s^2, \varepsilon_{1:N}^*)$ be the complete-data log-likelihood and $\ell^{\text{marg}}(\beta, s^2)$ the marginal log-likelihood integrating out the latent effects. The following hold:

- (i) **Posterior Equivalence:** For any fixed hyperparameters (β, s^2) , the joint maximizer of the complete-data log-likelihood over the latent estimands $\varepsilon_{1:N}^*$ is exactly the vector of Stage 3 empirical Bayes posterior means:

$$\arg \max_{\varepsilon_{1:N}^*} \ell_c(\beta, s^2, \varepsilon_{1:N}^*) = \{w_i^M \hat{\varepsilon}_i + (1 - w_i^M) \mu_i(\beta)\}_{i=1}^N. \quad (\text{A.11})$$

- (ii) **Coefficient Equivalence:** Concentrating $\varepsilon_{1:N}^*$ out of the complete-data likelihood yields the profile log-likelihood $\ell^{\text{prof}}(\beta, s^2) \equiv \max_{\varepsilon_{1:N}^*} \ell_c$. This profile log-likelihood differs from the marginal log-likelihood ℓ^{marg} strictly by an additive function of s^2 . Consequently, for any fixed $s^2 > 0$, both objectives share the identical generalized least squares estimator $\hat{\beta}(s^2)$, given by

$$\hat{\beta}(s^2) = \left(\sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{s^2 + \sigma_i^2} \right)^{-1} \sum_{i=1}^N \frac{\mathbf{Z}_i^* \hat{\varepsilon}_i}{s^2 + \sigma_i^2}. \quad (\text{A.12})$$

- (iii) **Degeneracy of Joint MLE:** The complete-data likelihood is unbounded above. As $s^2 \downarrow 0$, the profile log-likelihood $\ell^{\text{prof}}(\hat{\beta}(s^2), s^2)$ diverges to $+\infty$.

Proposition A.6 precisely diagnoses why the three-stage estimator must separate the estimation of the variance component (Stage 2) from the point estimates of the country elasticities (Stage 3). Treating the N

country-specific elasticities ε_i^* as free parameters while simultaneously attempting to estimate their shared variance s^2 introduces an incidental parameters problem: because the number of parameters grows proportionally with the sample size N , classical joint maximum likelihood breaks down—here in the strong form of an infinite likelihood spike at $s^2 = 0$, rather than the finite bias of the textbook case. Integrating out the latent effects via the marginal likelihood, as performed in our Stage 2, is mathematically required to obtain a non-degenerate estimate of cross-country heterogeneity. Once s^2 is safely identified from the marginal likelihood, substituting it back into the latent score equations perfectly recovers the optimal shrinkage weights of Stage 3 (as shown in part (i)).

Throughout this proof, the “data” consist of the Stage 1 estimates $\widehat{\varepsilon}_i$ and their known sampling variances σ_i^2 . The reduction of the panel data to the sufficient statistics $(\widehat{\varepsilon}_i, \sigma_i^2)$ is exact by Theorem 2.3, whose conditional Gaussianity requires Assumption 2.2(iv). Let $\mu_i \equiv (\mathbf{Z}_i^*)' \beta$ denote the structural prior mean and $r_i \equiv \widehat{\varepsilon}_i - \mu_i$ the marginal residual.

Proof. The proof proceeds in six steps. Steps 1 and 2 solve the conditional maximization over the latent effects $\varepsilon_{1:N}^*$. Step 3 constructs the profile likelihood. Step 4 constructs the marginal likelihood. Step 5 proves the exact equivalence of the β estimators. Step 6 rigorously establishes the divergence in estimating s^2 .

Step 1: Conditional maximization over latent effects. Treat the full vector $(\beta, s^2, \varepsilon_1^*, \dots, \varepsilon_N^*)$ as parameters to be estimated. Under the measurement equation $\widehat{\varepsilon}_i = \varepsilon_i^* + v_i$ with conditionally independent sampling errors $v_i \mid \mathbf{R}, \mathbf{Z} \sim \mathcal{N}(0, \sigma_i^2)$ and the structural prior $\varepsilon_i^* \mid \mathbf{Z}_i^* \sim \mathcal{N}(\mu_i, s^2)$, the complete-data likelihood factorizes across jurisdictions:

$$L(\beta, s^2, \varepsilon_{1:N}^*) = \underbrace{\prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(\widehat{\varepsilon}_i - \varepsilon_i^*)^2}{2\sigma_i^2}\right)}_{\text{measurement density}} \times \underbrace{\frac{1}{\sqrt{2\pi s^2}} \exp\left(-\frac{(\varepsilon_i^* - \mu_i)^2}{2s^2}\right)}_{\text{prior density}}. \quad (\text{A.13})$$

Taking the natural logarithm yields the complete-data log-likelihood:

$$\ell_c(\beta, s^2, \varepsilon_{1:N}^*) = -\frac{1}{2} \sum_{i=1}^N \left[\log(2\pi\sigma_i^2) + \frac{(\widehat{\varepsilon}_i - \varepsilon_i^*)^2}{\sigma_i^2} \right] - \frac{1}{2} \sum_{i=1}^N \left[\log(2\pi s^2) + \frac{(\varepsilon_i^* - \mu_i)^2}{s^2} \right]. \quad (\text{A.14})$$

Because (A.14) is completely separable in the latent effects ε_i^* , fixing the hyperparameters (β, s^2) allows us to maximize with respect to each ε_i^* independently. The first-order condition (score) is:

$$\frac{\partial \ell_c}{\partial \varepsilon_i^*} = \frac{\widehat{\varepsilon}_i - \varepsilon_i^*}{\sigma_i^2} - \frac{\varepsilon_i^* - \mu_i}{s^2} = 0 \implies \varepsilon_i^* \left(\frac{1}{\sigma_i^2} + \frac{1}{s^2} \right) = \frac{\widehat{\varepsilon}_i}{\sigma_i^2} + \frac{\mu_i}{s^2}.$$

Multiplying both sides by the harmonic weight $\frac{\sigma_i^2 s^2}{s^2 + \sigma_i^2}$ yields the unique solution:

$$\widehat{\varepsilon}_i^*(\beta, s^2) = \frac{s^2 \widehat{\varepsilon}_i + \sigma_i^2 \mu_i}{s^2 + \sigma_i^2} = \left(\frac{s^2}{s^2 + \sigma_i^2} \right) \widehat{\varepsilon}_i + \left(\frac{\sigma_i^2}{s^2 + \sigma_i^2} \right) \mu_i \equiv w_i^M \widehat{\varepsilon}_i + (1 - w_i^M) \mu_i, \quad (\text{A.15})$$

where $w_i^M \equiv s^2 / (s^2 + \sigma_i^2)$. This is analytically identical to the Stage 3 empirical Bayes posterior mean evaluated with the Morris weight, establishing part (i) of the proposition.

Step 2: Strict concavity and global maximizer. Checking the second derivative verifies global strict concavity:

$$\frac{\partial^2 \ell_c}{\partial (\varepsilon_i^*)^2} = - \left(\frac{1}{\sigma_i^2} + \frac{1}{s^2} \right) = - \frac{s^2 + \sigma_i^2}{\sigma_i^2 s^2} < 0.$$

Since the cross-partials $\partial^2 \ell_c / (\partial \varepsilon_i^* \partial \varepsilon_j^*)$ vanish entirely for $i \neq j$, the Hessian over $\varepsilon_{1:N}^*$ is diagonal and strictly negative definite. The posterior mean (A.15) is thus the unique global maximizer of the latent variables.

Step 3: Profile log-likelihood construction. To obtain the profile (concentrated) log-likelihood for the structural parameters, we substitute the optimal latent estimates $\widehat{\varepsilon}_i^*$ back into (A.14). The residuals evaluate to:

$$\widehat{\varepsilon}_i - \widehat{\varepsilon}_i^* = \frac{\sigma_i^2 (\widehat{\varepsilon}_i - \mu_i)}{s^2 + \sigma_i^2} = \frac{\sigma_i^2 r_i}{s^2 + \sigma_i^2}, \quad \widehat{\varepsilon}_i^* - \mu_i = \frac{s^2 (\widehat{\varepsilon}_i - \mu_i)}{s^2 + \sigma_i^2} = \frac{s^2 r_i}{s^2 + \sigma_i^2}.$$

Summing the weighted squared residuals inside the likelihood simplifies algebraically:

$$\frac{(\widehat{\varepsilon}_i - \widehat{\varepsilon}_i^*)^2}{\sigma_i^2} + \frac{(\widehat{\varepsilon}_i^* - \mu_i)^2}{s^2} = \frac{\sigma_i^4 r_i^2}{\sigma_i^2 (s^2 + \sigma_i^2)^2} + \frac{s^4 r_i^2}{s^2 (s^2 + \sigma_i^2)^2} = \frac{(\sigma_i^2 + s^2) r_i^2}{(s^2 + \sigma_i^2)^2} = \frac{r_i^2}{s^2 + \sigma_i^2}.$$

Thus, the profile log-likelihood for the hyperparameters collapses to:

$$\ell^{\text{prof}}(\beta, s^2) = -\frac{1}{2} \sum_{i=1}^N \left[\log(2\pi\sigma_i^2) + \log(2\pi s^2) + \frac{r_i^2}{s^2 + \sigma_i^2} \right]. \quad (\text{A.16})$$

Step 4: Marginal log-likelihood construction. Instead of maximizing over $\varepsilon_{1:N}^*$, the Bayesian approach integrates them out to form the marginal density of the observed estimates $\widehat{\varepsilon}_i$. Let $Q(\varepsilon_i^*) = \frac{(\widehat{\varepsilon}_i - \varepsilon_i^*)^2}{\sigma_i^2} + \frac{(\varepsilon_i^* - \mu_i)^2}{s^2}$. Completing the square around the minimizer $\widehat{\varepsilon}_i^*$ yields:

$$Q(\varepsilon_i^*) = \frac{s^2 + \sigma_i^2}{\sigma_i^2 s^2} (\varepsilon_i^* - \widehat{\varepsilon}_i^*)^2 + \frac{r_i^2}{s^2 + \sigma_i^2}.$$

Integrating the complete-data density with respect to ε_i^* yields a standard Gaussian integral calculation

$$\begin{aligned} p(\hat{\varepsilon}_i \mid \mathbf{Z}_i^*, \beta, s^2) &= \int_{\mathbb{R}} \frac{1}{2\pi\sqrt{\sigma_i^2 s^2}} \exp\left(-\frac{1}{2}Q(\varepsilon_i^*)\right) d\varepsilon_i^* \\ &= \frac{1}{2\pi\sqrt{\sigma_i^2 s^2}} \exp\left(-\frac{r_i^2}{2(s^2 + \sigma_i^2)}\right) \sqrt{\frac{2\pi\sigma_i^2 s^2}{s^2 + \sigma_i^2}} = \frac{1}{\sqrt{2\pi(s^2 + \sigma_i^2)}} \exp\left(-\frac{r_i^2}{2(s^2 + \sigma_i^2)}\right). \end{aligned}$$

This formally shows that $\hat{\varepsilon}_i \mid \mathbf{Z}_i^* \sim \mathcal{N}(\mu_i, s^2 + \sigma_i^2)$, matching the marginal model (9). Taking the logarithm gives the marginal log-likelihood optimized in Stage 2, i.e.

$$\ell^{\text{marg}}(\beta, s^2) = -\frac{1}{2} \sum_{i=1}^N \left[\log(2\pi(s^2 + \sigma_i^2)) + \frac{r_i^2}{s^2 + \sigma_i^2} \right]. \quad (\text{A.17})$$

Step 5: Exact equivalence for the structural mean β . Comparing the profile log-likelihood (A.16) and the marginal log-likelihood (A.17), both objectives depend on the coefficient vector β exclusively through the weighted sum of squared residuals:

$$-\frac{1}{2} \sum_{i=1}^N \frac{r_i^2}{s^2 + \sigma_i^2} = -\frac{1}{2} \sum_{i=1}^N \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)^2}{s^2 + \sigma_i^2}.$$

For any fixed $s^2 > 0$, taking the gradient with respect to β and setting it to zero yields identical first-order conditions for both objectives:

$$\sum_{i=1}^N \frac{\mathbf{Z}_i^* (\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)}{s^2 + \sigma_i^2} = 0 \implies \hat{\beta}(s^2) = \left(\sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{s^2 + \sigma_i^2} \right)^{-1} \sum_{i=1}^N \frac{\mathbf{Z}_i^* \hat{\varepsilon}_i}{s^2 + \sigma_i^2}.$$

Thus, whether one uses joint maximization or marginal maximization, the generalized least squares (GLS) estimator for β is mathematically identical conditional on s^2 , establishing part (ii).

Step 6: The incidental parameters problem and the variance component. Despite their exact agreement on ε_i^* and β , the two objectives diverge fundamentally when estimating s^2 . Subtracting the marginal log-likelihood from the profile log-likelihood isolates their difference:

$$\ell^{\text{prof}}(\beta, s^2) - \ell^{\text{marg}}(\beta, s^2) = -\frac{1}{2} \sum_{i=1}^N \log(2\pi) - \frac{1}{2} \sum_{i=1}^N \log\left(\frac{\sigma_i^2 s^2}{s^2 + \sigma_i^2}\right). \quad (\text{A.18})$$

This difference represents the Jacobian of the Gaussian integration. It is strictly independent of β (hence the shared GLS estimator), but depends critically on s^2 . To see why joint maximization fails, observe the

behavior of the complete-data profile likelihood as $s^2 \downarrow 0$:

$$\ell^{\text{prof}}(\beta, s^2) = \underbrace{-\frac{1}{2} \sum_{i=1}^N \log(2\pi\sigma_i^2)}_{\equiv \kappa(\sigma)} - \frac{N}{2} \log(s^2) - \frac{1}{2} \sum_{i=1}^N \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}(s^2))^2}{s^2 + \sigma_i^2}. \quad (\text{A.19})$$

As $s^2 \downarrow 0$, the GLS estimator $\hat{\beta}(s^2)$ converges smoothly to the precision-weighted OLS estimator, $\hat{\beta}(0) = (\sum \sigma_i^{-2} \mathbf{Z}_i^* (\mathbf{Z}_i^*)')^{-1} \sum \sigma_i^{-2} \mathbf{Z}_i^* \hat{\varepsilon}_i$. Thus, the residual sum of squares converges to a finite positive constant:

$$\lim_{s^2 \downarrow 0} \sum_{i=1}^N \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}(s^2))^2}{s^2 + \sigma_i^2} = \sum_{i=1}^N \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}(0))^2}{\sigma_i^2} < \infty. \quad (\text{A.20})$$

Because the residual penalty remains strictly bounded, the unpenalized log-determinant term $-\frac{N}{2} \log(s^2)$ dominates, driving the entire profile likelihood to $+\infty$.

Consequently, the complete-data likelihood is *unbounded above*, reaching its supremum at $s^2 = 0$. Naive joint maximum likelihood estimation will continuously push $\hat{s}^2 \rightarrow 0$, forcing $\hat{\varepsilon}_i^* \rightarrow \mu_i$. This degeneracy is a classical incidental parameters problem (Neyman and Scott, 1948). By treating the N latent estimands $\varepsilon_1^*, \dots, \varepsilon_N^*$ as fixed parameters to be estimated alongside the structural hyperparameters, the dimension of the parameter space grows linearly with the sample size N , rendering maximum likelihood inconsistent.

By contrast, the marginal likelihood integrates out these incidental parameters. The problematic normalizers $\prod_i (2\pi s^2)^{-1/2}$ that cause the infinite spike are analytically absorbed by the integration process, replaced by the well-behaved factor $\prod_i (2\pi(s^2 + \sigma_i^2))^{-1/2}$. As $s^2 \downarrow 0$, this factor smoothly converges to a finite constant. The marginal log-likelihood (A.17) is strictly bounded above on $[0, \infty)$ and possesses a valid, estimable interior maximum.

Therefore, when s^2 must be estimated, a one-stage joint maximum likelihood estimator is mathematically ill-posed due to unboundedness. The statistically principled one-stage analogue is to maximize the marginal likelihood for (β, s^2) and plug these into the posterior mean formula (A.15). This corresponds precisely to the sequential three-stage structure: collect sufficient statistics, estimate hyperparameters from the marginal likelihood, and apply empirical Bayes shrinkage. $\square$

Remark A.7 (GLS vs. CRC Implementations). While Proposition A.6 proves the structural equivalence of the sequential procedure to joint marginal MLE using the GLS estimator $\hat{\beta}(s^2)$, the empirical application in the paper utilizes the pooled CRC estimator $\hat{\beta}^{\text{CRC}}$ in Stage 2. Note that $\hat{\beta}^{\text{CRC}}$ implicitly weights observations by their identifying variation $\omega_i \propto 1/\sigma_i^2$, which makes it mathematically equivalent to the exact GLS estimator evaluated strictly at $s^2 = 0$. While both estimators are consistent, substituting the CRC estimator bypasses the need for iterative numerical optimization between β and s^2 . $\diamond$

B Estimation Details

The shrinkage weight of Theorems 2.5 depends on three unknown variance objects: the variance σ_η^2 (which delivers the sampling variances $\sigma_i^2 = \sigma_\eta^2/\omega_i$), the residual structural heterogeneity s^2 , and $\mathbf{V}_\beta$ (the uncertainty quantification of CRC estimates $\hat{\beta}^{CRC}$). This subsection develops consistent estimators for all three and establishes their asymptotic properties.

Estimating σ_η^2

A natural first impulse is to estimate σ_η^2 from the pooled Stage 2 residuals $e_{js} \equiv y_{js}^\perp - \mathbf{W}_{js}' \hat{\beta}^{CRC}$. This turns out to be incorrect. The pooled residual contains the random-coefficient component in addition to the micro-shock, i.e.

$$e_{js} = \tilde{u}_j^\varepsilon R_{js}^\perp + \eta_{js}^\perp - \mathbf{W}_{js}'(\hat{\beta}^{CRC} - \beta),$$

so that $\mathbb{E}[e_{js}^2 \mid \mathbf{R}, \mathbf{Z}] = s^2(R_{js}^\perp)^2 + \sigma_\eta^2 + O(1/N)$. Consequently, the naive estimator $(NT)^{-1} \sum_{j,s} e_{js}^2$ converges to $\sigma_\eta^2 + s^2 \times \lim_N(\bar{\omega}_N/T)$, which strictly exceeds σ_η^2 whenever $s^2 > 0$. As such, any estimation procedure for σ_η^2 must come from a procedure that estimates (σ_η^2, s^2) jointly.

We therefore propose a method-of-moments estimator built from two aggregate second moments that load on (σ_η^2, s^2) with linearly independent coefficients. Let $r_i \equiv \hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}^{CRC}$ denote the country-level residuals, and define

$$\hat{\mu}_1 \equiv \frac{\sum_{j=1}^N \sum_{s=1}^T e_{js}^2}{NT}, \quad \hat{\mu}_2 \equiv \frac{1}{N} \sum_{i=1}^N \omega_i r_i^2. \quad (\text{B.1})$$

Writing $\bar{\omega}_N \equiv N^{-1} \sum_j \omega_j$ and assuming a balanced panel with $T_i = T$ for all i (so that $\sum_j T_j = NT$), the population analogues satisfy, up to $O_p(N^{-1/2})$ terms,

$$\mu_1 = \sigma_\eta^2 \left(1 - \frac{2}{T}\right) + \frac{\bar{\omega}_N}{T} s^2, \quad \mu_2 = \sigma_\eta^2 + \bar{\omega}_N s^2, \quad (\text{B.2})$$

a linear system whose determinant is $\bar{\omega}_N(T-3)/T > 0$ whenever the common panel length exceeds three. Solving,

$$\hat{\sigma}_\eta^2 = \frac{\hat{\mu}_1 T - \hat{\mu}_2}{T-3}, \quad \hat{s}_{MM}^2 = \frac{\hat{\mu}_2 - \hat{\sigma}_\eta^2}{\bar{\omega}_N}, \quad (\text{B.3})$$

with negative values truncated at zero. Lemma D.6 establishes that both estimators are $\sqrt{N}$ -consistent. The method-of-moments value $\hat{s}_{MM}^2$ may either be used directly or, as we recommend, as the starting value for the likelihood-based estimator below; the key output of this step is $\hat{\sigma}_\eta^2$, which delivers the country-level sampling variances $\hat{\sigma}_i^2 = \hat{\sigma}_\eta^2/\omega_i$.

Estimating (β, s^2)

Given the sampling variances, the hierarchical model (9) implies that the observed estimates are independent Gaussians, $\hat{\varepsilon}_i \mid \mathbf{Z}_i^* \sim \mathcal{N}((\mathbf{Z}_i^*)'\beta, s^2 + \sigma_i^2)$, with marginal log-likelihood

$$L_N(\beta, s^2) = -\frac{1}{2} \sum_{i=1}^N \left[\ln(s^2 + \sigma_i^2) + \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)^2}{s^2 + \sigma_i^2} \right] + \text{constant}. \quad (\text{B.4})$$

The score equations have a transparent structure. Differentiating with respect to β and equating to zero yields, for any fixed s^2 , the precision-weighted (GLS) estimator

$$\hat{\beta}(s^2) = \left[\sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{s^2 + \sigma_i^2} \right]^{-1} \sum_{i=1}^N \frac{\mathbf{Z}_i^* \hat{\varepsilon}_i}{s^2 + \sigma_i^2}, \quad (\text{B.5})$$

and differentiating with respect to s^2 yields the variance score

$$\frac{\partial L_N}{\partial s^2} = \frac{1}{2} \sum_{i=1}^N \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)^2 - (s^2 + \sigma_i^2)}{(s^2 + \sigma_i^2)^2}, \quad (\text{B.6})$$

which balances squared residuals against their model-implied magnitudes, precision-weighting countries with small total variance. Substituting (B.5) into L_N gives the concentrated likelihood $L_N^c(s^2) \equiv L_N(\hat{\beta}(s^2), s^2)$, a smooth function of a scalar that is maximized by a one-dimensional Newton or golden-section search on $[0, \bar{s}^2]$. Equivalently, the maximizer can be computed by an Expectation-Maximization (EM) iteration that treats the latent ε_i^* as missing data: at iterate $(\beta^{(k)}, s^{2(k)})$, with $w_i^{(k)} = s^{2(k)} / (s^{2(k)} + \sigma_i^2)$,

$$\text{E-step: } \tilde{\varepsilon}_i^{(k)} = w_i^{(k)} \hat{\varepsilon}_i + (1 - w_i^{(k)}) (\mathbf{Z}_i^*)' \beta^{(k)}, \quad \tilde{V}_i^{(k)} = w_i^{(k)} \sigma_i^2, \quad (\text{B.7})$$

$$\text{M-step: } \beta^{(k+1)} = \left(\sum_{i=1}^N \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \right)^{-1} \sum_{i=1}^N \mathbf{Z}_i^* \tilde{\varepsilon}_i^{(k)}, \quad s^{2(k+1)} = \frac{1}{N} \sum_{i=1}^N \left[\left(\tilde{\varepsilon}_i^{(k)} - (\mathbf{Z}_i^*)' \beta^{(k+1)} \right)^2 + \tilde{V}_i^{(k)} \right]. \quad (\text{B.8})$$

Each iteration increases L_N , and the algorithm converges to a stationary point of the marginal likelihood from any interior starting value.

The asymptotic behavior of the resulting estimator is summarized by the following theorem, whose information matrix is block-diagonal between location and scale—the same structural fact (Lemma D.3) that decouples the sequential updates in Section 4.1.

Theorem B.1 (Asymptotics of the Marginal MLE). Let $\hat{\theta}_N = (\hat{\beta}_N', \hat{s}_N^2)'$ maximize (B.4) over a compact parameter space containing the true $\theta_0 = (\beta_0', s_0^2)'$ in its interior. Under Assumptions 2.2, 2.4, and A.3, as $N \rightarrow \infty$:

- (i) **Consistency:** $\hat{\theta}_N \xrightarrow{p} \theta_0$.

(ii) **Asymptotic Normality:** $\sqrt{N}(\hat{\theta}_N - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}(\theta_0)^{-1})$, where the Fisher information is block-diagonal,

$$\mathbf{I}(\theta_0) = \begin{pmatrix} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{\mathbf{s}_0^2 + \sigma_i^2} & \mathbf{0} \\ \mathbf{0} & \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{i=1}^N \frac{1}{(\mathbf{s}_0^2 + \sigma_i^2)^2} \end{pmatrix} \succ 0. \quad (\text{B.9})$$

(iii) **Plug-in Behavior:** Replacing the known σ_i^2 by the Stage 1 estimates $\hat{\sigma}_i^2 = \hat{\sigma}_\eta^2 / \omega_i$ preserves consistency and the $\sqrt{N}$ rate.

Finally, the feasible weights $\hat{w}_i^*$ replace $(\mathbf{s}^2, \sigma_i^2, \sigma_\eta^2)$ in Theorem 2.5 by their estimates. Since the weight formula is a smooth function of the hyperparameters with bounded derivatives (the denominators are bounded away from zero when $\hat{\sigma}_i^2 > 0$), the feasible empirical Bayes estimator differs from the oracle-weighted one by $O_p(N^{-1/2})$ uniformly in i , so the compound risk of the feasible rule converges to the oracle compound risk.

Proof of Theorem B.1. Write $V_i(\mathbf{s}^2) \equiv \mathbf{s}^2 + \sigma_i^2$ and let $\ell_i(\theta) = -\frac{1}{2}[\ln V_i(\mathbf{s}^2) + (\hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta)^2 / V_i(\mathbf{s}^2)]$ denote the marginal log-likelihood contribution. Under Assumption A.3(ii), $V_i(\mathbf{s}^2)$ is bounded above and away from zero uniformly over i and over Θ . We prove the three parts in turn.

Part (i): Consistency Under the hierarchical model, $\hat{\varepsilon}_i \mid \mathbf{Z}_i^* \sim \mathcal{N}((\mathbf{Z}_i^*)'\beta_0, V_i(\mathbf{s}_0^2))$, so for any θ ,

$$\mathbb{E}[\ell_i(\theta)] = -\frac{1}{2} \left[\ln V_i(\mathbf{s}^2) + \frac{V_i(\mathbf{s}_0^2) + ((\mathbf{Z}_i^*)'(\beta - \beta_0))^2}{V_i(\mathbf{s}^2)} \right].$$

Comparing with the value at θ_0 and writing $x_i \equiv V_i(\mathbf{s}^2) / V_i(\mathbf{s}_0^2)$,

$$\mathbb{E}[\ell_i(\theta_0)] - \mathbb{E}[\ell_i(\theta)] = \frac{1}{2} \left[\ln x_i + \frac{1}{x_i} - 1 \right] + \frac{((\mathbf{Z}_i^*)'(\beta - \beta_0))^2}{2V_i(\mathbf{s}^2)} \geq 0,$$

where both terms are non-negative (the first by the inequality $\ln x + 1/x \geq 1$ for all $x > 0$, the second trivially). Equality for asymptotically all i requires $(\mathbf{Z}_i^*)'(\beta - \beta_0) = 0$ for asymptotically all i —which forces $\beta = \beta_0$ by the full-rank condition of Assumption A.3(iii)—and $V_i(\mathbf{s}^2) = V_i(\mathbf{s}_0^2)$, i.e., $\mathbf{s}^2 = \mathbf{s}_0^2$ since the difference $\mathbf{s}^2 - \mathbf{s}_0^2$ is common across i . Thus θ_0 uniquely maximizes the limit criterion, and the separation is strict on $\{\|\theta - \theta_0\| \geq \epsilon\}$ by compactness and continuity. Uniform convergence of $N^{-1} \sum_i \ell_i(\theta)$ to its expectation over the compact Θ follows from the Lipschitz-envelope argument used in Step 2 of the proof of Theorem D.11 (the gradients in β and $\mathbf{s}^2$ are bounded by $C(1 + \hat{\varepsilon}_i^2)$ uniformly on Θ). The argmax theorem then yields $\hat{\theta}_N \xrightarrow{p} \theta_0$.

Part (ii): Asymptotic normality and block-diagonal information The score components at θ_0 , with $r_i^0 = \hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta_0$, are

$$\nabla_{\beta} \ell_i(\theta_0) = \frac{r_i^0 \mathbf{Z}_i^*}{V_i(\mathbf{s}_0^2)}, \quad \frac{\partial \ell_i}{\partial \mathbf{s}^2}(\theta_0) = \frac{(r_i^0)^2 - V_i(\mathbf{s}_0^2)}{2V_i(\mathbf{s}_0^2)^2}.$$

Both have mean zero ($\mathbb{E}[r_i^0] = 0$ and $\mathbb{E}[(r_i^0)^2] = V_i(\mathbf{s}_0^2)$). Their variances are $\mathbf{Z}_i^* (\mathbf{Z}_i^*)' / V_i(\mathbf{s}_0^2)$ and, using $\text{Var}((r_i^0)^2) = 2V_i(\mathbf{s}_0^2)^2$ for a Gaussian, $1/(2V_i(\mathbf{s}_0^2)^2)$. The cross moment is

$$\mathbb{E} \left[\nabla_{\beta} \ell_i \cdot \frac{\partial \ell_i}{\partial \mathbf{s}^2} \right] = \frac{\mathbf{Z}_i^*}{2V_i^3} \mathbb{E} \left[r_i^0 \left((r_i^0)^2 - V_i \right) \right] = \frac{\mathbf{Z}_i^*}{2V_i^3} \left(\mathbb{E}[(r_i^0)^3] - V_i \mathbb{E}[r_i^0] \right) = \mathbf{0},$$

because all odd moments of the centered Gaussian r_i^0 vanish. Hence the information matrix is block-diagonal between β and $\mathbf{s}^2$, with blocks exactly as displayed in (B.9); this is the same computation as Lemma D.3 in Appendix C, recorded there in expected-Hessian form. The information equality $\mathbb{E}[-\nabla^2 \ell_i(\theta_0)] = \text{Var}(\nabla \ell_i(\theta_0))$ holds because the marginal model is a correctly specified Gaussian likelihood. The summands are independent with uniformly bounded third absolute moments (Gaussianity plus the uniform bounds on V_i and $\|\mathbf{Z}_i^*\|$), so the Lyapunov CLT applies to the normalized score; combined with the uniform Hessian convergence of Assumption A.3(iv) and the interiority of θ_0 , the standard mean-value expansion of the first-order condition delivers $\sqrt{N}(\hat{\theta}_N - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}(\theta_0)^{-1})$.

Part (iii): Plug-in The feasible criterion replaces σ_i^2 by $\hat{\sigma}_i^2 = \hat{\sigma}_\eta^2 / \omega_i$, i.e., perturbs the parameter σ_η^2 entering $V_i = \mathbf{s}^2 + \sigma_\eta^2 / \omega_i$ by $\hat{\sigma}_\eta^2 - \sigma_\eta^2 = O_p(N^{-1/2})$ (Lemma D.6). Consistency is preserved by the same uniform-perturbation argument as Step 3 of the proof of Theorem D.11, since the criterion is uniformly Lipschitz in σ_η^2 on a neighborhood of the truth.

For the first-order effect, we compute the expected cross-derivatives of the two score components with respect to σ_η^2 at the truth. For the location score, $\partial^2 \ell_i / \partial \beta \partial \sigma_\eta^2 = -r_i^0 \mathbf{Z}_i^* / (\omega_i V_i^2)$, whose expectation vanishes because r_i^0 enters linearly; thus $\hat{\beta}_N$ is Neyman-orthogonal to the first step, and its asymptotic distribution is unchanged.

We now turn to the variance score. It turns out that the asymptotic variance of $\hat{\mathbf{s}}_N^2$ acquires an additional term reflecting the first-step uncertainty,

$$\begin{aligned} \sqrt{N}(\hat{\mathbf{s}}_{N,\text{feas}}^2 - \mathbf{s}_0^2) &= \sqrt{N}(\hat{\mathbf{s}}_N^2 - \mathbf{s}_0^2) - \varrho_N \cdot \sqrt{N}(\hat{\sigma}_\eta^2 - \sigma_\eta^2) + o_p(1), \\ \text{where } \varrho_N &\equiv \frac{\sum_{i=1}^N \omega_i^{-1} (\mathbf{s}_0^2 + \sigma_i^2)^{-2}}{\sum_{i=1}^N (\mathbf{s}_0^2 + \sigma_i^2)^{-2}} \in \left(0, \frac{1}{\min_i \omega_i} \right), \end{aligned} \quad (\text{B.10})$$

so that valid standard errors for $\hat{\mathbf{s}}^2$ must include some correction terms. While we do not conduct inference on $\hat{\mathbf{s}}^2$, we still present the entire required set of corrections.

First, direct calculations reveal that

$$\mathbb{E} \left[\frac{\partial^2 \ell_i}{\partial s^2 \partial \sigma_\eta^2} \right] = \frac{1}{\omega_i} \mathbb{E} \left[\frac{\partial}{\partial V_i} \left(\frac{(r_i^0)^2 - V_i}{2V_i^2} \right) \right] = \frac{1}{\omega_i} \cdot \left(-\frac{1}{2V_i^2} \right) \neq 0 ,$$

so we would be using $\mathbb{E}[(r_i^0)^2] = V_i$ to kill the term proportional to $(r_i^0)^2 - V_i$. The implicit function theorem applied to the s^2 first-order condition, profiling β out (which is innocuous by the block-diagonality just established), gives the sensitivity term

$$\frac{\partial \hat{s}^2}{\partial \sigma_\eta^2} = - \frac{\sum_i \mathbb{E}[\partial^2 \ell_i / \partial s^2 \partial \sigma_\eta^2]}{\sum_i \mathbb{E}[\partial^2 \ell_i / \partial (s^2)^2]} = - \frac{\sum_i \omega_i^{-1} V_i^{-2}}{\sum_i V_i^{-2}} = -\varrho_N ,$$

which is the leading coefficient in the influence decomposition (B.10); the remainder is $o_p(1)$ by the smoothness of the score in (θ, σ_η^2) and the $\sqrt{N}$ -rates of both estimators. Finally, writing $\sqrt{N}(\hat{\sigma}_\eta^2 - \sigma_\eta^2) = N^{-1/2} \sum_i \phi_i + o_p(1)$ for the influence function ϕ_i obtained by applying the inverse of the linear system (B.2) to the centered moment contributions of country i , and $\sqrt{N}(\hat{s}_N^2 - s_0^2) = N^{-1/2} \sum_i \psi_i + o_p(1)$ with $\psi_i = [(r_i^0)^2 - V_i] / (2V_i^2) \cdot [\lim_N N^{-1} \sum_j (2V_j^2)^{-1}]^{-1}$, the feasible estimator has influence function $\psi_i - \varrho_N \phi_i$ and asymptotic variance $\lim_N N^{-1} \sum_i \mathbb{E}[(\psi_i - \varrho_N \phi_i)^2]$, which can be estimated by plugging in sample analogues. This is the sandwich form referenced in the theorem, and this completes the proof. $\square$

C Proofs of Results from Section 4

C.1 Learning from New Data

Proof of Proposition 4.2. We proceed in two steps.

Step 1 Under the hierarchical model (Assumption 4.1), the country-specific estimands ε_i^* are conditionally independent across countries given the hyperparameters $\theta_0 = (\beta_0', s_0^2)'$. Likewise, the measurement errors v_i are independent conditional on the latent parameters and covariates. Consequently, the joint density of all latent and observed estimates for the first $N + 1$ countries factorizes as

$$\mathbb{P}(\varepsilon_{1:N+1}^*, \hat{\varepsilon}_{1:N+1} \mid \mathbf{Z}_{1:N+1}^*, \theta_0) = \prod_{i=1}^{N+1} \mathbb{P}(\hat{\varepsilon}_i \mid \varepsilon_i^*, \sigma_i^2) \mathbb{P}(\varepsilon_i^* \mid \mathbf{Z}_i^*, \theta_0) . \quad (\text{C.1})$$

Step 2 We evaluate the posterior distribution of an existing country's parameter ε_i^* ($i \leq N$) conditional on the expanded information set $\mathcal{F}_{N+1} = \sigma(\mathcal{D}_{N+1})$, where all covariate profiles and sampling variances are part of the conditioning information. The posterior density is proportional to the joint density of $(\varepsilon_i^*, \hat{\varepsilon}_{1:N+1})$ with the latent parameters of the *other* countries integrated out. Carrying out that integration in the factorization (C.1), every factor with index $j \neq i$ integrates to the marginal density $\mathbb{P}(\hat{\varepsilon}_j \mid \mathbf{Z}_j^*, \sigma_j^2, \theta_0)$, which does

not involve ε_i^* :

$$\mathbb{P}(\varepsilon_i^*, \widehat{\varepsilon}_{1:N+1} \mid \mathbf{Z}_{1:N+1}^*, \sigma_{1:N+1}^2, \theta_0) = \underbrace{\mathbb{P}(\widehat{\varepsilon}_i \mid \varepsilon_i^*, \sigma_i^2) \mathbb{P}(\varepsilon_i^* \mid \mathbf{Z}_i^*, \theta_0)}_{\text{depends on } \varepsilon_i^*} \times \prod_{j \neq i}^{N+1} \mathbb{P}(\widehat{\varepsilon}_j \mid \mathbf{Z}_j^*, \sigma_j^2, \theta_0).$$

Dividing by the marginal of the observables, the second product cancels identically from numerator and denominator, leaving

$$\mathbb{P}(\varepsilon_i^* \mid \mathcal{F}_{N+1}, \theta_0) = \frac{\mathbb{P}(\widehat{\varepsilon}_i \mid \varepsilon_i^*, \sigma_i^2) \mathbb{P}(\varepsilon_i^* \mid \mathbf{Z}_i^*, \theta_0)}{\int \mathbb{P}(\widehat{\varepsilon}_i \mid e, \sigma_i^2) \mathbb{P}(e \mid \mathbf{Z}_i^*, \theta_0) de} = \mathbb{P}(\varepsilon_i^* \mid \mathcal{D}_i^{\text{own}}, \theta_0),$$

a function of country i 's own data $\mathcal{D}_i^{\text{own}} = (\widehat{\varepsilon}_i, \mathbf{Z}_i^*, \sigma_i^2)$ alone. In particular, the right-hand side is the same whether we condition on $\mathcal{F}_N$ or on $\mathcal{F}_{N+1}$, which establishes the conditional independence. $\square$

Proof of Proposition 4.3. The proof proceeds in a few steps.

Step 1 We define the joint log-likelihood at stage $N+1$ as $L_{N+1}(\theta) = L_N(\theta) + \ell_{N+1}(\theta)$, where $\ell_{N+1}(\theta)$ is the marginal log-likelihood contribution of country $N+1$, given by

$$\ell_{N+1}(\beta, \mathbf{s}^2) = -\frac{1}{2} \ln(\mathbf{s}^2 + \sigma_{N+1}^2) - \frac{(\widehat{\varepsilon}_{N+1} - (\mathbf{Z}_{N+1}^*)' \beta)^2}{2(\mathbf{s}^2 + \sigma_{N+1}^2)}. \quad (\text{C.2})$$

Let $\widehat{\theta}_N = (\widehat{\beta}_N', \widehat{\mathbf{s}}_N^2)'$ maximize $L_N(\theta)$, which implies $\nabla_{\theta} L_N(\widehat{\theta}_N) = 0$. Evaluating the joint score vector at $\widehat{\theta}_N$ yields

$$\nabla_{\beta} L_{N+1}(\widehat{\theta}_N) = \frac{\mathbf{Z}_{N+1}^* r_{N+1}^N}{\widehat{\mathbf{s}}_N^2 + \sigma_{N+1}^2}, \quad \frac{\partial L_{N+1}}{\partial \mathbf{s}^2}(\widehat{\theta}_N) = \frac{1}{2} \left[\frac{(r_{N+1}^N)^2 - (\widehat{\mathbf{s}}_N^2 + \sigma_{N+1}^2)}{(\widehat{\mathbf{s}}_N^2 + \sigma_{N+1}^2)^2} \right], \quad (\text{C.3})$$

where we denote the residual by $r_{N+1}^N \equiv \widehat{\varepsilon}_{N+1} - (\mathbf{Z}_{N+1}^*)' \widehat{\beta}_N$.

Step 2 We apply a Taylor expansion to the score equation $\nabla_{\theta} L_{N+1}(\widehat{\theta}_{N+1}) = 0$ around $\widehat{\theta}_N$, yielding

$$0 = \nabla_{\theta} L_{N+1}(\widehat{\theta}_N) + \left[\sum_{i=1}^{N+1} \nabla_{\theta\theta}^2 \ell_i(\widehat{\theta}_N) \right] \Delta \widehat{\theta} + \mathcal{R}_2, \quad (\text{C.4})$$

where $\mathcal{R}_2 = \frac{1}{2} \sum_{i=1}^{N+1} \nabla_{\theta\theta\theta}^3 \ell_i(\bar{\theta})(\Delta \widehat{\theta}, \Delta \widehat{\theta})$. By Lemma D.4, we have $\Delta \widehat{\theta} = O_p(N^{-1})$. Under Assumptions 4.1 and A.4, the third-order derivatives are uniformly bounded in expectation, which implies $\sum_{i=1}^{N+1} \nabla^3 \ell_i = O_p(N)$, and the remainder term satisfies $\mathcal{R}_2 = O_p(N^{-1})$. Thus, we have the simplified equation

$$0 = \nabla_{\theta} \ell_{N+1}(\widehat{\theta}_N) + \left[\sum_{i=1}^{N+1} \nabla_{\theta\theta}^2 \ell_i(\widehat{\theta}_N) \right] \Delta \widehat{\theta} + O_p(N^{-1}). \quad (\text{C.5})$$

Step 3 We decouple the updates for β and s^2 . By Lemma D.3, the average expected Fisher Information matrix $\mathbf{I}_N(\theta)$ is block-diagonal between the location and variance components. Combining this block structure with the uniform convergence of the Hessian from Assumption A.4(iv), we have

$$\sum_{i=1}^{N+1} \nabla_{\theta\theta}^2 \ell_i(\hat{\theta}_N) = -N\mathbf{I}_N(\hat{\theta}_N) + O_p(N^{1/2}). \quad (\text{C.6})$$

Substituting this relation back into (C.5) and solving for the update requires inverting the bracketed matrix; since $\lambda_{\min}(N\mathbf{I}_N) \geq cN$ by Assumption A.4(iii), the resolvent expansion gives $[N\mathbf{I}_N + O_p(N^{1/2})]^{-1} = [N\mathbf{I}_N]^{-1} + O_p(N^{-3/2})$ in operator norm (the correction is $[N\mathbf{I}_N]^{-1} \cdot O_p(N^{1/2}) \cdot [N\mathbf{I}_N]^{-1}$ up to higher order). Applying this inverse to the $O_p(1)$ score,

$$\Delta\hat{\theta} = \left[N\mathbf{I}_N(\hat{\theta}_N) \right]^{-1} \nabla_{\theta} \ell_{N+1}(\hat{\theta}_N) + O_p(N^{-3/2}), \quad (\text{C.7})$$

and the block-diagonality of $\mathbf{I}_N$ means the inverse acts separately on the β -block and the s^2 -block of the score: to leading order, the location update is driven only by the location score and the variance update only by the variance score. This result is not an assumption but rather a consequence of Lemma D.3, accurate up to the stated $O_p(N^{-3/2})$ remainder.

Step 4 We evaluate the independent Fisher information components, which are given by

$$N\mathcal{I}_{N,\beta\beta} = \sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{\hat{s}_N^2 + \sigma_i^2}, \quad N\mathcal{I}_{N,s^2s^2} = \frac{1}{2} \sum_{i=1}^N \frac{1}{(\hat{s}_N^2 + \sigma_i^2)^2}. \quad (\text{C.8})$$

Solving the decoupled equations yields the updates

$$\Delta\hat{\beta} = \left[\sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{\hat{s}_N^2 + \sigma_i^2} \right]^{-1} \frac{\mathbf{Z}_{N+1}^* \cdot r_{N+1}^N}{\hat{s}_N^2 + \sigma_{N+1}^2} + O_p(N^{-3/2}), \quad (\text{C.9})$$

and

$$\Delta\hat{s}^2 = \left[\sum_{i=1}^N \frac{1}{(\hat{s}_N^2 + \sigma_i^2)^2} \right]^{-1} \frac{(r_{N+1}^N)^2 - (\hat{s}_N^2 + \sigma_{N+1}^2)}{(\hat{s}_N^2 + \sigma_{N+1}^2)^2} + O_p(N^{-3/2}). \quad (\text{C.10})$$

Because the sum $\sum_{i=1}^N (\hat{s}_N^2 + \sigma_i^2)^{-2}$ and the denominator $(\hat{s}_N^2 + \sigma_{N+1}^2)^2$ are strictly positive, the sign of the variance component update satisfies $\Delta\hat{s}^2 > 0$ under the diagonalized average expected Hessian approximation if and only if $(r_{N+1}^N)^2 > \hat{s}_N^2 + \sigma_{N+1}^2$. This completes the proof of Proposition 4.3. $\square$

Proof of Proposition 4.4. There are a few important calculations.

Step 1 We expand the difference between the empirical Bayes estimates before and after incorporating country $N + 1$. Let $\mu_i^N \equiv (\mathbf{Z}_i^*)' \hat{\beta}_N$ and $\mu_i^{N+1} \equiv (\mathbf{Z}_i^*)' \hat{\beta}_{N+1} = \mu_i^N + (\mathbf{Z}_i^*)' \Delta \hat{\beta}$, which yields the expansion

$$\begin{aligned} \Delta \hat{\varepsilon}_i^{EB} &\equiv \hat{\varepsilon}_{i,N+1}^{EB} - \hat{\varepsilon}_{i,N}^{EB} \\ &= \left[(1 - w_i^{N+1}) \mu_i^{N+1} + w_i^{N+1} \hat{\varepsilon}_i \right] - \left[(1 - w_i^N) \mu_i^N + w_i^N \hat{\varepsilon}_i \right] \\ &= (1 - w_i^{N+1}) (\mu_i^N + (\mathbf{Z}_i^*)' \Delta \hat{\beta}) - (1 - w_i^N) \mu_i^N + (w_i^{N+1} - w_i^N) \hat{\varepsilon}_i \\ &= (1 - w_i^{N+1}) (\mathbf{Z}_i^*)' \Delta \hat{\beta} + (w_i^{N+1} - w_i^N) \hat{\varepsilon}_i + (w_i^N - w_i^{N+1}) \mu_i^N \\ &= (1 - w_i^{N+1}) (\mathbf{Z}_i^*)' \Delta \hat{\beta} + \Delta w_i \left(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}_N \right). \end{aligned}$$

Step 2 We evaluate the exact change in the country-specific shrinkage weight $w_i = s^2 / (s^2 + \sigma_i^2)$, yielding

$$\begin{aligned} \Delta w_i &= \frac{\hat{s}_{N+1}^2}{\hat{s}_{N+1}^2 + \sigma_i^2} - \frac{\hat{s}_N^2}{\hat{s}_N^2 + \sigma_i^2} \\ &= \frac{\hat{s}_{N+1}^2 (\hat{s}_N^2 + \sigma_i^2) - \hat{s}_N^2 (\hat{s}_{N+1}^2 + \sigma_i^2)}{(\hat{s}_N^2 + \sigma_i^2) (\hat{s}_{N+1}^2 + \sigma_i^2)} \\ &= \frac{\sigma_i^2 (\hat{s}_{N+1}^2 - \hat{s}_N^2)}{(\hat{s}_N^2 + \sigma_i^2) (\hat{s}_{N+1}^2 + \sigma_i^2)} = \frac{\sigma_i^2 \Delta \hat{s}^2}{(\hat{s}_N^2 + \sigma_i^2) (\hat{s}_{N+1}^2 + \sigma_i^2)}. \end{aligned}$$

Since the sampling variance satisfies $\sigma_i^2 > 0$ and the denominator terms are strictly positive, the weight change Δw_i necessarily shares the sign of the variance update $\Delta \hat{s}^2$.

Step 3 We approximate the point revision using the stage N weight. Expanding the first term yields

$$(1 - w_i^{N+1}) (\mathbf{Z}_i^*)' \Delta \hat{\beta} = (1 - w_i^N) (\mathbf{Z}_i^*)' \Delta \hat{\beta} - \Delta w_i (\mathbf{Z}_i^*)' \Delta \hat{\beta}. \quad (\text{C.11})$$

By Lemma D.4, we have $\Delta \hat{\beta} = O_p(N^{-1})$ and $\Delta \hat{s}^2 = O_p(N^{-1})$, which implies $\Delta w_i = O_p(N^{-1})$. Consequently, the product is of order $\Delta w_i (\mathbf{Z}_i^*)' \Delta \hat{\beta} = O_p(N^{-2})$, which gives the decomposition

$$\Delta \hat{\varepsilon}_i^{EB} = (1 - w_i^N) (\mathbf{Z}_i^*)' \Delta \hat{\beta} + \Delta w_i \left(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}_N \right) + O_p(N^{-2}). \quad (\text{C.12})$$

This completes the proof of Proposition 4.4. $\square$

Proof of Proposition 4.5. The posterior variance is $s_i^2 = \frac{s^2 \sigma_i^2}{s^2 + \sigma_i^2} = w_i \sigma_i^2$. We calculate the exact change in the posterior variance, yielding

$$\begin{aligned} \Delta s_i^2 &= \frac{\hat{s}_{N+1}^2 \sigma_i^2}{\hat{s}_{N+1}^2 + \sigma_i^2} - \frac{\hat{s}_N^2 \sigma_i^2}{\hat{s}_N^2 + \sigma_i^2} \\ &= \sigma_i^2 \left[\frac{\hat{s}_{N+1}^2}{\hat{s}_{N+1}^2 + \sigma_i^2} - \frac{\hat{s}_N^2}{\hat{s}_N^2 + \sigma_i^2} \right] = \sigma_i^2 \Delta w_i = \frac{\sigma_i^4 \Delta \hat{s}^2}{(\hat{s}_N^2 + \sigma_i^2) (\hat{s}_{N+1}^2 + \sigma_i^2)}. \end{aligned}$$

Since the sampling variance σ_i^4 and the denominator terms are strictly positive, the change in posterior variance Δs_i^2 is directly proportional to, and shares the sign of, the variance update $\Delta \hat{s}^2$. This completes the proof of Proposition 4.5. $\square$

C.2 ETW and Non-Adopters

Proof of Theorem 4.7 (Selection-Corrected EB Shrinkage). Parts of this proof relies on properties of the conditional expectations in multivariate Gaussian models under truncation. Those details are left to Appendix D.

Step 1 We characterize the joint distribution of the structural error $\tilde{u}_i^\varepsilon$, the observed estimation residual $e_i \equiv \hat{\varepsilon}_i - (\mathbf{Z}_i^*)'\beta$, and the selection shock ν_i . Under our Gaussian assumptions, they are jointly normal,

$$\begin{pmatrix} \tilde{u}_i^\varepsilon \\ e_i \\ \nu_i \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} s^2 & s^2 & \rho s \\ s^2 & s^2 + \sigma_i^2 & \rho s \\ \rho s & \rho s & 1 \end{pmatrix} \right). \quad (\text{C.13})$$

We derive the conditional distribution of $(\tilde{u}_i^\varepsilon, \nu_i)$ given the observed residual e_i , which yields

$$\mathbb{E} \left[\begin{pmatrix} \tilde{u}_i^\varepsilon \\ \nu_i \end{pmatrix} \middle| e_i \right] = \frac{e_i}{s^2 + \sigma_i^2} \begin{pmatrix} s^2 \\ \rho s \end{pmatrix} = \begin{pmatrix} (1 - B_i)e_i \\ \mu_{\nu|e,i} \end{pmatrix}, \quad (\text{C.14})$$

where $B_i = \sigma_i^2/(s^2 + \sigma_i^2)$ and $\mu_{\nu|e,i} \equiv \frac{\rho s}{s^2 + \sigma_i^2} e_i$. The conditional covariance matrix is

$$\text{Var} \left(\begin{pmatrix} \tilde{u}_i^\varepsilon \\ \nu_i \end{pmatrix} \middle| e_i \right) = \begin{pmatrix} s^2 & \rho s \\ \rho s & 1 \end{pmatrix} - \frac{1}{s^2 + \sigma_i^2} \begin{pmatrix} s^4 & \rho s^3 \\ \rho s^3 & \rho^2 s^2 \end{pmatrix} = \begin{pmatrix} s^2 B_i & \rho s B_i \\ \rho s B_i & 1 - \rho^2(1 - B_i) \end{pmatrix}.$$

Step 2 We evaluate the expectation of $\tilde{u}_i^\varepsilon$ conditional on both e_i and the selection event $\nu_i > -(\mathbf{q}_i)'\gamma$. The conditional distribution of $\tilde{u}_i^\varepsilon$ given e_i and ν_i is normal,

$$\mathbb{E}[\tilde{u}_i^\varepsilon | e_i, \nu_i] = (1 - B_i)e_i + \frac{\rho s B_i}{1 - \rho^2(1 - B_i)}(\nu_i - \mu_{\nu|e,i}), \quad (\text{C.15})$$

with conditional variance

$$\text{Var}(\tilde{u}_i^\varepsilon | e_i, \nu_i) = s^2 B_i \left(1 - \frac{\rho^2 B_i}{1 - \rho^2(1 - B_i)} \right). \quad (\text{C.16})$$

Using the law of iterated expectations, we integrate over the truncated distribution of $\nu_i \mid e_i, D_i = 1$. Write $V_{\nu|e} \equiv 1 - \rho^2(1 - B_i)$ for the conditional variance of ν_i given e_i from Step 1, and note that the selection event is $\nu_i > c_i$ with threshold $c_i \equiv -(\mathbf{q}_i)' \gamma$. Applying the location-scale truncated-mean formula of Lemma D.7(iv) to $\nu_i \mid e_i \sim \mathcal{N}(\mu_{\nu|e,i}, V_{\nu|e})$,

$$\mathbb{E}[\nu_i \mid e_i, \nu_i > c_i] - \mu_{\nu|e,i} = \sqrt{V_{\nu|e}} \lambda^c \left(\frac{c_i - \mu_{\nu|e,i}}{\sqrt{V_{\nu|e}}} \right) = \sqrt{V_{\nu|e}} \lambda \left(\frac{(\mathbf{q}_i)' \gamma + \mu_{\nu|e,i}}{\sqrt{V_{\nu|e}}} \right) = \sqrt{V_{\nu|e}} \lambda(\alpha_i(e_i)) ,$$

where the middle equality uses the reflection identity $\lambda^c(x) = \lambda(-x)$ established in the proof of Lemma D.7, and $\alpha_i(e_i) \equiv [(\mathbf{q}_i)' \gamma + \mu_{\nu|e,i}] / \sqrt{V_{\nu|e}}$. Substituting into the conditional-mean formula from Step 1,

$$\begin{aligned} \mathbb{E}[\tilde{u}_i^\varepsilon \mid e_i, \nu_i > -(\mathbf{q}_i)' \gamma] &= (1 - B_i)e_i + \frac{\rho s B_i}{V_{\nu|e}} \left(\mathbb{E}[\nu_i \mid e_i, \nu_i > -(\mathbf{q}_i)' \gamma] - \mu_{\nu|e,i} \right) \\ &= (1 - B_i)e_i + \frac{\rho s B_i}{V_{\nu|e}} \cdot \sqrt{V_{\nu|e}} \lambda(\alpha_i(e_i)) = (1 - B_i)e_i + \frac{\rho s B_i}{\sqrt{1 - \rho^2(1 - B_i)}} \lambda(\alpha_i(e_i)) . \end{aligned}$$

Step 3 We derive the posterior variance using the law of total variance, where

$$\text{Var}(\tilde{u}_i^\varepsilon \mid e_i, D_i = 1) = \mathbb{E}[\text{Var}(\tilde{u}_i^\varepsilon \mid e_i, \nu_i) \mid e_i, D_i = 1] + \text{Var}(\mathbb{E}[\tilde{u}_i^\varepsilon \mid e_i, \nu_i] \mid e_i, D_i = 1) . \quad (\text{C.17})$$

The first term is constant and equals the conditional variance from Step 2. The second term is

$$\begin{aligned} \text{Var}(\mathbb{E}[\tilde{u}_i^\varepsilon \mid e_i, \nu_i] \mid e_i, D_i = 1) &= \left(\frac{\rho s B_i}{1 - \rho^2(1 - B_i)} \right)^2 \text{Var}(\nu_i \mid e_i, \nu_i > -(\mathbf{q}_i)' \gamma) \\ &= \frac{\rho^2 s^2 B_i^2}{1 - \rho^2(1 - B_i)} (1 - \delta(\alpha_i(e_i))) . \end{aligned}$$

Summing these two components and simplifying yields

$$\text{Var}(\varepsilon_i^* \mid \hat{\varepsilon}_i, \mathbf{Z}_i^*, \mathbf{q}_i, D_i = 1) = s^2 B_i \left(1 - \frac{\rho^2 B_i}{1 - \rho^2(1 - B_i)} \delta(\alpha_i(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta)) \right) . \quad (\text{C.18})$$

This completes the proof of Theorem 4.7. □

C.2.1 Joint Asymptotics of the Selection-Model Estimator

The parameters $\theta = (\gamma', \beta', \theta_{\rho s}, s^2)'$ are estimated in steps—a probit for adoption on the full sample, a Heckman-type mean regression on the adopter sample, and a variance-equation moment on the adopter sample—but the steps can be stacked into a single exactly-identified system of estimating equations. Define

the residual function $m_i(\theta) \equiv \widehat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta - \theta_{\rho s} \lambda(\mathbf{q}_i' \gamma)$ and the stacked moment vector

$$g_i(\theta) = \begin{pmatrix} [D_i - \Phi(\mathbf{q}_i' \gamma)] \frac{\phi(\mathbf{q}_i' \gamma)}{\Phi(\mathbf{q}_i' \gamma) [1 - \Phi(\mathbf{q}_i' \gamma)]} \mathbf{q}_i \\ D_i m_i(\theta) \mathbf{Z}_i^* \\ D_i m_i(\theta) \lambda(\mathbf{q}_i' \gamma) \\ D_i [m_i(\theta)^2 - \sigma_i^2 - s^2 + \theta_{\rho s}^2 \delta(\mathbf{q}_i' \gamma)] \end{pmatrix}, \quad (\text{C.19})$$

whose first block is the probit score and whose last block is linear in s^2 given $\theta_{\rho s}$.

Proposition C.1 (Joint Asymptotic Distribution of the Selection-Model Estimator). Suppose the tuples $(D_i, \widehat{\varepsilon}_i, \mathbf{Z}_i^*, \mathbf{q}_i, \sigma_i^2)$ are independent across i , the parameter θ_0 lies in the interior of a compact set, $\mathbb{E}[\sup_{\theta} \|g_i(\theta)\|^2] < \infty$, the Jacobian $\mathbf{G} \equiv \lim_N N^{-1} \sum_i \mathbb{E}[\nabla_{\theta'} g_i(\theta_0)]$ is non-singular, and

$$\mathbf{\Omega} \equiv \lim_N N^{-1} \sum_i \mathbb{E}[g_i(\theta_0) g_i(\theta_0)']$$

is finite and positive definite. Then the Z-estimator $\widehat{\theta}$ solving $\sum_i g_i(\widehat{\theta}) = 0$ satisfies

$$\sqrt{N}(\widehat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\theta}), \quad \Sigma_{\theta} = \mathbf{G}^{-1} \mathbf{\Omega} (\mathbf{G}^{-1})', \quad (\text{C.20})$$

and Σ_{θ} is consistently estimated by replacing $\mathbf{G}$ and $\mathbf{\Omega}$ with their sample analogues at $\widehat{\theta}$.

Proof of Proposition C.1. The estimator is a Z-estimator built from the exactly-identified system $\bar{g}_N(\theta) \equiv N^{-1} \sum_i g_i(\theta) = 0$ with g_i given in (C.19). We verify the population zeros, identification, consistency, and asymptotic normality in turn.

Step 1: Population moment conditions The first block is the probit score, which has conditional mean zero at γ_0 because $\mathbb{E}[D_i \mid \mathbf{q}_i] = \Phi(\mathbf{q}_i' \gamma_0)$. The second and third blocks have conditional mean zero at θ_0 by Proposition 4.6(ii): $\mathbb{E}[m_i(\theta_0) \mid \mathbf{Z}_i^*, \mathbf{q}_i, D_i = 1] = 0$, and multiplying by the $\sigma(\mathbf{Z}_i^*, \mathbf{q}_i)$ -measurable instruments $\mathbf{Z}_i^*$ and $\lambda(\mathbf{q}_i' \gamma_0)$ preserves the zero. The fourth block has conditional mean zero by the variance decomposition in Proposition 4.6(iii): $\mathbb{E}[m_i(\theta_0)^2 \mid \cdot, D_i = 1] = \sigma_i^2 + s_0^2(1 - \rho_0^2 \delta(\mathbf{q}_i' \gamma_0)) + \theta_{\rho s, 0}^2 \delta(\mathbf{q}_i' \gamma_0) - \theta_{\rho s, 0}^2 \delta(\mathbf{q}_i' \gamma_0)$, and since $s_0^2 \rho_0^2 = \theta_{\rho s, 0}^2$ the conditional second moment equals $\sigma_i^2 + s_0^2 - \theta_{\rho s, 0}^2 \delta(\mathbf{q}_i' \gamma_0)$, which is precisely what the moment subtracts.

Step 2: Identification through the triangular structure The probit block depends only on γ and identifies γ_0 uniquely (the probit population criterion is strictly concave in γ under any non-degenerate design for $\mathbf{q}_i$). Given γ_0 , the second and third blocks are linear in $(\beta, \theta_{\rho s})$ with coefficient matrix $\mathbf{M}_{\text{select}}$, which is non-singular by hypothesis (Proposition 4.6(ii)), so $(\beta_0, \theta_{\rho s, 0})$ is the unique zero. Given those, the fourth

block is linear in s^2 with coefficient $-\mathbb{P}(D_i = 1) \neq 0$, identifying s_0^2 . Hence θ_0 is the unique zero of the limiting moment map, and well-separatedness on the compact parameter set follows from continuity.

Step 3: Consistency and asymptotic normality The summands $g_i(\theta)$ are independent across i (countries are sampled independently) with square-integrable envelopes by hypothesis, and each component is continuous in θ with derivatives dominated by integrable envelopes (λ , λ' , and δ are smooth with at-most-linear growth, and the parameter set is compact). A uniform law of large numbers then gives $\sup_{\theta} \|\bar{g}_N(\theta) - \mathbb{E} \bar{g}_N(\theta)\| \rightarrow_p 0$, and consistency follows from Step 2 by the standard Z-estimator consistency theorem. For normality, a mean-value expansion of $0 = \bar{g}_N(\hat{\theta})$ around θ_0 gives

$$\sqrt{N}(\hat{\theta} - \theta_0) = - \left[\frac{1}{N} \sum_i \nabla_{\theta'} g_i(\bar{\theta}) \right]^{-1} \frac{1}{\sqrt{N}} \sum_i g_i(\theta_0),$$

where the bracketed Jacobian converges in probability to $\mathbf{G}$ (uniform law of large numbers plus consistency) and is invertible with probability approaching one, and the normalized sum converges to $\mathcal{N}(0, \mathbf{\Omega})$ by the Lindeberg–Lévy CLT for independent arrays with $\mathbb{E} \|g_i(\theta_0)\|^2 < \infty$. Slutsky’s theorem delivers $\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathbf{G}^{-1} \mathbf{\Omega} (\mathbf{G}^{-1})')$.

Step 4: Block-triangularity and sequential equivalence Direct differentiation shows $\nabla_{(\beta, \theta_{ps}, s^2)} g_i^{(1)} = 0$ (the probit score involves only γ) and $\nabla_{s^2} g_i^{(2,3)} = 0$ (the mean residuals do not involve s^2), so $\mathbf{G}$ is block lower-triangular in the stated ordering. Solving the system blockwise therefore reproduces the sequential estimator. Consistent estimation of Σ_{θ} by sample analogues follows from the uniform law of large numbers and the continuous mapping theorem. This completes the proof. $\square$

C.2.2 Prediction Intervals under Selection

Conditioning a jointly normal pair on the truncation of one coordinate skews the distribution of the other. The resulting law is the *extended skew-normal* (ESN) distribution (e.g. [Azzalini, 2005](#); [Azzalini and Capitanio, 2013](#); [Arnold et al., 2002](#); [Ghosh and Ng, 2025](#)).

Definition C.2 (Extended Skew-Normal Distribution). A random variable U follows the standardized extended skew-normal law $\text{ESN}(\alpha_0, \alpha_1)$ if its density is

$$f_{\text{ESN}}(u; \alpha_0, \alpha_1) = \phi(u) \frac{\Phi(\alpha_0 + \alpha_1 u)}{\Phi(\alpha_0 / \sqrt{1 + \alpha_1^2})}, \quad u \in \mathbb{R},$$

with cumulative distribution function $F_{\text{ESN}}(u; \alpha_0, \alpha_1) = \Phi_2\left(u, \alpha_0 / \sqrt{1 + \alpha_1^2}; -\alpha_1 / \sqrt{1 + \alpha_1^2}\right) / \Phi\left(\alpha_0 / \sqrt{1 + \alpha_1^2}\right)$, where $\Phi_2(\cdot, \cdot; r)$ is the bivariate standard normal distribution function with correlation r and $\Phi(\cdot)$ is the distribution function of the standard normal distribution.

Theorem C.3 (External Validity and Prediction Intervals under Selection). Let $\hat{\theta} = (\hat{\gamma}', \hat{\beta}', \hat{\theta}_{\rho s}, \hat{s}^2)'$ be the stacked estimator of Proposition C.1 with estimated covariance $\hat{\Sigma}_{\theta}$, and let j index a non-adopter with covariates $\mathbf{W}_j^* \equiv (\mathbf{Z}_j^*, \mathbf{q}_j)$.

(i) **Unconditional (random-assignment) interval:** The feasible interval

$$PI_j^{\text{uncond}}(\alpha) = \left[(\mathbf{Z}_j^*)' \hat{\beta} \pm z_{1-\alpha/2} \sqrt{\hat{s}^2 + (\mathbf{Z}_j^*)' \hat{\Sigma}_{\beta} \mathbf{Z}_j^*} \right] \quad (\text{C.21})$$

achieves asymptotic coverage $\mathbb{P}(\varepsilon_j^* \in PI_j^{\text{uncond}}(\alpha) \mid \mathbf{Z}_j^*) \rightarrow 1 - \alpha$, where the probability integrates over the *unconditional* population distribution $\tilde{u}_j^\varepsilon \sim \mathcal{N}(0, s_0^2)$. It is *not* valid conditional on $D_j = 0$.

(ii) **Conditional law:** Conditional on non-adoption, the potential elasticity is extended skew-normal, i.e.

$$\frac{\varepsilon_j^* - (\mathbf{Z}_j^*)' \beta_0}{s_0} \mid D_j = 0 \sim \text{ESN}(\alpha_{0j}, \alpha_{1j}), \quad \alpha_{0j} = \frac{-\mathbf{q}_j' \gamma_0}{\sqrt{1 - \rho_0^2}}, \quad \alpha_{1j} = \frac{-\rho_0}{\sqrt{1 - \rho_0^2}}, \quad (\text{C.22})$$

with mean $-\rho_0 s_0 \lambda^c(\mathbf{q}_j' \gamma_0)$ and variance $s_0^2(1 - \rho_0^2 \delta^c(\mathbf{q}_j' \gamma_0))$, where $\lambda^c(x) = \phi(x)/(1 - \Phi(x))$ and $\delta^c(x) = \lambda^c(x)(\lambda^c(x) - x)$.

(iii) **Conditional interval:** The feasible equal-tailed interval built from the plug-in ESN quantiles

$$PI_j^{\text{cond}}(\alpha) = \left[(\mathbf{Z}_j^*)' \hat{\beta} + \hat{s} F_{ESN}^{-1}\left(\frac{\alpha}{2}; \hat{\alpha}_{0j}, \hat{\alpha}_{1j}\right), (\mathbf{Z}_j^*)' \hat{\beta} + \hat{s} F_{ESN}^{-1}\left(1 - \frac{\alpha}{2}; \hat{\alpha}_{0j}, \hat{\alpha}_{1j}\right) \right], \quad (\text{C.23})$$

achieves asymptotic conditional coverage $\mathbb{P}(\varepsilon_j^* \in PI_j^{\text{cond}}(\alpha) \mid \mathbf{Z}_j^*, \mathbf{q}_j, D_j = 0) \rightarrow 1 - \alpha$.

(iv) **Gaussian intervals are incorrect:** A normal-quantile interval centered at $\hat{\varepsilon}_j^{\text{cond}}$ has asymptotic conditional coverage equal to $1 - \alpha$ for all α if and only if $\rho_0 = 0$.

Proof of Theorem C.3. First, note that the stacked estimator $\hat{\theta}$ depends on non-adopter j only through its adoption indicator D_j in the probit block. By the same leave-one-out algebra as Theorem D.9 and all the limiting distribution functions involved are continuous, the influence of any single observation on $\hat{\theta}$ is $O_p(1/N)$, and we may treat $\hat{\theta}$ as independent of country j 's latent variables throughout.

Part (i): Unconditional coverage Under the unconditional (random-assignment) experiment, the potential estimand is $\varepsilon_j^* = (\mathbf{Z}_j^*)' \beta_0 + \tilde{u}_j^\varepsilon$ with $\tilde{u}_j^\varepsilon \sim \mathcal{N}(0, s_0^2)$ *not* conditioned on D_j . The standardized prediction error is

$$T_N \equiv \frac{\varepsilon_j^* - (\mathbf{Z}_j^*)' \hat{\beta}}{\sqrt{\hat{s}^2 + (\mathbf{Z}_j^*)' \hat{\Sigma}_{\beta} \mathbf{Z}_j^*}} = \frac{\tilde{u}_j^\varepsilon - (\mathbf{Z}_j^*)' (\hat{\beta} - \beta_0)}{\sqrt{\hat{s}^2 + (\mathbf{Z}_j^*)' \hat{\Sigma}_{\beta} \mathbf{Z}_j^*}}.$$

By Proposition C.1 and the observation before this paragraph, the numerator is the sum of a random variable distributed $\mathcal{N}(0, s_0^2)$ and an independent asymptotically Gaussian term, and the denominator converges in probability to $s_0 > 0$; therefore $T_N \xrightarrow{d} \mathcal{N}(0, 1)$ by Slutsky's theorem, and $\mathbb{P}(|T_N| \leq z_{1-\alpha/2} \mid \mathbf{Z}_j^*) \rightarrow 1 - \alpha$

by Pólya's theorem (convergence of distribution functions to a continuous limit is uniform). The final claim of part (i)—invalidity conditional on $D_j = 0$ —follows from part (ii): conditional on non-adoption, $\tilde{u}_j^\varepsilon$ has the non-degenerate ESN law with mean $-\rho_0 \mathbf{s}_0 \lambda^c(\mathbf{q}'_j \gamma_0) \neq 0$ whenever $\rho_0 \neq 0$, so the interval is mis-centered and its conditional coverage differs from $1 - \alpha$.

Part (ii): The conditional law is extended skew-normal Non-adoption is the event $\nu_j \leq c_j$ with $c_j \equiv -\mathbf{q}'_j \gamma_0$. By Bayes' rule, the conditional density of the standardized structural shock $T \equiv \tilde{u}_j^\varepsilon / \mathbf{s}_0$ given $\nu_j \leq c_j$ is

$$f_{T|\nu_j \leq c_j}(t) = \phi(t) \frac{\mathbb{P}(\nu_j \leq c_j \mid \tilde{u}_j^\varepsilon = \mathbf{s}_0 t)}{\mathbb{P}(\nu_j \leq c_j)} = \phi(t) \frac{\Phi\left(\frac{c_j - \rho_0 t}{\sqrt{1 - \rho_0^2}}\right)}{\Phi(c_j)},$$

where the conditional probability in the numerator uses $\nu_j \mid \tilde{u}_j^\varepsilon \sim \mathcal{N}(\rho_0 \tilde{u}_j^\varepsilon / \mathbf{s}_0, 1 - \rho_0^2)$ from the joint normality (29). Writing the Φ argument as $\alpha_{0j} + \alpha_{1j}t$ with $\alpha_{0j} = c_j / \sqrt{1 - \rho_0^2} = -\mathbf{q}'_j \gamma_0 / \sqrt{1 - \rho_0^2}$ and $\alpha_{1j} = -\rho_0 / \sqrt{1 - \rho_0^2}$, and noting that $\alpha_{0j} / \sqrt{1 + \alpha_{1j}^2} = c_j$ (so the normalizing constant $\Phi(c_j)$ agrees with Definition C.2) shows $T \mid D_j = 0 \sim \text{ESN}(\alpha_{0j}, \alpha_{1j})$, which is (C.22).⁹ The mean and variance follow from the orthogonal decomposition $\tilde{u}_j^\varepsilon = \rho_0 \mathbf{s}_0 \nu_j + e_j^\perp$ with $e_j^\perp \perp \nu_j$: by Lemma D.7(ii) and the reflections $\lambda(-x) = \lambda^c(x)$, $\delta(-x) = \delta^c(x)$, we have

$$\mathbb{E}[\tilde{u}_j^\varepsilon \mid \nu_j \leq c_j] = \rho_0 \mathbf{s}_0 \mathbb{E}[\nu_j \mid \nu_j \leq c_j] = -\rho_0 \mathbf{s}_0 \lambda(c_j) = -\rho_0 \mathbf{s}_0 \lambda^c(\mathbf{q}'_j \gamma_0),$$

$$\text{Var}(\tilde{u}_j^\varepsilon \mid \nu_j \leq c_j) = \rho_0^2 \mathbf{s}_0^2 [1 - \delta(c_j)] + \mathbf{s}_0^2 (1 - \rho_0^2) = \mathbf{s}_0^2 \left(1 - \rho_0^2 \delta^c(\mathbf{q}'_j \gamma_0)\right).$$

Part (iii): Coverage of the ESN-quantile interval Let $q_p(\theta) \equiv F_{ESN}^{-1}(p; \alpha_{0j}(\theta), \alpha_{1j}(\theta))$ denote the plug-in quantile as a function of the parameters. The ESN distribution function is continuous and strictly increasing in its argument (the density is everywhere positive) and continuous in (α_0, α_1) , so the quantile map $\theta \mapsto q_p(\theta)$ is continuous at θ_0 ; by Proposition C.1, $q_p(\hat{\theta}) \xrightarrow{p} q_p(\theta_0)$. Standardizing the event $\{\varepsilon_j^* \in PI_j^{\text{cond}}\}$ by $(\cdot - (\mathbf{Z}_j^*)' \beta_0) / \mathbf{s}_0$, the interval endpoints become

$$\hat{b}_p \equiv \frac{(\mathbf{Z}_j^*)'(\hat{\beta} - \beta_0) + \hat{\mathbf{s}} q_p(\hat{\theta})}{\mathbf{s}_0} \xrightarrow{p} q_p(\theta_0), \quad p \in \{\alpha/2, 1 - \alpha/2\},$$

and these endpoints are asymptotically independent of the target variable $T \mid D_j = 0$ by Step 0. Hence, by the continuity of F_{ESN} and dominated convergence,

$$\mathbb{P}\left(\varepsilon_j^* \in PI_j^{\text{cond}}(\alpha) \mid \mathbf{Z}_j^*, \mathbf{q}_j, D_j = 0\right) = \mathbb{E}\left[F_{ESN}(\hat{b}_{1-\alpha/2}) - F_{ESN}(\hat{b}_{\alpha/2})\right] \rightarrow (1 - \frac{\alpha}{2}) - \frac{\alpha}{2} = 1 - \alpha.$$

⁹The normalization identity itself is the classical Gaussian convolution formula $\mathbb{E}[\Phi(\alpha_0 + \alpha_1 Z)] = \Phi(\alpha_0 / \sqrt{1 + \alpha_1^2})$ for $Z \sim \mathcal{N}(0, 1)$, which follows by writing $\Phi(\alpha_0 + \alpha_1 Z) = \mathbb{P}(W \leq \alpha_0 + \alpha_1 Z \mid Z)$ for an independent standard normal W and noting $W - \alpha_1 Z \sim \mathcal{N}(0, 1 + \alpha_1^2)$; the cdf expression in Definition C.2 follows from the same bivariate representation.

Part (iv): Mis-calibration of the Gaussian interval Let $\tilde{\mu}_j$ and $\tilde{\sigma}_j^2 = 1 - \rho_0^2 \delta^c(\mathbf{q}'_j \gamma_0)$ denote the mean and variance of the standardized conditional law from part (ii). The Gaussian interval standardizes to $[\tilde{\mu}_j - z_{1-\alpha/2} \tilde{\sigma}_j, \tilde{\mu}_j + z_{1-\alpha/2} \tilde{\sigma}_j]$, whose asymptotic conditional coverage is the stated F_{ESN} -difference by the same argument as part (iii).

If $\rho_0 = 0$, then $\alpha_{1j} = 0$ and the ESN density collapses to $\phi(t)\Phi(\alpha_{0j})/\Phi(\alpha_{0j}) = \phi(t)$: the conditional law is exactly standard normal, $\tilde{\mu}_j = 0$, $\tilde{\sigma}_j = 1$, and the coverage equals $\Phi(z_{1-\alpha/2}) - \Phi(-z_{1-\alpha/2}) = 1 - \alpha$ for every α . Conversely, suppose $\rho_0 \neq 0$ and, for concreteness, $\rho_0 > 0$ (so $\alpha_{1j} < 0$; the other sign is symmetric). Then $\delta^c > 0$ implies $\tilde{\sigma}_j < 1$ strictly. Consider the standardized variable $\tilde{T} = (T - \tilde{\mu}_j)/\tilde{\sigma}_j$, whose density at x is $\tilde{\sigma}_j f_{ESN}(\tilde{\mu}_j + \tilde{\sigma}_j x)$.

On the non-truncated side ($x \rightarrow -\infty$), the factor $\Phi(\alpha_{0j} + \alpha_{1j}(\tilde{\mu}_j + \tilde{\sigma}_j x)) \rightarrow 1$, so the density behaves as $\tilde{\sigma}_j \phi(\tilde{\mu}_j + \tilde{\sigma}_j x)/\Phi(c_j) \asymp \exp(-\tilde{\sigma}_j^2 x^2/2 + O(|x|))$, which, because $\tilde{\sigma}_j < 1$, decays *strictly more slowly* than the standard normal density $\phi(x)$. Consequently $\mathbb{P}(|\tilde{T}| > z)/[2(1 - \Phi(z))] \rightarrow \infty$ as $z \rightarrow \infty$, so for all sufficiently small α the Gaussian interval under-covers: its conditional coverage is strictly below $1 - \alpha$. Hence exact calibration for all α forces $\rho_0 = 0$.

Finally, by the two-sided Chebyshev inequality applied to the conditional law, $\mathbb{P}(|T - \tilde{\mu}_j| \geq k\tilde{\sigma}_j \mid D_j = 0) \leq 1/k^2$ for any $k > 0$; choosing $k = 1/\sqrt{\alpha}$ bounds the miss probability by α regardless of the value of ρ_0 , and plugging in consistent parameter estimates preserves the bound asymptotically by the same continuity arguments as part (iii). This completes the proof of Theorem C.3. $\square$

C.3 Nested Empirical Bayes Estimator (Repeated Observations)

Proof of Proposition 4.8. The proof proceeds by a number of direct calculations.

Step 1 We derive the marginal distribution and pooled likelihood of the country-level estimand ε_i^* from the set of study-level estimates within country i . The study-level estimands are $\varepsilon_{ij}^* \mid \varepsilon_i^* \sim \mathcal{N}(\varepsilon_i^*, \varsigma^2)$, and we observe $\hat{\varepsilon}_{ij} \mid \varepsilon_{ij}^* \sim \mathcal{N}(\varepsilon_{ij}^*, \sigma_{ij}^2)$. Integrating out the study-level parameters yields

$$\hat{\varepsilon}_{ij} \mid \varepsilon_i^* \sim \mathcal{N}(\varepsilon_i^*, \varsigma^2 + \sigma_{ij}^2). \quad (\text{C.24})$$

Conditional on ε_i^* , the estimates $\hat{\varepsilon}_{i1}, \dots, \hat{\varepsilon}_{ik_i}$ are independent. The joint likelihood is

$$\mathcal{L}(\varepsilon_i^* \mid \hat{\varepsilon}_i) \propto \exp \left(-\frac{1}{2} \sum_{j=1}^{k_i} \frac{(\hat{\varepsilon}_{ij} - \varepsilon_i^*)^2}{\varsigma^2 + \sigma_{ij}^2} \right) \propto \exp \left(-\frac{(\bar{\varepsilon}_i - \varepsilon_i^*)^2}{2V_i} \right), \quad (\text{C.25})$$

where the precision-weighted average and pooled variance are

$$\bar{\varepsilon}_i \equiv \frac{\sum_{j=1}^{k_i} \frac{\hat{\varepsilon}_{ij}}{\varsigma^2 + \sigma_{ij}^2}}{\sum_{j=1}^{k_i} \frac{1}{\varsigma^2 + \sigma_{ij}^2}}, \quad V_i \equiv \left(\sum_{j=1}^{k_i} \frac{1}{\varsigma^2 + \sigma_{ij}^2} \right)^{-1}. \quad (\text{C.26})$$

Step 2 We combine the pooled country-level likelihood with the global cross-country prior distribution ε_i^* from $\mathcal{N}((\mathbf{Z}_i^*)'\beta, s^2)$. Under Gaussian conjugacy, the posterior distribution of ε_i^* is normal with mean

$$\mathbb{E}[\varepsilon_i^* \mid \widehat{\varepsilon}_i, \mathbf{Z}_i^*] = \frac{s^2}{s^2 + V_i} \bar{\varepsilon}_i + \frac{V_i}{s^2 + V_i} (\mathbf{Z}_i^*)' \beta = W_i \bar{\varepsilon}_i + (1 - W_i) (\mathbf{Z}_i^*)' \beta, \quad (\text{C.27})$$

where $W_i = \frac{s^2}{s^2 + V_i}$.

Step 3 We derive the posterior expectation of the study-specific parameter ε_{ij}^* . We write the joint distribution of ε_{ij}^* and $\widehat{\varepsilon}_{ij}$ conditional on ε_i^* ,

$$\varepsilon_{ij}^* \mid \varepsilon_i^*, \widehat{\varepsilon}_{ij} \sim \mathcal{N} \left((1 - w_{ij}) \varepsilon_i^* + w_{ij} \widehat{\varepsilon}_{ij}, \varsigma^2 (1 - w_{ij}) \right), \quad (\text{C.28})$$

where $w_{ij} = \frac{\varsigma^2}{\varsigma^2 + \sigma_{ij}^2}$. Before iterating expectations, we record the conditional-independence fact that licenses the step: given the country-level parameter ε_i^* , the pair $(\varepsilon_{ij}^*, \widehat{\varepsilon}_{ij})$ is independent of the *other* studies $\{\widehat{\varepsilon}_{il}\}_{l \neq j}$, because the study-level deviations and sampling errors are drawn independently across studies. Hence conditioning on the full within-country vector adds nothing beyond $(\varepsilon_i^*, \widehat{\varepsilon}_{ij})$:

$$\mathbb{E}[\varepsilon_{ij}^* \mid \varepsilon_i^*, \widehat{\varepsilon}_i, \mathbf{Z}_i^*] = \mathbb{E}[\varepsilon_{ij}^* \mid \varepsilon_i^*, \widehat{\varepsilon}_{ij}].$$

Using the law of iterated expectations through ε_i^* , and the linearity of the inner conditional expectation,

$$\begin{aligned} \widehat{\varepsilon}_{ij}^{EB} &\equiv \mathbb{E}[\varepsilon_{ij}^* \mid \widehat{\varepsilon}_i, \mathbf{Z}_i^*] = \mathbb{E} \left[\mathbb{E}[\varepsilon_{ij}^* \mid \varepsilon_i^*, \widehat{\varepsilon}_{ij}] \mid \widehat{\varepsilon}_i, \mathbf{Z}_i^* \right] \\ &= (1 - w_{ij}) \mathbb{E}[\varepsilon_i^* \mid \widehat{\varepsilon}_i, \mathbf{Z}_i^*] + w_{ij} \widehat{\varepsilon}_{ij} = (1 - w_{ij}) \widehat{\varepsilon}_i^{EB} + w_{ij} \widehat{\varepsilon}_{ij}. \end{aligned}$$

This completes the proof of Proposition 4.8. □

Finally, the three variance components (s^2, ς^2) are estimated by marginal maximum likelihood on the country blocks. Integrating out both latent layers, the within-country vector $\widehat{\varepsilon}_i = (\widehat{\varepsilon}_{i1}, \dots, \widehat{\varepsilon}_{ik_i})'$ is jointly Gaussian with mean $(\mathbf{Z}_i^*)' \beta \cdot \boldsymbol{\iota}_{k_i}$ (where $\boldsymbol{\iota}_{k_i}$ is a vector of ones) and block covariance

$$\boldsymbol{\Sigma}_i(s^2, \varsigma^2) = s^2 \boldsymbol{\iota}_{k_i} \boldsymbol{\iota}_{k_i}' + \text{diag} \left(\varsigma^2 + \sigma_{i1}^2, \dots, \varsigma^2 + \sigma_{ik_i}^2 \right), \quad (\text{C.29})$$

in which the rank-one term reflects the shared country-level draw and the diagonal reflects study-level heterogeneity and noise. The marginal log-likelihood is computed efficiently via the rank-one Sherman–Morrison identity.

Proposition C.4 (Estimation of the Nested Variance Components). Suppose the number of countries $C \rightarrow \infty$ with study counts $2 \leq k_i \leq \bar{k} < \infty$ for a non-vanishing fraction of countries, bounded covariates and sampling variances as in Assumption A.3, and true components (s_0^2, ς_0^2) interior to a compact parameter

space. Then the marginal MLE $(\hat{\beta}, \hat{s}^2, \hat{\varsigma}^2)$ is consistent and $\sqrt{C}$ -asymptotically normal with sandwich covariance equal to the inverse Fisher information of the block-Gaussian model; identification of ς^2 separately from s^2 requires that at least an asymptotically non-vanishing fraction of countries contribute $k_i \geq 2$ studies, since single-study countries reveal only the sum $s^2 + \varsigma^2 + \sigma_{i1}^2$.

D Additional Results

D.1 Technical Results

Lemma D.1 (FWL Reduction). Let $\mathbf{X}_i$ be the $T \times 2$ matrix with rows $(1, t)$, let $\mathbf{P}_i^\perp = \mathbf{I} - \mathbf{X}_i(\mathbf{X}_i' \mathbf{X}_i)^{-1} \mathbf{X}_i'$, and stack country i 's variables as $\mathbf{y}_i, \mathbf{R}_i, \boldsymbol{\eta}_i \in \mathbb{R}^T$. Under the panel model (2) and Assumption 2.1:

(i) **Reduced form:** The projected data satisfies

$$\mathbf{y}_i^\perp = \mathbf{W}_i \beta + \tilde{u}_i^\varepsilon \mathbf{R}_i^\perp + \boldsymbol{\eta}_i^\perp,$$

where $\mathbf{W}_i$ is the $T \times K$ matrix with rows $\mathbf{W}_{it}' = \left(R_{it}^\perp \quad ((\ddot{Z}_{it} R_{it})^\perp)' \quad \bar{Z}_i' R_{it}^\perp \right)$.

(ii) **Weighted covariate identity:** $\mathbf{W}_i' \mathbf{R}_i^\perp = \omega_i \mathbf{Z}_i^*$ with $\mathbf{Z}_i^* = (1, (\ddot{Z}_i^*)', \bar{Z}_i')'$; in particular, the first coordinate of $\mathbf{Z}_i^*$ equals one and the last block equals the simple time average $\bar{Z}_i$.

(iii) **Projection absorption:** For any fixed vector $\mathbf{c} \in \mathbb{R}^T$ lying in the range of $\mathbf{P}_i^\perp$ (in particular, for $\mathbf{c} = \mathbf{R}_i^\perp$ and for each column of $\mathbf{W}_i$),

$$\mathbf{c}' \boldsymbol{\eta}_i^\perp = \mathbf{c}' \boldsymbol{\eta}_i.$$

Consequently, every weighted sum of projected shocks in the paper equals the corresponding weighted sum of shocks, and under Assumption 2.2(ii) such sums have exact variance $\sigma_\eta^2 \|\mathbf{c}\|^2$.

Proof. We prove the three parts in turn.

(i) Substituting the Mundlak-projected elasticity $\varepsilon_{it} = \varepsilon + b' \ddot{Z}_{it} + \pi' \bar{Z}_i + \tilde{u}_i^\varepsilon$ into the outcome equation (2) and stacking over t gives

$$\mathbf{y}_i = \mathbf{X}_i \begin{pmatrix} \alpha_i \\ \xi_i \end{pmatrix} + \varepsilon \mathbf{R}_i + (\ddot{\mathbf{Z}}_i \circ \mathbf{R}_i) b + (\bar{Z}_i' \pi) \mathbf{R}_i + \tilde{u}_i^\varepsilon \mathbf{R}_i + \boldsymbol{\eta}_i,$$

where $\ddot{\mathbf{Z}}_i \circ \mathbf{R}_i$ denotes the matrix whose t -th row is $(\ddot{Z}_{it} R_{it})'$. The operator $\mathbf{P}_i^\perp$ is linear and annihilates the column space of $\mathbf{X}_i$, so applying it to both sides eliminates the intercept and trend exactly. Since $\bar{Z}_i' \pi$ is a country-level constant, $\mathbf{P}_i^\perp [(\bar{Z}_i' \pi) \mathbf{R}_i] = (\bar{Z}_i' \pi) \mathbf{R}_i^\perp$, and collecting the projected regressors into $\mathbf{W}_i$ with $\beta = (\varepsilon, b', \pi')'$ yields the displayed reduced form.

(ii) We compute the three blocks of $\mathbf{W}_i' \mathbf{R}_i^\perp$ directly. The first block is $\sum_t (R_{it}^\perp)^2 = \omega_i$. The middle block is $\sum_t R_{it}^\perp (\ddot{Z}_{it} R_{it})^\perp = \omega_i \ddot{Z}_i^*$ by the definition of $\ddot{Z}_i^*$. For the last block, note that $\sum_t (\bar{Z}_i R_{it}^\perp) R_{it}^\perp = \bar{Z}_i \omega_i$.

Dividing by ω_i delivers $\mathbf{Z}_i^* = (1, (\ddot{Z}_i^*)', \bar{Z}_i')'$ as claimed.

(iii) Since $\mathbf{c}$ lies in the range of $\mathbf{P}_i^\perp$, we may write $\mathbf{c} = \mathbf{P}_i^\perp \mathbf{c}$. Because $\mathbf{P}_i^\perp$ is symmetric and idempotent,

$$\mathbf{c}' \boldsymbol{\eta}_i^\perp = (\mathbf{P}_i^\perp \mathbf{c})' \mathbf{P}_i^\perp \boldsymbol{\eta}_i = \mathbf{c}' (\mathbf{P}_i^\perp)' \mathbf{P}_i^\perp \boldsymbol{\eta}_i = \mathbf{c}' \mathbf{P}_i^\perp \boldsymbol{\eta}_i = \mathbf{c}' \boldsymbol{\eta}_i.$$

The variance claim follows because, under Assumption 2.2(ii), the shocks η_i have conditional covariance $\sigma_\eta^2 \mathbf{I}$, therefore $\text{Var}(\mathbf{c}' \boldsymbol{\eta}_i \mid \mathbf{R}, \mathbf{Z}) = \sigma_\eta^2 \mathbf{c}' \mathbf{c}$. $\square$

Lemma D.2 (Leverage Properties). Let $h_{ij} = \omega_j (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^*$, $\mathbf{H}_{ii} = \mathbf{W}_i \mathbf{M}^{-1} \mathbf{W}_i'$, and $A_i = (1 - h_{ii})^2 + \sum_{j \neq i} h_{ij}^2$. Then:

- (i) **Range:** $0 \leq h_{ii} \leq 1$ for every i , and $h_{ii} = \omega_i^{-1} (\mathbf{R}_i^\perp)' \mathbf{H}_{ii} \mathbf{R}_i^\perp$.
- (ii) **Variance identities:** $(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* = s^2 \sum_{j=1}^N h_{ij}^2$ and $s^2 A_i = (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* + s^2 (1 - 2h_{ii})$.
- (iii) **Weight range:** $A_i \geq (1 - h_{ii})^2 \geq 0$ and $w_i^* = s^2 A_i / [s^2 A_i + \sigma_i^2 (1 - h_{ii})] \in [0, 1]$.
- (iv) **Extrapolation penalty bounds:** $\sum_{j \neq i} (\omega_i / \omega_j) h_{ij}^2 \leq h_{ii} (1 - h_{ii})$. Consequently, letting $\bar{r}_i \equiv \max_{j \neq i} \omega_j / \omega_i$ denote the largest precision ratio between another country and country i , we have

$$(1 - h_{ii})^2 \leq A_i \leq (1 - h_{ii})^2 + \bar{r}_i h_{ii} (1 - h_{ii}) \leq (1 - h_{ii}) \max\{1, \bar{r}_i\}.$$

In particular: (a) if $\omega_i \geq \max_{j \neq i} \omega_j$, then $A_i \leq 1 - h_{ii} \leq 1$; (b) if $\bar{r}_i$ remains bounded, then $A_i \rightarrow 0$ as $h_{ii} \rightarrow 1$; and (c) $A_i \rightarrow \infty$ requires $\bar{r}_i h_{ii} (1 - h_{ii}) \rightarrow \infty$, and in particular $\bar{r}_i \rightarrow \infty$.

- (v) **Rates:** Under Assumption A.2(v)–(vi), $\max_{i,j} |h_{ij}| \leq \bar{\omega} C^2 / (cN) = O_p(1/N)$, $\sum_{j \neq i} h_{ij}^2 = O_p(1/N)$ uniformly in i , and $A_i = 1 + O_p(1/N)$ uniformly in i .

Proof. (i) Non-negativity is immediate: $\mathbf{M}$ is positive definite under Assumption A.1(i), so $h_{ii} = \omega_i (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_i^* \geq 0$. For the upper bound, Lemma D.1(ii) gives $\omega_i \mathbf{Z}_i^* = \mathbf{W}_i' \mathbf{R}_i^\perp$, hence

$$h_{ii} = \frac{1}{\omega_i} (\mathbf{R}_i^\perp)' \mathbf{W}_i \mathbf{M}^{-1} \mathbf{W}_i' \mathbf{R}_i^\perp = \frac{1}{\omega_i} (\mathbf{R}_i^\perp)' \mathbf{H}_{ii} \mathbf{R}_i^\perp.$$

Now let $\mathbb{W}$ denote the $(NT) \times K$ design matrix stacking all countries, so that $\mathbf{M} = \mathbb{W}' \mathbb{W}$ and $\mathbf{P} \equiv \mathbb{W} \mathbf{M}^{-1} \mathbb{W}'$ is the orthogonal projection onto the column space of $\mathbb{W}$: it is symmetric by construction and idempotent because $\mathbf{P}^2 = \mathbb{W} \mathbf{M}^{-1} (\mathbb{W}' \mathbb{W}) \mathbf{M}^{-1} \mathbb{W}' = \mathbf{P}$. The block $\mathbf{H}_{ii}$ is the i -th diagonal block of $\mathbf{P}$. For any $\mathbf{a} \in \mathbb{R}^T$, let $\tilde{\mathbf{a}}$ embed $\mathbf{a}$ into the stacked space with zeros elsewhere; then, since orthogonal projections are contractions,

$$\mathbf{a}' \mathbf{H}_{ii} \mathbf{a} = \tilde{\mathbf{a}}' \mathbf{P} \tilde{\mathbf{a}} = \|\mathbf{P} \tilde{\mathbf{a}}\|^2 \leq \|\tilde{\mathbf{a}}\|^2 = \|\mathbf{a}\|^2.$$

Applying this with $\mathbf{a} = \mathbf{R}_i^\perp$ yields $(\mathbf{R}_i^\perp)' \mathbf{H}_{ii} \mathbf{R}_i^\perp \leq \omega_i$ and therefore $h_{ii} \leq 1$.

(ii) From the definition $\mathbf{V}_\beta^{\text{struct}} = \mathbf{s}^2 \mathbf{M}^{-1} (\sum_j \omega_j^2 \mathbf{Z}_j^* (\mathbf{Z}_j^*)') \mathbf{M}^{-1}$,

$$(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{\text{struct}} \mathbf{Z}_i^* = \mathbf{s}^2 \sum_{j=1}^N \omega_j^2 \left((\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^* \right)^2 = \mathbf{s}^2 \sum_{j=1}^N h_{ij}^2.$$

Expanding the square in A_i gives $A_i = 1 - 2h_{ii} + h_{ii}^2 + \sum_{j \neq i} h_{ij}^2 = 1 - 2h_{ii} + \sum_{j=1}^N h_{ij}^2$, and multiplying by $\mathbf{s}^2$ and substituting the first identity yields the second.

(iii) The bound $A_i \geq (1 - h_{ii})^2$ holds because $\sum_{j \neq i} h_{ij}^2 \geq 0$. Both the numerator $\mathbf{s}^2 A_i$ and the increment $\sigma_i^2(1 - h_{ii})$ in the denominator of w_i^* are non-negative by part (i), so the ratio lies in $[0, 1]$.

(iv) Retain the notation of part (i): $\mathbb{W}$ stacks the country design blocks, $\mathbf{M} = \mathbb{W}'\mathbb{W}$, and $\mathbf{P} = \mathbb{W}\mathbf{M}^{-1}\mathbb{W}'$ is the orthogonal projection onto the column space of $\mathbb{W}$. Let $\tilde{\mathbf{a}}_i \in \mathbb{R}^{NT}$ embed $\mathbf{R}_i^\perp$ into country i 's block with zeros elsewhere, and define $g_{ij} \equiv \tilde{\mathbf{a}}_i' \mathbf{P} \tilde{\mathbf{a}}_j$. By Lemma D.1(ii), $\mathbf{W}_j' \mathbf{R}_j^\perp = \omega_j \mathbf{Z}_j^*$, so

$$g_{ij} = (\mathbf{R}_i^\perp)' \mathbf{W}_i \mathbf{M}^{-1} \mathbf{W}_j' \mathbf{R}_j^\perp = \omega_i \omega_j (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_j^* = \omega_i h_{ij}.$$

Partition $\mathbf{P} \tilde{\mathbf{a}}_i = (\mathbf{p}_1', \dots, \mathbf{p}_N')'$ into country blocks $\mathbf{p}_j \in \mathbb{R}^T$. Then $g_{ij} = \tilde{\mathbf{a}}_j' (\mathbf{P} \tilde{\mathbf{a}}_i) = (\mathbf{R}_j^\perp)' \mathbf{p}_j$, so the Cauchy–Schwarz inequality applied blockwise gives $g_{ij}^2 \leq \omega_j \|\mathbf{p}_j\|^2$ for every j , including $j = i$. Because $\mathbf{P}$ is a symmetric idempotent projection, $\sum_{j=1}^N \|\mathbf{p}_j\|^2 = \|\mathbf{P} \tilde{\mathbf{a}}_i\|^2 = \tilde{\mathbf{a}}_i' \mathbf{P} \tilde{\mathbf{a}}_i = g_{ii} = \omega_i h_{ii}$. Therefore

$$\sum_{j \neq i} \frac{g_{ij}^2}{\omega_j} \leq \sum_{j \neq i} \|\mathbf{p}_j\|^2 = \omega_i h_{ii} - \|\mathbf{p}_i\|^2 \leq \omega_i h_{ii} - \frac{g_{ii}^2}{\omega_i} = \omega_i h_{ii} (1 - h_{ii}),$$

where the second inequality uses $g_{ii}^2 \leq \omega_i \|\mathbf{p}_i\|^2$. Substituting $g_{ij} = \omega_i h_{ij}$ and dividing by ω_i yields $\sum_{j \neq i} (\omega_i / \omega_j) h_{ij}^2 \leq h_{ii} (1 - h_{ii})$. The upper bound on A_i follows from

$$\sum_{j \neq i} h_{ij}^2 = \sum_{j \neq i} \frac{\omega_j}{\omega_i} \cdot \frac{\omega_i}{\omega_j} h_{ij}^2 \leq \bar{r}_i \sum_{j \neq i} \frac{\omega_i}{\omega_j} h_{ij}^2 \leq \bar{r}_i h_{ii} (1 - h_{ii}),$$

and the final inequality in the displayed chain from $(1 - h_{ii})^2 + \bar{r}_i h_{ii} (1 - h_{ii}) = (1 - h_{ii}) [(1 - h_{ii}) + \bar{r}_i h_{ii}] \leq (1 - h_{ii}) \max\{1, \bar{r}_i\}$, since $(1 - h_{ii}) + \bar{r}_i h_{ii}$ is a convex combination of 1 and $\bar{r}_i$. The lower bound is part (iii). Claims (a) and (b) are immediate from the upper bound. For (c), $A_i \leq 1 + \bar{r}_i h_{ii} (1 - h_{ii})$ because $(1 - h_{ii})^2 \leq 1$, so $A_i \rightarrow \infty$ forces $\bar{r}_i h_{ii} (1 - h_{ii}) \rightarrow \infty$; since $h_{ii} (1 - h_{ii}) \leq 1/4$ by part (i), this in turn forces $\bar{r}_i \rightarrow \infty$, and the divergence must load through $\sum_{j \neq i} h_{ij}^2$ because the diagonal component $(1 - h_{ii})^2$ is bounded by one.

(v) By the Cauchy–Schwarz inequality and Assumption A.2(v)–(vi), $|h_{ij}| \leq \omega_j \|\mathbf{Z}_i^*\| \|\mathbf{Z}_j^*\| / \lambda_{\min}(\mathbf{M}) \leq \bar{\omega} C^2 / (cN)$ with probability approaching one, uniformly in (i, j) . Summing $N - 1$ squared terms each of order N^{-2} gives $\sum_{j \neq i} h_{ij}^2 \leq (N - 1) \bar{\omega}^2 C^4 / (cN)^2 = O_p(1/N)$, and then $A_i = 1 - 2h_{ii} + \sum_j h_{ij}^2 = 1 + O_p(1/N)$, with all bounds uniform in i . $\square$

Lemma D.3 (Block-Diagonalization of average Expected Fisher Information). Let the marginal log-likelihood contribution of country i be $\ell_i(\beta, s^2) = -\frac{1}{2} \ln(s^2 + \sigma_i^2) - \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta)^2}{2(s^2 + \sigma_i^2)}$. Under Assumptions 4.1, A.3(i)(ii) and A.4, the expected cross-derivative of the log-likelihood satisfies

$$\mathbb{E} \left[\frac{\partial^2 \ell_i(\beta, s^2)}{\partial \beta \partial s^2} \middle| \mathbf{Z}_i^*, \sigma_i^2 \right] = \mathbf{0} . \quad (\text{D.1})$$

Consequently, the joint expected Fisher Information matrix $\mathbf{I}_N(\theta)$ is block-diagonal with respect to the location parameter β and the variance parameter s^2 ,

$$\mathbf{I}_N(\theta) = \begin{pmatrix} \mathbf{I}_{N,\beta\beta}(\theta) & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{N,s^2s^2}(\theta) \end{pmatrix} . \quad (\text{D.2})$$

Proof of Lemma D.3. The score component for the location parameter is

$$\nabla_\beta \ell_i(\beta, s^2) = \frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta) \mathbf{Z}_i^*}{s^2 + \sigma_i^2} . \quad (\text{D.3})$$

Differentiating with respect to the variance parameter s^2 yields

$$\frac{\partial^2 \ell_i(\beta, s^2)}{\partial \beta \partial s^2} = -\frac{(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta) \mathbf{Z}_i^*}{(s^2 + \sigma_i^2)^2} . \quad (\text{D.4})$$

Evaluating the conditional expectation at the true parameter $\theta_0 = (\beta_0', s_0^2)'$, we have

$$\mathbb{E} \left[\frac{\partial^2 \ell_i(\beta_0, s_0^2)}{\partial \beta \partial s^2} \middle| \mathbf{Z}_i^*, \sigma_i^2 \right] = -\frac{\mathbf{Z}_i^*}{(s_0^2 + \sigma_i^2)^2} \mathbb{E} \left[\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta_0 \middle| \mathbf{Z}_i^*, \sigma_i^2 \right] = \mathbf{0} , \quad (\text{D.5})$$

where the inner expectation is evaluated by iterating through the latent layer via LIE, i.e. $\mathbb{E}[\hat{\varepsilon}_i \mid \mathbf{Z}_i^*, \sigma_i^2] = \mathbb{E}[\mathbb{E}[\hat{\varepsilon}_i \mid \varepsilon_i^*, \mathbf{Z}_i^*, \sigma_i^2] \mid \mathbf{Z}_i^*, \sigma_i^2] = \mathbb{E}[\varepsilon_i^* \mid \mathbf{Z}_i^*, \sigma_i^2] = (\mathbf{Z}_i^*)' \beta_0$. Thus, the expected cross-derivative is zero. The block-diagonal structure of $\mathbf{I}_N(\theta)$ follows immediately, with diagonal blocks

$$\mathbf{I}_{N,\beta\beta}(\theta_0) = \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{Z}_i^* (\mathbf{Z}_i^*)'}{s_0^2 + \sigma_i^2} , \quad \mathbf{I}_{N,s^2s^2}(\theta_0) = \frac{1}{2N} \sum_{i=1}^N \frac{1}{(s_0^2 + \sigma_i^2)^2} ,$$

obtained by direct second differentiation of ℓ_i and the moment identities $\mathbb{E}[(\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta_0)^2 \mid \mathbf{Z}_i^*, \sigma_i^2] = s_0^2 + \sigma_i^2$. The same vanishing-cross-moment computation, in score (rather than Hessian) form, appears in the proof of Theorem B.1, where it rests on the vanishing of the Gaussian third moment. $\square$

Lemma D.4 (Asymptotic Order of Hyperparameter Updates). Under Assumptions 4.1, A.3(i)(ii) and A.4,

the joint Maximum Likelihood updates $\Delta\hat{\beta} \equiv \hat{\beta}_{N+1} - \hat{\beta}_N$ and $\Delta\hat{s}^2 \equiv \hat{s}_{N+1}^2 - \hat{s}_N^2$ satisfy

$$\|\Delta\hat{\beta}\| = O_p(N^{-1}) \quad \text{and} \quad |\Delta\hat{s}^2| = O_p(N^{-1}). \quad (\text{D.6})$$

Proof of Lemma D.4. The proof proceeds in three steps: first we establish that the update is $o_p(1)$ by consistency of both estimators; then we control all the derivative arrays appearing in a second-order expansion; finally we apply a self-bounding inequality that upgrades $o_p(1)$ to the exact rate $O_p(N^{-1})$. The two-step structure is necessary: the self-bounding inequality below involves a quadratic term in $\|\Delta\hat{\theta}\|$, and only after we know the update is small can the quadratic term be absorbed.

Step 1: The update is $o_p(1)$ Under Assumptions 4.1, A.3(i)(ii) and A.4, the marginal MLE is consistent at every stage—this is part (i) of Theorem B.1, whose proof uses exactly these conditions (compactness, interiority, identification, uniform convergence). Applying it at stages N and $N+1$ gives $\hat{\theta}_N \xrightarrow{p} \theta_0$ and $\hat{\theta}_{N+1} \xrightarrow{p} \theta_0$, hence $\Delta\hat{\theta} = \hat{\theta}_{N+1} - \hat{\theta}_N = o_p(1)$ by the triangle inequality.

Step 2: Expansion with controlled remainders Let $\hat{\theta}_N$ maximize $\sum_{i=1}^N \ell_i(\theta)$, so that $\sum_{i=1}^N \nabla_{\theta} \ell_i(\hat{\theta}_N) = \mathbf{0}$. Since θ_0 is interior and both estimators are consistent, the stage- $(N+1)$ first-order condition $\sum_{i=1}^{N+1} \nabla_{\theta} \ell_i(\hat{\theta}_{N+1}) = \mathbf{0}$ holds with probability approaching one, and a second-order Taylor expansion around $\hat{\theta}_N$ gives

$$\mathbf{0} = \nabla_{\theta} \ell_{N+1}(\hat{\theta}_N) + \left[\sum_{i=1}^{N+1} \nabla_{\theta\theta}^2 \ell_i(\hat{\theta}_N) \right] \Delta\hat{\theta} + \mathcal{R}_2, \quad \mathcal{R}_2 = \frac{1}{2} \sum_{i=1}^{N+1} \nabla_{\theta\theta\theta}^3 \ell_i(\bar{\theta})(\Delta\hat{\theta}, \Delta\hat{\theta}), \quad (\text{D.7})$$

where $\bar{\theta}$ lies on the segment joining $\hat{\theta}_N$ and $\hat{\theta}_{N+1}$, and we used $\sum_{i=1}^N \nabla_{\theta} \ell_i(\hat{\theta}_N) = \mathbf{0}$ to reduce the score term to the single new contribution. We now bound each array. The third derivatives of $\ell_i(\theta) = -\frac{1}{2} \ln(s^2 + \sigma_i^2) - (\hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta)^2 / [2(s^2 + \sigma_i^2)]$ are rational functions whose denominators are powers of $s^2 + \sigma_i^2 \geq \underline{\sigma}^2 > 0$ (Assumption A.4(ii)) and whose numerators are at most quadratic in $\hat{\varepsilon}_i$ with coefficients bounded by powers of $\|\mathbf{Z}_i^*\| \leq C$; hence $\sup_{\theta \in \Theta} \|\nabla_{\theta\theta\theta}^3 \ell_i(\theta)\| \leq C'(1 + \hat{\varepsilon}_i^2)$, which has bounded expectation, so $\sup_{\theta} \|\sum_{i=1}^{N+1} \nabla_{\theta\theta\theta}^3 \ell_i(\theta)\| = O_p(N)$ and $\|\mathcal{R}_2\| \leq O_p(N) \|\Delta\hat{\theta}\|^2$.

For the Hessian, Assumption A.4(iv) and the continuity of $\mathbf{I}_N(\cdot)$ give

$$-\frac{1}{N} \sum_{i=1}^{N+1} \nabla_{\theta\theta}^2 \ell_i(\hat{\theta}_N) = \mathbf{I}_N(\hat{\theta}_N) + O_p(N^{-1/2}), \quad (\text{D.8})$$

where the single extra summand $\nabla_{\theta\theta}^2 \ell_{N+1}/N = O_p(N^{-1})$ is absorbed into the remainder. Finally, the new score is bounded in probability, $\nabla_{\theta} \ell_{N+1}(\hat{\theta}_N) = O_p(1)$, since it is a fixed smooth function of $(\hat{\varepsilon}_{N+1}, \mathbf{Z}_{N+1}^*, \sigma_{N+1}^2)$ evaluated at a consistent point.

Step 3: Self-bounding inequality Dividing the expansion by N and rearranging,

$$\left[\mathbf{I}_N(\hat{\theta}_N) + O_p(N^{-1/2}) \right] \Delta\hat{\theta} = \frac{1}{N} \nabla_{\theta} \ell_{N+1}(\hat{\theta}_N) + O_p(1) \cdot \|\Delta\hat{\theta}\|^2. \quad (\text{D.9})$$

By Assumption A.4(iii), $\lambda_{\min}(\mathbf{I}_N(\hat{\theta}_N)) \geq c > 0$ with probability approaching one (using consistency of $\hat{\theta}_N$ and the uniform lower bound on Θ), so taking norms,

$$(c - o_p(1)) \|\Delta\hat{\theta}\| \leq O_p(N^{-1}) + K_N \|\Delta\hat{\theta}\|^2,$$

where $K_N = O_p(1)$. By Step 1, $\|\Delta\hat{\theta}\| = o_p(1)$, so on an event of probability approaching one we have $K_N \|\Delta\hat{\theta}\| \leq c/2$; on that event the quadratic term is at most $(c/2) \|\Delta\hat{\theta}\|$ and may be moved to the left-hand side:

$$\frac{c}{2} \|\Delta\hat{\theta}\| \leq O_p(N^{-1}) \implies \|\Delta\hat{\theta}\| = O_p(N^{-1}).$$

Both components inherit the rate, which proves Lemma D.4. $\square$

Lemma D.5 (Asymptotic Normality of the Stage 1 Sampling Error). Suppose Assumptions 2.2(i)–(iii) and A.1 hold. Then, conditional on $(\mathbf{R}, \mathbf{Z})$, as $T \rightarrow \infty$,

$$\frac{v_i}{\sigma_i} = \frac{1}{\sigma_{\eta} \sqrt{\omega_i}} \sum_{t=1}^T R_{it}^{\perp} \eta_{it} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof. By Lemma D.1(iii), the projected-shock representation of v_i equals the raw-shock weighted sum displayed above, so we may work with the independent summands $X_{T,t} \equiv R_{it}^{\perp} \eta_{it} / (\sigma_{\eta} \sqrt{\omega_i})$, $t = 1, \dots, T$, which form a triangular array of conditionally independent, mean-zero random variables with $\sum_t \text{Var}(X_{T,t} | \mathbf{R}, \mathbf{Z}) = \sum_t (R_{it}^{\perp})^2 / \omega_i = 1$ for every T . Therefore, to prove the statement, it remains to verify the Lindeberg condition.

Fix $\epsilon > 0$ and set $\tilde{\iota} \equiv \min(\iota, 2) > 0$, where ι is the moment of Assumption A.1(iii). By the elementary inequality $\mathbb{E}[X^2 \mathbf{1}\{|X| > \epsilon\}] \leq \epsilon^{-\tilde{\iota}} \mathbb{E}[|X|^{2+\tilde{\iota}}]$ and the moment bound $\sup_{i,t} \mathbb{E}[|\eta_{it}/\sigma_{\eta}|^{2+\tilde{\iota}} | \mathbf{R}, \mathbf{Z}] \leq C' < \infty$ (which follows from Assumption A.1(iii) by Lyapunov's inequality, since $2 + \tilde{\iota} \leq 4 + \iota$), we have

$$\sum_{t=1}^T \mathbb{E} \left[X_{T,t}^2 \mathbf{1}\{|X_{T,t}| > \epsilon\} \right] \leq \frac{C'}{\epsilon^{\tilde{\iota}}} \sum_{t=1}^T \left(\frac{|R_{it}^{\perp}|}{\sqrt{\omega_i}} \right)^{2+\tilde{\iota}} \leq \frac{C'}{\epsilon^{\tilde{\iota}}} \left(\max_{t \leq T} \frac{(R_{it}^{\perp})^2}{\omega_i} \right)^{\tilde{\iota}/2} \underbrace{\sum_{t=1}^T \frac{(R_{it}^{\perp})^2}{\omega_i}}_{=1},$$

where the last inequality bounds each summand's excess power by the maximal normalized weight. The no-dominance condition of Assumption A.1(ii) states precisely that $\max_t (R_{it}^{\perp})^2 / \omega_i \rightarrow 0$, so the right-hand side vanishes as $T \rightarrow \infty$. Applying the Lindeberg-Feller central limit theorem then completes the proof. $\square$

Lemma D.6 (Consistency of the Method-of-Moments Variance Estimators). Suppose Assumptions 2.2(i)–

(iii), 2.4, A.1, and A.2 hold, $T > 3$, and $\liminf_N \bar{\omega}_N > 0$. Then the moments in (B.1) satisfy

$$\hat{\mu}_1 = \sigma_\eta^2 \left(1 - \frac{2}{T}\right) + \frac{\bar{\omega}_N}{T} s^2 + O_p(N^{-1/2}), \quad \hat{\mu}_2 = \sigma_\eta^2 + \bar{\omega}_N s^2 + O_p(N^{-1/2}),$$

and consequently the estimators in (B.3) obey $\hat{s}_{MM}^2 - s^2 = O_p(N^{-1/2})$ and $\hat{\sigma}_\eta^2 - \sigma_\eta^2 = O_p(N^{-1/2})$.

Proof. Write $\Delta \equiv \hat{\beta}^{CRC} - \beta = O_p(N^{-1/2})$ (Assumption A.2(i)–(iv)). We treat the two moments in turn.

First moment. By Lemma D.1(i), the pooled residual decomposes as $e_{js} = \tilde{u}_j^\varepsilon R_{js}^\perp + \eta_{js}^\perp - \mathbf{W}_{js}' \Delta$. Squaring and averaging over the NT observations produces three squared terms and three cross terms. The leading squared term has mean

$$\frac{1}{NT} \sum_{j,s} \mathbb{E} \left[(\tilde{u}_j^\varepsilon R_{js}^\perp + \eta_{js}^\perp)^2 \right] = \frac{1}{NT} \left(s^2 \sum_j \omega_j + \sigma_\eta^2 \sum_j (T-2) \right) = \sigma_\eta^2 \left(1 - \frac{2}{T}\right) + \frac{\bar{\omega}_N}{T} s^2,$$

where we used $\sum_s (R_{js}^\perp)^2 = \omega_j$, the identity of Lemma D.1(iii) applied with $\mathbf{c} = \mathbf{R}_j^\perp$ (the cross term $2\tilde{u}_j \sum_s R_{js}^\perp \eta_{js}^\perp$ has conditional mean zero, and the exact trace identity $\mathbb{E}[\|\boldsymbol{\eta}_j^\perp\|^2] = \sigma_\eta^2(T-2)$ accounts for the degrees of freedom lost by the within-country FWL projection). The fluctuation of this average around its mean is $O_p(N^{-1/2})$ by Chebyshev's inequality, since the summands are independent across j with uniformly bounded fourth moments (Assumptions A.1(iii), A.2(ii), and the Gaussian or sub-fourth-moment structure of $\tilde{u}_j$). The estimation-error terms are negligible: the pure term satisfies $(NT)^{-1} \sum_{j,s} (\mathbf{W}_{js}' \Delta)^2 \leq \|\Delta\|^2 \cdot (NT)^{-1} \sum_{j,s} \|\mathbf{W}_{js}\|^2 = O_p(N^{-1})$, and the cross term is bounded by Cauchy–Schwarz by the geometric mean of the two squared averages, which is $O_p(N^{-1/2})$.

Second moment. By Theorem 2.3 and the definition of r_i , $r_i = \tilde{u}_i^\varepsilon + v_i - (\mathbf{Z}_i^*)' \Delta$. Then $\omega_i r_i^2 = \omega_i (\tilde{u}_i + v_i)^2 - 2\omega_i (\tilde{u}_i + v_i) (\mathbf{Z}_i^*)' \Delta + \omega_i ((\mathbf{Z}_i^*)' \Delta)^2$. The first term has mean $\omega_i s^2 + \sigma_\eta^2$ (using $\text{Var}(v_i) = \sigma_\eta^2 / \omega_i$ and the conditional independence of $\tilde{u}_i$ and v_i), so its cross-sectional average equals $\bar{\omega}_N s^2 + \sigma_\eta^2 + O_p(N^{-1/2})$, again by an independent-array law of large numbers with uniformly bounded second moments (Assumption A.2(v) bounds ω_i). The remaining terms are $O_p(N^{-1/2})$ and $O_p(N^{-1})$ respectively, by the same Cauchy–Schwarz argument as above.

Solving the system. The map from (μ_1, μ_2) to (σ_η^2, s^2) defined by (B.2) is linear with determinant $\bar{\omega}_N(T-3)/T$, which is bounded away from zero by hypothesis. A linear map with uniformly invertible coefficient matrix transports $O_p(N^{-1/2})$ perturbations of the inputs to $O_p(N^{-1/2})$ perturbations of the outputs, which proves the final claim. Truncation at zero can only reduce the error when the true variances are positive, since the truncated value is closer to the (positive) truth whenever truncation binds. $\square$

Lemma D.7 (Truncated Normal Moments). Let $X \sim \mathcal{N}(0, 1)$, let $\lambda(x) = \phi(x)/\Phi(x)$ and $\lambda^c(x) = \phi(x)/(1 - \Phi(x))$ denote the lower- and upper-tail inverse Mills ratios, and define $\delta(x) = \lambda(x)(\lambda(x) + x)$ and $\delta^c(x) = \lambda^c(x)(\lambda^c(x) - x)$.

(i) **Upper truncation:** $\mathbb{E}[X \mid X > a] = \lambda^c(a)$ and $\text{Var}(X \mid X > a) = 1 - \delta^c(a)$.

(ii) **Lower truncation:** $\mathbb{E}[X \mid X < a] = -\lambda(a)$ and $\text{Var}(X \mid X < a) = 1 - \delta(a)$.

(iii) **Range:** $0 < \delta(x) < 1$ and $0 < \delta^c(x) < 1$ for every $x \in \mathbb{R}$.

(iv) **Location-scale version:** If $Y \sim \mathcal{N}(\mu, \tau^2)$, then $\mathbb{E}[Y \mid Y > a] = \mu + \tau \lambda^c\left(\frac{a-\mu}{\tau}\right)$ and $\text{Var}(Y \mid Y > a) = \tau^2 \left[1 - \delta^c\left(\frac{a-\mu}{\tau}\right)\right]$.

Proof of Lemma D.7. (i) Both moments follow from integration by parts against the Gaussian density, using $\phi'(x) = -x\phi(x)$. For the mean, $\int_a^\infty x\phi(x)dx = [-\phi(x)]_a^\infty = \phi(a)$, so $\mathbb{E}[X \mid X > a] = \phi(a)/(1 - \Phi(a)) = \lambda^c(a)$. For the second moment, integrating $x^2\phi(x) = -x\phi'(x)$ by parts gives $\int_a^\infty x^2\phi(x)dx = a\phi(a) + (1 - \Phi(a))$, so $\mathbb{E}[X^2 \mid X > a] = 1 + a\lambda^c(a)$ and therefore

$$\text{Var}(X \mid X > a) = 1 + a\lambda^c(a) - \lambda^c(a)^2 = 1 - \lambda^c(a)(\lambda^c(a) - a) = 1 - \delta^c(a).$$

(ii) Apply part (i) to $-X$, which is also standard normal, using $\{X < a\} = \{-X > -a\}$. By symmetry of ϕ , $\lambda^c(-a) = \phi(-a)/(1 - \Phi(-a)) = \phi(a)/\Phi(a) = \lambda(a)$, so $\mathbb{E}[X \mid X < a] = -\mathbb{E}[-X \mid -X > -a] = -\lambda^c(-a) = -\lambda(a)$. For the variance, $\text{Var}(X \mid X < a) = \text{Var}(-X \mid -X > -a) = 1 - \delta^c(-a)$, and $\delta^c(-a) = \lambda^c(-a)(\lambda^c(-a) + a) = \lambda(a)(\lambda(a) + a) = \delta(a)$.

(iii) Strict positivity: $\lambda^c(a) > \max(a, 0)$ for every a —the conditional mean of a random variable supported on (a, ∞) strictly exceeds a , and it is strictly positive since the truncated variable is positive with positive probability—hence $\delta^c(a) = \lambda^c(a)(\lambda^c(a) - a) > 0$. Proving strict upper bound $\delta^c(a) < 1$ is equivalent to showing that $\text{Var}(X \mid X > a) > 0$, which holds because the truncated distribution is non-degenerate. The claims for δ follow from the reflection identity $\delta(x) = \delta^c(-x)$.

(iv) Standardize $Y = \mu + \tau X$ and apply (i) yields the desired result. $\square$

D.2 Leave-One-Out Estimation for Stage 2

As we noted in the main text, because the CRC estimator $\hat{\beta}^{CRC}$ utilizes the data of country i , the measurement error v_i and the prediction error E_i are mechanically correlated, hence the optimal weights formula in the main text required an in-sample correction via the leverage. To completely bypass the in-sample correlation, we can construct the structural prediction using a leave-one-out (LOO) estimator

$$\hat{\varepsilon}_i^{CRC, LOO} = (\mathbf{Z}_i^*)' \hat{\beta}^{(-i)}, \quad (\text{D.10})$$

where $\hat{\beta}^{(-i)}$ drops country i from the Stage 2 pooled OLS regression. Because country i 's data is excluded from $\hat{\beta}^{(-i)}$, the estimation error $\hat{\beta}^{(-i)} - \beta$ is strictly independent of $\tilde{u}_i^\varepsilon$ and v_i .

Theorem D.8 (LOO Shrinkage Weight). Under LOO estimation, the optimal shrinkage weight collapses to:

$$w_i^{*, LOO} = \frac{s^2 + (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^*}{s^2 + (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^* + \sigma_i^2}, \quad (\text{D.11})$$

where $\mathbf{V}_\beta^{(-i)} \equiv \text{Var}(\hat{\beta}^{(-i)})$.

Theorem D.8 establishes that the LOO formulation completely eliminates the in-sample correlation by omitting the target country i from the estimation of the structural parameter vector $\hat{\beta}^{(-i)}$. Consequently, the covariance term h_{ii} vanishes, and the weight $w_i^{*,LOO}$ recovers the standard Morris (1983)-like structure, but with a crucial correction for estimation uncertainty around $\hat{\beta}^{CRC}$. The significance of the term $(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^*$ is that it formally propagates the finite-sample estimation error of the structural model into the shrinkage rule. Therefore, these weights have the same properties and behaviour as we described in the main text.

Proof of Theorem D.8. We proceed in three steps.

Step 1: Independence of the LOO estimator from country i 's data The leave-one-out estimator and its exact conditional covariance matrix are

$$\hat{\beta}^{(-i)} = \mathbf{M}_{(-i)}^{-1} \sum_{j \neq i} \sum_{s=1}^T \mathbf{W}_{js} y_{js}^\perp, \quad \mathbf{V}_\beta^{(-i)} = s^2 \mathbf{M}_{(-i)}^{-1} \left(\sum_{j \neq i} \omega_j^2 \mathbf{Z}_j^* (\mathbf{Z}_j^*)' \right) \mathbf{M}_{(-i)}^{-1} + \sigma_\eta^2 \mathbf{M}_{(-i)}^{-1},$$

where $\mathbf{M}_{(-i)} = \mathbf{M} - \mathbf{W}_i' \mathbf{W}_i$; the covariance formula follows from the same decomposition as in Step 1 of the proof of Theorem 2.5, applied to the sample without country i , together with the projection absorption identity of Lemma D.1(iii). The estimation error $\hat{\beta}^{(-i)} - \beta$ is a function of $\{\tilde{u}_j^\varepsilon, \boldsymbol{\eta}_j\}_{j \neq i}$ alone, while v_i is a function of $\boldsymbol{\eta}_i$ and $\tilde{u}_i^\varepsilon$ is country i 's own random effect. Assumption 2.2(iii) makes these three blocks mutually independent conditional on $(\mathbf{R}, \mathbf{Z})$, so

$$\text{Cov} \left(v_i, (\mathbf{Z}_i^*)' (\hat{\beta}^{(-i)} - \beta) \right) = 0, \quad \text{Cov} \left(\tilde{u}_i^\varepsilon, (\mathbf{Z}_i^*)' (\hat{\beta}^{(-i)} - \beta) \right) = 0. \quad (\text{D.12})$$

Step 2: Moments of the LOO prediction error Define $E_i^{LOO} \equiv \hat{\varepsilon}_i^{CRC, LOO} - \varepsilon_i^* = (\mathbf{Z}_i^*)' (\hat{\beta}^{(-i)} - \beta) - \tilde{u}_i^\varepsilon$. By Step 1, $\text{Cov}(v_i, E_i^{LOO}) = 0$, and the two components of E_i^{LOO} are themselves uncorrelated, so the variance is the sum

$$\text{Var}(E_i^{LOO}) = \text{Var} \left((\mathbf{Z}_i^*)' (\hat{\beta}^{(-i)} - \beta) \right) + \text{Var}(\tilde{u}_i^\varepsilon) = (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^* + s^2. \quad (\text{D.13})$$

Both signals remain conditionally unbiased: $\mathbb{E}[E_i^{LOO} \mid \mathbf{R}, \mathbf{Z}] = 0$ because $\hat{\beta}^{(-i)}$ is unbiased for β on the reduced sample and $\mathbb{E}[\tilde{u}_i^\varepsilon] = 0$.

Step 3: Risk minimization with uncorrelated errors The risk of the combination $w\hat{\varepsilon}_i + (1-w)\hat{\varepsilon}_i^{CRC, LOO}$ is $R_i^{LOO}(w) = w^2 \sigma_i^2 + (1-w)^2 \text{Var}(E_i^{LOO})$, with no cross term by Step 1. Setting the derivative

$2w\sigma_i^2 - 2(1-w)\text{Var}(E_i^{LOO})$ to zero (the second derivative $2[\sigma_i^2 + \text{Var}(E_i^{LOO})]$ is strictly positive) gives

$$w_i^{*,LOO} = \frac{\text{Var}(E_i^{LOO})}{\text{Var}(v_i) + \text{Var}(E_i^{LOO})} = \frac{s^2 + (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^*}{s^2 + (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^* + \sigma_i^2}. \quad (\text{D.14})$$

This is the oracle empirical Bayes weight with the parameter-uncertainty correction $(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^*$ added to the structural variance, completing the proof of Theorem D.8. $\square$

D.3 Equivalence and Asymptotic Similarity of LOO and Full-Sample EB

In this section, we prove and quantify the exact difference between the Leave-One-Out estimator $\hat{\beta}^{(-i)}$ and the full-sample estimator $\hat{\beta}^{CRC}$, and subsequently prove that their corresponding empirical Bayes shrinkage weights are asymptotically similar to each other.

Theorem D.9 (Estimation Difference and Weight Equivalence). Under the regularity conditions in Assumption A.2, as $N \rightarrow \infty$:

- (i) **Estimator Update Formula:** The exact algebraic difference between the full-sample OLS estimator $\hat{\beta}^{CRC}$ and the LOO estimator $\hat{\beta}^{(-i)}$ is:

$$\hat{\beta}^{CRC} - \hat{\beta}^{(-i)} = \mathbf{M}^{-1} \mathbf{W}_i' \mathbf{r}_i^{LOO} = O_p(1/N), \quad (\text{D.15})$$

where $\mathbf{r}_i^{LOO} \equiv \mathbf{y}_i^\perp - \mathbf{W}_i \hat{\beta}^{(-i)}$ is the $T \times 1$ vector of leave-one-out residuals for country i , and $\mathbf{W}_i$ is the $T \times K$ matrix of stacked regressors $\mathbf{W}_{it}'$ (with $K = 1 + 2 \dim(Z_{it})$).

- (ii) **Prediction Update Formula:** The difference in their structural predictions for country i is:

$$\hat{\varepsilon}_i^{CRC} - \hat{\varepsilon}_i^{CRC,LOO} = \frac{1}{\omega_i} (\mathbf{R}_i^\perp)' \mathbf{H}_{ii} \mathbf{r}_i^{LOO} = O_p(1/N), \quad (\text{D.16})$$

where $\mathbf{H}_{ii} \equiv \mathbf{W}_i \mathbf{M}^{-1} \mathbf{W}_i'$ is the $T \times T$ diagonal block of the pooled hat matrix corresponding to country i 's observations.

- (iii) **Weight Equivalence:** The difference between the Leave-One-Out shrinkage weight $w_i^{*,LOO}$ and the full-sample weight w_i^* is:

$$w_i^{*,LOO} - w_i^* = h_{ii} \left(\frac{\sigma_i^2}{s^2 + \sigma_i^2} \right) + O_p(1/N^2) = O_p(1/N), \quad (\text{D.17})$$

where the leverage entry $h_{ii} \equiv \omega_i (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{Z}_i^* = O_p(1/N)$ by Lemma D.2.

This theorem establishes that while the full-sample and LOO approaches handle in-sample correlation differently, they converge to the same shrinkage weights and predictions at a rate of $O_p(1/N)$. The practical

message is reassuring: with a moderately large cross-section, the computationally cheaper full-sample procedure with the analytic leverage correction of Theorem 2.5 and the N separate leave-one-out regressions deliver numerically close answers, and the sign of the gap is known—the LOO weight always leans slightly more on the country’s own estimate, because excluding country i makes the cross-country witness genuinely independent but also slightly noisier.

Proof of Theorem D.9. We proceed in four steps.

Step 1: Exact estimator update and its rate The two normal equations are $\mathbf{M}\hat{\beta}^{CRC} = \sum_j \mathbf{W}'_j \mathbf{y}_j^\perp$ and $\mathbf{M}_{(-i)}\hat{\beta}^{(-i)} = \sum_{j \neq i} \mathbf{W}'_j \mathbf{y}_j^\perp$ with $\mathbf{M}_{(-i)} = \mathbf{M} - \mathbf{W}'_i \mathbf{W}_i$. Subtracting the second from the first and adding and subtracting $\mathbf{W}'_i \mathbf{W}_i \hat{\beta}^{(-i)}$,

$$\mathbf{M} \left(\hat{\beta}^{CRC} - \hat{\beta}^{(-i)} \right) = \mathbf{W}'_i \mathbf{y}_i^\perp - \mathbf{W}'_i \mathbf{W}_i \hat{\beta}^{(-i)} = \mathbf{W}'_i \mathbf{r}_i^{LOO},$$

which yields the exact update formula $\hat{\beta}^{CRC} - \hat{\beta}^{(-i)} = \mathbf{M}^{-1} \mathbf{W}'_i \mathbf{r}_i^{LOO}$.

For the rate of convergence, we have $\|\mathbf{M}^{-1}\| \leq 1/(cN)$ by Assumption A.2(vi), while $\|\mathbf{W}'_i \mathbf{r}_i^{LOO}\| = O_p(1)$ because T is bounded, the regressors have bounded fourth moments, and the LOO residuals $\mathbf{r}_i^{LOO} = \mathbf{W}_i(\beta - \hat{\beta}^{(-i)}) + \tilde{u}_i^\varepsilon \mathbf{R}_i^\perp + \boldsymbol{\eta}_i^\perp$ are $O_p(1)$ coordinate-wise. Hence $\hat{\beta}^{CRC} - \hat{\beta}^{(-i)} = O_p(1/N)$.

Step 2: Prediction update Premultiplying the update formula by $(\mathbf{Z}_i^*)'$ and using $\omega_i \mathbf{Z}_i^* = \mathbf{W}'_i \mathbf{R}_i^\perp$ (Lemma D.1(ii)),

$$\hat{\varepsilon}_i^{CRC} - \hat{\varepsilon}_i^{CRC, LOO} = (\mathbf{Z}_i^*)' \mathbf{M}^{-1} \mathbf{W}'_i \mathbf{r}_i^{LOO} = \frac{1}{\omega_i} (\mathbf{R}_i^\perp)' \mathbf{W}_i \mathbf{M}^{-1} \mathbf{W}'_i \mathbf{r}_i^{LOO} = \frac{1}{\omega_i} (\mathbf{R}_i^\perp)' \mathbf{H}_{ii} \mathbf{r}_i^{LOO}.$$

The operator norm of the block satisfies $\|\mathbf{H}_{ii}\| \leq \text{tr}(\mathbf{H}_{ii}) = \text{tr}(\mathbf{M}^{-1} \mathbf{W}'_i \mathbf{W}_i) \leq K \|\mathbf{W}'_i \mathbf{W}_i\|/(cN) = O_p(1/N)$, so the prediction difference is $O_p(1/N)$ by Cauchy–Schwarz.

Step 3: Covariance matrices differ at second order We claim $(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^* = (\mathbf{Z}_i^*)' \mathbf{V}_\beta \mathbf{Z}_i^* + O_p(N^{-2})$. By the Sherman–Morrison–Woodbury identity,

$$\mathbf{M}_{(-i)}^{-1} = \mathbf{M}^{-1} + \mathbf{M}^{-1} \mathbf{W}'_i (\mathbf{I} - \mathbf{H}_{ii})^{-1} \mathbf{W}_i \mathbf{M}^{-1},$$

where $\mathbf{I} - \mathbf{H}_{ii}$ is invertible with probability approaching one because $\|\mathbf{H}_{ii}\| = O_p(1/N) < 1$ eventually. The correction term has norm $O_p(N^{-1}) \cdot O_p(1) \cdot O_p(N^{-1}) = O_p(N^{-2})$, so $\mathbf{M}_{(-i)}^{-1} = \mathbf{M}^{-1} + O_p(N^{-2})$ in norm. The sampling components of the two covariance matrices are $\sigma_\eta^2 \mathbf{M}^{-1}$ and $\sigma_\eta^2 \mathbf{M}_{(-i)}^{-1}$, which therefore differ by $O_p(N^{-2})$. The structural components differ through both the outer matrices ($O_p(N^{-2})$ by the same argument, since the inner sum is $O_p(N)$ and each outer perturbation contributes $O_p(N^{-2}) \cdot O_p(N) \cdot O_p(N^{-1}) = O_p(N^{-2})$) and the removal of the $j = i$ term, which contributes $s^2 \mathbf{M}^{-1} \omega_i^2 \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \mathbf{M}^{-1} =$

$O_p(N^{-2})$ since the middle factor is $O_p(1)$. Projecting along the bounded direction $\mathbf{Z}_i^*$ preserves the rate, which proves the claim. Combining with the projection identity (A.9),

$$(\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^* = s^2 \sum_{j=1}^N h_{ij}^2 + h_{ii} \sigma_i^2 + O_p(N^{-2}). \quad (\text{D.18})$$

Step 4: Exact comparison of the two weight ratios Write the two weights as $w_i^* = n_i^F / D_i^F$ and $w_i^{*,LOO} = n_i^L / D_i^L$ with

$$n_i^F = s^2 A_i, \quad D_i^F = s^2 A_i + \sigma_i^2 (1 - h_{ii}), \quad n_i^L = s^2 + (\mathbf{Z}_i^*)' \mathbf{V}_\beta^{(-i)} \mathbf{Z}_i^*, \quad D_i^L = n_i^L + \sigma_i^2.$$

Using $s^2 A_i = s^2 (1 - 2h_{ii}) + s^2 \sum_j h_{ij}^2$ (Lemma D.2(ii)) and the expansion (D.18), the numerators differ by

$$n_i^L - n_i^F = \left[s^2 + s^2 \sum_j h_{ij}^2 + h_{ii} \sigma_i^2 \right] - \left[s^2 (1 - 2h_{ii}) + s^2 \sum_j h_{ij}^2 \right] + O_p(N^{-2}) = h_{ii} (2s^2 + \sigma_i^2) + O_p(N^{-2}),$$

and consequently the denominators differ by

$$D_i^L - D_i^F = (n_i^L - n_i^F) + \sigma_i^2 - \sigma_i^2 (1 - h_{ii}) = h_{ii} (2s^2 + 2\sigma_i^2) + O_p(N^{-2}).$$

Note also the levels: $n_i^F = s^2 + O_p(1/N)$, $n_i^L = s^2 + O_p(1/N)$, and $D_i^F, D_i^L = s^2 + \sigma_i^2 + O_p(1/N)$, all by Lemma D.2(v). The difference of the two ratios is then

$$w_i^{*,LOO} - w_i^* = \frac{n_i^L D_i^F - n_i^F D_i^L}{D_i^L D_i^F} = \frac{n_i^L (D_i^F - D_i^L) + D_i^L (n_i^L - n_i^F)}{D_i^L D_i^F}.$$

Substituting the computed differences and levels into the numerator,

$$\begin{aligned} n_i^L (D_i^F - D_i^L) + D_i^L (n_i^L - n_i^F) &= - \left(s^2 + O_p(N^{-1}) \right) \cdot 2h_{ii} (s^2 + \sigma_i^2) \\ &\quad + \left(s^2 + \sigma_i^2 + O_p(N^{-1}) \right) \cdot h_{ii} (2s^2 + \sigma_i^2) + O_p(N^{-2}) \\ &= h_{ii} (s^2 + \sigma_i^2) \left[(2s^2 + \sigma_i^2) - 2s^2 \right] + O_p(N^{-2}) = h_{ii} \sigma_i^2 (s^2 + \sigma_i^2) + O_p(N^{-2}), \end{aligned}$$

where the cross terms between $O_p(N^{-1})$ level corrections and $O_p(N^{-1})$ differences are absorbed into the remainder. Dividing by $D_i^L D_i^F = (s^2 + \sigma_i^2)^2 + O_p(N^{-1})$,

$$w_i^{*,LOO} - w_i^* = \frac{h_{ii} \sigma_i^2 (s^2 + \sigma_i^2)}{(s^2 + \sigma_i^2)^2} + O_p(N^{-2}) = h_{ii} \left(\frac{\sigma_i^2}{s^2 + \sigma_i^2} \right) + O_p(N^{-2}),$$

and since $h_{ii} = O_p(1/N)$ by Lemma D.2(v), the difference is $O_p(1/N)$. The leading term is non-negative, so the LOO weight always (weakly) exceeds the full-sample weight, by exactly the leverage times the noise share of the country's total variance. This completes the proof of Theorem D.9. $\square$

D.4 Heteroskedastic Residual Heterogeneity

We extend the baseline model by allowing the structural prior variance $s_i^2 \equiv \text{Var}(\tilde{u}_i^\varepsilon \mid \mathbf{Z}_i^*)$ to be heteroscedastic, taking a log-linear parametric form

$$s_i^2 = \exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa}), \quad (\text{D.19})$$

where $\boldsymbol{\kappa}$ is a parameter vector. The exponential link guarantees positivity for every parameter value and makes the log-variance linear in observables, so $\boldsymbol{\kappa}$ answers a transparent question: which structural characteristics make a country's elasticity *harder to predict* from observables, as opposed to merely larger or smaller? Under the Gaussian hierarchical model, the conditional variance of the observed estimate is $\text{Var}(\hat{\varepsilon}_i \mid \mathbf{Z}_i^*) = \exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa}) + \sigma_i^2$.

Let $r_i \equiv \hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}^{CRC}$ denote the Stage 2 residuals. We emphasize that this is a *two-step* procedure: the residuals are formed with the estimated $\hat{\beta}^{CRC}$, not the true β , and the asymptotic theory below explicitly accounts for the plug-in. The profile log-likelihood for $\boldsymbol{\kappa}$ is

$$\ell_{\text{prof}}(\boldsymbol{\kappa}) = -\frac{1}{2} \sum_{i=1}^N \left[\ln \left(\exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa}) + \sigma_i^2 \right) + \frac{r_i^2}{\exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa}) + \sigma_i^2} \right], \quad (\text{D.20})$$

and the profile MLE is $\hat{\boldsymbol{\kappa}} = \arg \max_{\boldsymbol{\kappa} \in \mathcal{K}} \ell_{\text{prof}}(\boldsymbol{\kappa})$.

We maintain the following regularity assumptions.

Assumption D.10 (Regularity for the Heteroscedastic Hyperparameter MLE). Let $\boldsymbol{\kappa}$ denote the parameter vector of the heteroscedastic structural variance model $s_i^2 = \exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa})$.

- (i) **Compact Parameter Space:** $\boldsymbol{\kappa} \in \mathcal{K} \subset \mathbb{R}^K$, where $\mathcal{K}$ is compact and the true value $\boldsymbol{\kappa}_0$ lies in its interior.
- (ii) **Identification:** The true parameter vector $\boldsymbol{\kappa}_0$ is the unique maximizer over $\mathcal{K}$ of the limiting expected log-likelihood $Q_0(\boldsymbol{\kappa}) = \lim_N N^{-1} \sum_i \mathbb{E}[q_i(\boldsymbol{\kappa})]$.
- (iii) **Full Rank Covariates:** The matrix $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbf{Z}_i^* (\mathbf{Z}_i^*)'$ is positive definite, and $\sup_i \|\mathbf{Z}_i^*\| \leq C < \infty$.
- (iv) **Uniform Boundedness:** The eigenvalues of the Fisher Information matrix $\mathbf{I}(\boldsymbol{\kappa})$ are bounded away from 0 and ∞ uniformly on $\mathcal{K}$, and the sampling variances satisfy $0 < \underline{\sigma}^2 \leq \sigma_i^2 \leq \bar{\sigma}^2 < \infty$.
- (v) **First-Step Rate:** The first-step estimator satisfies $\hat{\beta}^{CRC} - \beta = O_p(N^{-1/2})$.

Under Assumption D.10(i), (iii) and (iv), the variance functions $V_i(\boldsymbol{\kappa}) \equiv \exp((\mathbf{Z}_i^*)' \boldsymbol{\kappa}) + \sigma_i^2$ are bounded above and away from zero uniformly in $(i, \boldsymbol{\kappa}) \in \{1, \dots\} \times \mathcal{K}$, which provides the integrable Lipschitz envelope used in the uniform law of large numbers.

Theorem D.11 (Asymptotic Properties of Heteroskedastic Residual Heterogeneity). Under the regularity conditions in Assumption D.10, as $N \rightarrow \infty$, the two-step profile estimator in (D.20) satisfies

- (i) **Consistency:** $\hat{\kappa} \xrightarrow{p} \kappa_0$.
- (ii) **First-Step Orthogonality:** The estimation error of the first step, $\hat{\beta}^{CRC} - \beta = O_p(N^{-1/2})$, has no first-order effect on $\hat{\kappa}$: the expected cross-derivative of the variance score with respect to β vanishes at the truth, so the plug-in perturbs the score only at order $O_p(N^{-1})$ after normalization.
- (iii) **Asymptotic Normality:** $\sqrt{N}(\hat{\kappa} - \kappa_0) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}(\kappa_0)^{-1})$, where the asymptotic Fisher Information matrix is:

$$\mathbf{I}(\kappa_0) = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{i=1}^N \frac{\exp(2(\mathbf{Z}_i^*)' \kappa_0)}{[\exp((\mathbf{Z}_i^*)' \kappa_0) + \sigma_i^2]^2} \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \succ 0. \quad (\text{D.21})$$

Once consistent estimates $\hat{\kappa}$ are obtained, the predicted structural variance $\hat{s}_i^2 = \exp((\mathbf{Z}_i^*)' \hat{\kappa})$ simply replaces the pooled s^2 in the optimal shrinkage weight formulas. Substantively, heteroscedasticity reallocates trust country by country: a jurisdiction whose characteristics predict a tight structural distribution (small $\hat{s}_i^2$) is shrunk aggressively toward the cross-country prediction, while a jurisdiction in a high-variance region of the covariate space retains its own estimate even if its sampling noise is substantial.

To prove Theorem D.11, we note that this is a two-step M-estimation problem: the criterion is evaluated at residuals $r_i = \hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \hat{\beta}^{CRC}$ that depend on the first-step estimator. We therefore (1) analyze the *infeasible* criterion built from the true residuals $r_i^0 \equiv \hat{\varepsilon}_i - (\mathbf{Z}_i^*)' \beta = \tilde{u}_i^\varepsilon + v_i$, establishing identification, uniform convergence, and asymptotic normality there, and (2) show that the substitution $r_i^0 \mapsto r_i$ perturbs the criterion uniformly by $O_p(N^{-1/2})$ and the normalized score by $o_p(1)$ —the Neyman orthogonality step (which is part (ii) of the result—the technical contribution).

Write $V_i(\kappa) \equiv \exp((\mathbf{Z}_i^*)' \kappa) + \sigma_i^2$ and

$$q_i(\kappa; r) \equiv -\frac{1}{2} \left[\ln V_i(\kappa) + \frac{r^2}{V_i(\kappa)} \right], \quad Q_N(\kappa; \mathbf{r}) \equiv \frac{1}{N} \sum_{i=1}^N q_i(\kappa; r_i).$$

Under Assumption D.10(i), (iii)–(iv), there exist constants $0 < \underline{V} \leq \bar{V} < \infty$ with $\underline{V} \leq V_i(\kappa) \leq \bar{V}$ for all i and all $\kappa \in \mathcal{K}$; we use these bounds repeatedly.

Proof of Theorem D.11. We proceed in six steps.

Step 1: Identification of κ_0 from the infeasible criterion Under the heteroscedastic hierarchical model, $r_i^0 \mid \mathbf{Z}_i^* \sim \mathcal{N}(0, V_i(\kappa_0))$, so $\mathbb{E}[(r_i^0)^2 \mid \mathbf{Z}_i^*] = V_i(\kappa_0)$ and

$$\mathbb{E}[q_i(\kappa; r_i^0)] = -\frac{1}{2} \left[\ln V_i(\kappa) + \frac{V_i(\kappa_0)}{V_i(\kappa)} \right].$$

Subtracting the value at κ_0 and writing $x_i \equiv V_i(\kappa)/V_i(\kappa_0) \in [\underline{V}/\bar{V}, \bar{V}/\underline{V}]$,

$$\mathbb{E}[q_i(\kappa_0; r_i^0)] - \mathbb{E}[q_i(\kappa; r_i^0)] = \frac{1}{2} \left[\ln x_i + \frac{1}{x_i} - 1 \right] \geq 0,$$

where the inequality is the elementary fact that $g(x) = \ln x + 1/x - 1$ is non-negative for $x > 0$ with a unique zero at $x = 1$ (indeed $g'(x) = (x-1)/x^2$, so g decreases on $(0, 1)$ and increases on $(1, \infty)$). Averaging over i and passing to the limit, $Q_0(\kappa_0) - Q_0(\kappa) \geq 0$ with equality if and only if $V_i(\kappa) = V_i(\kappa_0)$ for asymptotically all i , i.e., $(\mathbf{Z}_i^*)'(\kappa - \kappa_0) = 0$ for asymptotically all i by injectivity of the exponential. The full-rank condition of Assumption D.10(iii) then forces $\kappa = \kappa_0$, and strict separation $\sup_{\|\kappa - \kappa_0\| \geq \epsilon} Q_0(\kappa) < Q_0(\kappa_0)$ follows from the compactness of $\mathcal{K}$, the continuity of Q_0 , and the strict positivity of g away from one—which is the identification requirement of Assumption D.10(ii).

Step 2: Uniform law of large numbers for the infeasible criterion The gradient bound

$$\|\nabla_{\kappa} q_i(\kappa; r)\| = \left\| \frac{1}{2} \left[\frac{r^2}{V_i(\kappa)} - 1 \right] \frac{\exp((\mathbf{Z}_i^*)'\kappa)}{V_i(\kappa)} \mathbf{Z}_i^* \right\| \leq \frac{C}{2} \left(\frac{r^2}{\underline{V}} + 1 \right) \equiv L_i(r)$$

holds uniformly on $\mathcal{K}$, because the variance ratio $\exp((\mathbf{Z}_i^*)'\kappa)/V_i(\kappa) \leq 1$ and $\|\mathbf{Z}_i^*\| \leq C$. The envelope $L_i(r_i^0)$ has uniformly bounded second moments since r_i^0 is Gaussian with bounded variance. Hence the family $\{q_i(\cdot; r_i^0)\}$ is Lipschitz in κ with integrable Lipschitz constants on the compact set $\mathcal{K}$; combined with the pointwise law of large numbers for independent arrays (Chebyshev, using the bounded second moments of q_i), a standard bracketing argument (e.g., van der Vaart, 2000, Theorem 19.4 and Example 19.7) yields

$$\sup_{\kappa \in \mathcal{K}} |Q_N(\kappa; \mathbf{r}^0) - Q_0(\kappa)| \xrightarrow{p} 0.$$

Step 3: The plug-in perturbs the criterion negligibly Write $\Delta \equiv \hat{\beta}^{CRC} - \beta = O_p(N^{-1/2})$ and $r_i = r_i^0 - (\mathbf{Z}_i^*)'\Delta$. Since q_i depends on r only through $r^2/V_i(\kappa)$ and $V_i(\kappa) \geq \underline{V}$,

$$\sup_{\kappa \in \mathcal{K}} |Q_N(\kappa; \mathbf{r}) - Q_N(\kappa; \mathbf{r}^0)| \leq \frac{1}{2\underline{V}N} \sum_{i=1}^N |r_i^2 - (r_i^0)^2| \leq \frac{1}{2\underline{V}} \left[2C\|\Delta\| \cdot \frac{1}{N} \sum_{i=1}^N |r_i^0| + C^2\|\Delta\|^2 \right] = O_p(N^{-1/2}),$$

using $|r_i^2 - (r_i^0)^2| = |2r_i^0(\mathbf{Z}_i^*)'\Delta - ((\mathbf{Z}_i^*)'\Delta)^2|$, the bound $\|\mathbf{Z}_i^*\| \leq C$, and the law of large numbers for $N^{-1} \sum |r_i^0|$. Combining Steps 1–3, the feasible criterion converges uniformly to Q_0 , whose unique and well-separated maximizer is κ_0 ; the standard argmax consistency theorem van der Vaart (2000, Theorem 5.7) gives $\hat{\kappa} \xrightarrow{p} \kappa_0$, proving part (i).

Step 4: Score and information at the truth Differentiating q_i with respect to κ and evaluating at (κ_0, r_i^0) ,

$$\nabla_{\kappa} q_i(\kappa_0; r_i^0) = \frac{1}{2} \left[\frac{(r_i^0)^2}{V_i(\kappa_0)} - 1 \right] \frac{\exp((\mathbf{Z}_i^*)' \kappa_0)}{V_i(\kappa_0)} \mathbf{Z}_i^* . \quad (\text{D.22})$$

Since $\mathbb{E}[(r_i^0)^2 \mid \mathbf{Z}_i^*] = V_i(\kappa_0)$, the score has conditional mean zero. Its conditional variance follows from the χ^2 structure, $(r_i^0)^2/V_i(\kappa_0) \sim \chi_1^2$, whose variance is 2, so

$$\text{Var} \left(\nabla_{\kappa} q_i(\kappa_0; r_i^0) \right) = \frac{2}{4} \cdot \frac{\exp(2(\mathbf{Z}_i^*)' \kappa_0)}{V_i(\kappa_0)^2} \mathbf{Z}_i^* (\mathbf{Z}_i^*)' = \frac{1}{2} \cdot \frac{\exp(2(\mathbf{Z}_i^*)' \kappa_0)}{V_i(\kappa_0)^2} \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \equiv \mathbf{I}_i(\kappa_0) .$$

A direct second differentiation shows $\mathbb{E}[\nabla_{\kappa\kappa}^2 q_i(\kappa_0; r_i^0)] = -\mathbf{I}_i(\kappa_0)$: the terms involving $(r_i^0)^2 - V_i$ average out, and the information equality holds because the model is a correctly specified Gaussian likelihood. The score summands are independent across i with $(2 + \iota')$ -th moments bounded uniformly (all moments of χ_1^2 are finite and V_i is bounded), so the Lyapunov central limit theorem gives

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \nabla_{\kappa} q_i(\kappa_0; r_i^0) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}(\kappa_0)) , \quad \mathbf{I}(\kappa_0) = \lim_N \frac{1}{N} \sum_i \mathbf{I}_i(\kappa_0) \succ 0 ,$$

where positive definiteness comes from Assumption D.10(iv).

Step 5: Neyman orthogonality of the plug-in (part (ii)) We compare the feasible and infeasible normalized scores at κ_0 . Since $\nabla_{\kappa} q_i$ depends on r smoothly, a second-order expansion in r around r_i^0 gives

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \left[\nabla_{\kappa} q_i(\kappa_0; r_i) - \nabla_{\kappa} q_i(\kappa_0; r_i^0) \right] = - \left[\frac{1}{N} \sum_{i=1}^N \frac{r_i^0 \exp((\mathbf{Z}_i^*)' \kappa_0)}{V_i(\kappa_0)^2} \mathbf{Z}_i^* (\mathbf{Z}_i^*)' \right] \sqrt{N} \Delta + O_p \left(\sqrt{N} \|\Delta\|^2 \right) ,$$

where we used $\partial_r \nabla_{\kappa} q_i = (r/V_i)(\exp((\mathbf{Z}_i^*)' \kappa_0)/V_i) \mathbf{Z}_i^*$ and bounded the second-order term by the uniform bounds on V_i and $\|\mathbf{Z}_i^*\|$. The bracketed average is a mean-zero independent array ($\mathbb{E}[r_i^0 \mid \mathbf{Z}_i^*] = 0$ enters linearly), so it is $O_p(N^{-1/2}) = o_p(1)$ by Chebyshev's inequality. Multiplied by $\sqrt{N} \Delta = O_p(1)$, the first term is $o_p(1)$. The second term is $O_p(\sqrt{N} \cdot N^{-1}) = O_p(N^{-1/2})$. Hence the feasible and infeasible normalized scores share the same limit distribution, which is the precise content of part (ii).

Step 6: Asymptotic normality The first-order condition $N^{-1} \sum_i \nabla_{\kappa} q_i(\hat{\kappa}; r_i) = 0$ holds with probability approaching one because $\hat{\kappa}$ is consistent for the interior point κ_0 . A mean-value expansion around κ_0 gives

$$0 = \frac{1}{\sqrt{N}} \sum_i \nabla_{\kappa} q_i(\kappa_0; r_i) + \left[\frac{1}{N} \sum_i \nabla_{\kappa\kappa}^2 q_i(\bar{\kappa}; r_i) \right] \sqrt{N} (\hat{\kappa} - \kappa_0) ,$$

with $\bar{\kappa}$ between $\hat{\kappa}$ and κ_0 componentwise. The Hessian average converges in probability to $-\mathbf{I}(\kappa_0)$ uniformly in a shrinking neighborhood of κ_0 , by the same Lipschitz-envelope argument as Step 2 applied to the second derivatives (whose envelopes are again of the form $C(1 + r_i^2)$) together with the consistency of $\bar{\kappa}$ and the negligibility of the plug-in (Step 3 applied to second derivatives). Inverting and applying Steps 4–5,

$$\sqrt{N}(\hat{\kappa} - \kappa_0) = \mathbf{I}(\kappa_0)^{-1} \frac{1}{\sqrt{N}} \sum_i \nabla_{\kappa} q_i(\kappa_0; r_i^0) + o_p(1) \xrightarrow{d} \mathcal{N}\left(0, \mathbf{I}(\kappa_0)^{-1}\right),$$

using the information equality to collapse the sandwich $\mathbf{I}^{-1} \mathbf{II}^{-1} = \mathbf{I}^{-1}$. This proves part (iii). $\square$

E Monte Carlo Simulation

E.1 Baseline Exercise

To verify that the correlated random coefficients (CRC) estimator recovers the mean ETI in the presence of policy endogeneity—while the country-by-country segmented estimator does not—we conduct a Monte Carlo study.

We use $N = 30$ countries, $T = 30$ periods, and 5,000 replications. The true mean elasticity is $\mathbb{E}[\varepsilon_i^*] = 0.5$. Each replication draws a country covariate level $\zeta_i \sim \mathcal{N}(0, 1)$, a time-varying covariate $Z_{it} = \zeta_i + \xi_{it}$ with $\xi_{it} \sim \mathcal{N}(0, 0.5^2)$, and its period-demeaned, standardized counterpart $\ddot{Z}_{it}$. Country intercepts $\alpha_i \sim \mathcal{N}(0, 1)$ and linear-trend coefficients $\lambda_i \sim \mathcal{N}(0, 0.02^2)$ are drawn once per country, and all random components are mutually independent and independent across i and t . The country-specific elasticity is

$$\varepsilon_{it} = \bar{\varepsilon} + b \ddot{Z}_{it} + u_i, \quad \bar{\varepsilon} = 0.5, \quad b = 0.5, \quad u_i \sim \mathcal{N}(0, 0.2^2),$$

so that $\mathbb{E}[\varepsilon_{it}] = \bar{\varepsilon}$. The endogeneity is generated by letting the net-of-tax rate respond both to the elasticity-shifting covariate and to the country's structural type:

$$R_{it} = \log(1 - \tau_{it}) = 0.5 + \theta Z_{it} + \phi u_i + \nu_{it}, \quad \nu_{it} \sim \mathcal{N}(0, 0.5^2),$$

and the outcome is $y_{it} = \alpha_i + \lambda_i t + \varepsilon_{it} R_{it} + \eta_{it}$, where $\eta_{it} \sim \mathcal{N}(0, 0.5^2)$ is the idiosyncratic income shock and α_i, λ_i are the country fixed effect and trend. The *segmented* estimator regresses y_{it} on R_{it} (with a country intercept and trend) separately for each country and averages the slopes; because it omits the $\ddot{Z}_{it} R_{it}$ interaction, it targets the tax-variation-weighted estimand and is biased for $\bar{\varepsilon}$ whenever Z_{it} and R_{it} are correlated. The *CRC* estimator is the pooled regression of y_{it} on R_{it} , the interaction $\ddot{Z}_{it} R_{it}$, the level $\ddot{Z}_{it}$, and country fixed effects and trends; its coefficient on R_{it} estimates $\bar{\varepsilon}$. We report four designs of increasing endogeneity, indexed by (θ, ϕ) :

- Model 1 (0.6, 0);
- Model 2 (1.0, 0);

- Model 3 (1.0, 1.0); and
- Model 4 (1.4, 1.5).

Table 6: Monte Carlo Simulation Results ($\mathbb{E}[\varepsilon_i^*] = 0.5$, $N = T = 30$, 5,000 replications).

DGP	Segmented (per-country, averaged)			Proposed CRC Model		
	Mean	S.D.	% Bias	Mean	S.D.	% Bias
Model 1	0.597	0.055	+19.4%	0.499	0.051	−0.2%
Model 2	0.612	0.064	+22.4%	0.500	0.051	−0.1%
Model 3	0.613	0.069	+22.7%	0.500	0.050	−0.1%
Model 4	0.604	0.078	+20.8%	0.499	0.051	−0.3%

Across all four designs the segmented estimator is biased upward for the mean ETI by roughly 19 to 23 percent, with the bias growing as the policy endogeneity strengthens. The CRC estimator is virtually unbiased (at most 0.3 percent) and is, in addition, somewhat more precise, confirming that the bias in the segmented estimates reflects policy endogeneity rather than sampling noise.

E.2 Credibility of the Feasible Estimates

The Monte Carlo exercise of Appendix E.1 establishes that, under policy endogeneity, the segmented estimator is biased for the mean ETI while the CRC estimator is not. A natural follow-up concern is one of interpretation: when the segmented and CRC estimates for a given country diverge, does the gap reflect *genuine bias correction*—the second stage removing the endogeneity between tax rates and elasticities—or is it an *artifact* of forcing each country onto the fitted cross-country relationship $(\mathbf{Z}_i^*)'\beta$? A related concern is whether the feasible empirical Bayes (EB) estimate is credible when the structural second stage is imperfect, given that richer within-country designs such as bunching or dynamic event studies are infeasible at the cross-country scale our exercise requires. Because these questions cannot be settled with the observational cross-country data alone, we address them with simulated data, where the true country-specific elasticities are known.

We build both experiments in this section on a single data-generating process (DGP) that makes the country-specific estimand explicit. For each country i we draw four time-invariant primitives: a covariate level $\zeta_i \sim \mathcal{N}(0, 1)$, an intercept $\alpha_i \sim \mathcal{N}(0, 1)$, a linear-trend coefficient $\lambda_i \sim \mathcal{N}(0, 0.02^2)$, and an unobserved structural type $u_i \sim \mathcal{N}(0, s^2)$. The observed covariate follows

$$Z_{it} = \zeta_i + \xi_{it}, \quad \xi_{it} \sim \mathcal{N}(0, 0.5^2), \quad (\text{E.1})$$

from which we form the Mundlak time-average $\bar{Z}_i = \frac{1}{T} \sum_{s=1}^T Z_{is}$ and the standardized within-country

deviation $\ddot{Z}_{it} = (Z_{it} - \bar{Z}_i) / \text{sd}(Z_{it} - \bar{Z}_i)$. The true, time-varying elasticity is

$$\varepsilon_{it} = \bar{\varepsilon} + b\ddot{Z}_{it} + \pi\bar{Z}_i + u_i, \quad \bar{\varepsilon} = 0.5, \quad (\text{E.2})$$

so that the *country-level estimand*—the time-average elasticity that Stages 2 and 3 target—is $\varepsilon_i^* = \bar{\varepsilon} + \pi\bar{Z}_i + u_i$. In (E.2), b is the within-country slope on the covariate deviation, π governs how much of the cross-country variation in ε_i^* the observable covariate $\bar{Z}_i$ explains, and $s^2 = \text{Var}(u_i)$ is the residual (unobserved) structural heterogeneity (from the hierarchical model (8)). The net-of-tax rate is chosen with policy endogeneity, responding both to the elasticity-shifting covariate (through θ) and to the country’s unobserved type (through ϕ):

$$R_{it} = 0.5 + \theta Z_{it} + \phi u_i + \nu_{it}, \quad \nu_{it} \sim \mathcal{N}(0, 0.5^2). \quad (\text{E.3})$$

Finally, the outcome is generated by the Stage-1 panel model (2),

$$y_{it} = \alpha_i + \lambda_i t + \varepsilon_{it} R_{it} + \eta_{it}, \quad \eta_{it} \sim \mathcal{N}(0, 0.5^2), \quad (\text{E.4})$$

where η_{it} is the idiosyncratic income shock. The two experiments below differ only in the parameter values $(\theta, \phi, \pi, b, \tau)$ assigned to (E.1)–(E.4); every other feature of the DGP is held fixed.

We compute three estimates of each country’s elasticity. The *segmented* estimate $\hat{\varepsilon}_i$ is the per-country OLS slope of y_{it} on R_{it} (with a country intercept and trend), together with its model-based sampling variance σ_i^2 . The *CRC* estimate is the country prediction $\hat{\varepsilon}_i^{CRC} = \hat{\varepsilon} + \hat{\pi}\bar{Z}_i$ from the pooled regression of y_{it} on R_{it} , the interactions $\ddot{Z}_{it}R_{it}$ and $\bar{Z}_iR_{it}$, the level $\ddot{Z}_{it}$, and country fixed effects and trends. The *empirical Bayes* estimate combines the two, $\hat{\varepsilon}_i^{EB} = w_i\hat{\varepsilon}_i + (1 - w_i)\hat{\varepsilon}_i^{CRC}$, with the residual heterogeneity s^2 estimated by marginal maximum likelihood from the N pairs $(\hat{\varepsilon}_i, \hat{\varepsilon}_i^{CRC})$ and the posterior-mean weight $w_i = s^2 / (s^2 + \sigma_i^2)$.¹⁰

Exercise 1: Is the gap signal or artifact? If the segmented–CRC gap were merely a mechanical consequence of imposing the cross-country specification, it would appear even when the segmented estimator is unbiased. To test this, we hold the covariates informative ($\pi = 0.30$, $b = 0.40$, $\tau = 0.10$) and vary only whether the tax rate is chosen endogenously. Table 7 reports the cross-country average of the segmented and CRC estimates and the gap between them.

When tax rates are exogenous the two estimators agree almost exactly and essentially unbiased for the true parameter. They diverge only when tax rates are chosen in anticipation of the elasticity, and it is then almost exactly the bias in the segmented estimator—which the CRC estimator removes.

¹⁰We use the posterior-mean (Morris) weight, which is the leave-one-out limit of the leverage-corrected weight of Theorem 2.5 once the mechanical correlation h_{ii} between $\hat{\varepsilon}_i$ and $\hat{\varepsilon}_i^{CRC}$ is removed; with $N = 30$ the two are numerically indistinguishable (Remark 2.12).

Table 7: Placebo test: the segmented–CRC gap appears only under policy endogeneity ($\bar{\varepsilon} = 0.5$, $N = T = 30$, 5,000 replications).

Tax-rate design	Segmented		CRC		Gap
	Mean	% Bias	Mean	% Bias	
Exogenous ($\theta = \phi = 0$)	0.500	+0.1%	0.500	−0.0%	0.000
Endogenous ($\theta = \phi = 1$)	0.705	+41.1%	0.500	+0.0%	0.205

Exercise 2: Is the feasible estimate credible when the structural model is weak? The credibility concern is that a misspecified or uninformative second stage could contaminate the EB estimate by pulling it toward a poor structural prediction. To isolate this channel we turn the policy endogeneity off ($\theta = \phi = 0$), so that the segmented estimator is an unbiased benchmark, and vary the share ρ of the cross-country variation in ε_i^* that the observable covariate explains, holding the total dispersion fixed at $\pi^2 \text{Var}(\bar{Z}_i) + s^2 = 0.09$. Table 8 reports the estimated residual heterogeneity, the average shrinkage weight, and the root mean squared error (RMSE) of each estimator for the country-level estimand ε_i^* .

Table 8: Conservatism and compound risk: the EB estimator tilts toward the within-country estimate as the covariates weaken and caps the risk of naive structural prediction ($N = T = 30$, 5,000 replications, endogeneity off).

Covariate informativeness ρ	Estimated $\hat{\varsigma}^2$	Mean weight $\bar{w}$	Country-level RMSE for ε_i^*		
			Segmented	CRC	EB
0.00	0.087	0.645	0.220	0.295	0.182
0.25	0.066	0.578	0.219	0.257	0.174
0.50	0.045	0.482	0.219	0.211	0.161
0.75	0.025	0.336	0.219	0.153	0.137
0.95	0.010	0.170	0.218	0.081	0.090

Three patterns stand out. *First, the estimated residual heterogeneity correctly detects how much the covariate explains.* Reading down the $\hat{\varsigma}^2$ column, the estimate is largest when the covariate is uninformative ($\hat{\varsigma}^2 = 0.087$ at $\rho = 0$) and falls monotonically toward zero as the covariate accounts for more of the cross-country variation ($\hat{\varsigma}^2 = 0.010$ at $\rho = 0.95$). The second stage therefore diagnoses its own explanatory power directly from the data, rather than having it asserted by the analyst.

Second, the shrinkage weight adjusts automatically, which is what makes the EB estimator conservative. Because the weight the empirical Bayes rule places on a country’s own segmented estimate is increasing in the residual heterogeneity, $\bar{w}$ rises as the covariate weakens—from $\bar{w} = 0.17$ when the covariate is nearly perfect ($\rho = 0.95$) to $\bar{w} = 0.65$ when it is uninformative ($\rho = 0$). A weak second stage thus cannot drag the estimate toward a spurious structural prediction: the data-driven weight discounts any prediction that fails to explain the observed dispersion, so the estimator leans on the credible within-country estimate precisely when the structural model has least to offer. This is the sense in which the feasible estimate is *conservative*.

by construction.

Third, and most directly on credibility, the EB estimator caps the risk of a poorly specified model. The final three columns report the RMSE of each estimator for the country-level estimand ε_i^* . Solely relying on the structural predictions of the CRC when the covariates are uninformative (i.e. low ρ) is bad for the risk: the CRC RMSE more than triples, from 0.081 at $\rho = 0.95$ to 0.295 at $\rho = 0$. The EB RMSE, by contrast, never exceeds that of the within-country segmented estimator (about 0.22 in every row) and is strictly below *both* inputs in every design except the near-perfect-covariate case ($\rho \leq 0.75$). The sole exception is the near-perfect-covariate design ($\rho = 0.95$), where the pure structural estimator edges it out (0.081 versus 0.090)—a case in which the structural model is essentially correct and there is almost no residual heterogeneity to protect against. The practical message is that the feasible estimate never does worse than the within-country estimate that the identification concern takes as the credible benchmark, while delivering substantial risk reductions whenever the structural model provides identifying content.